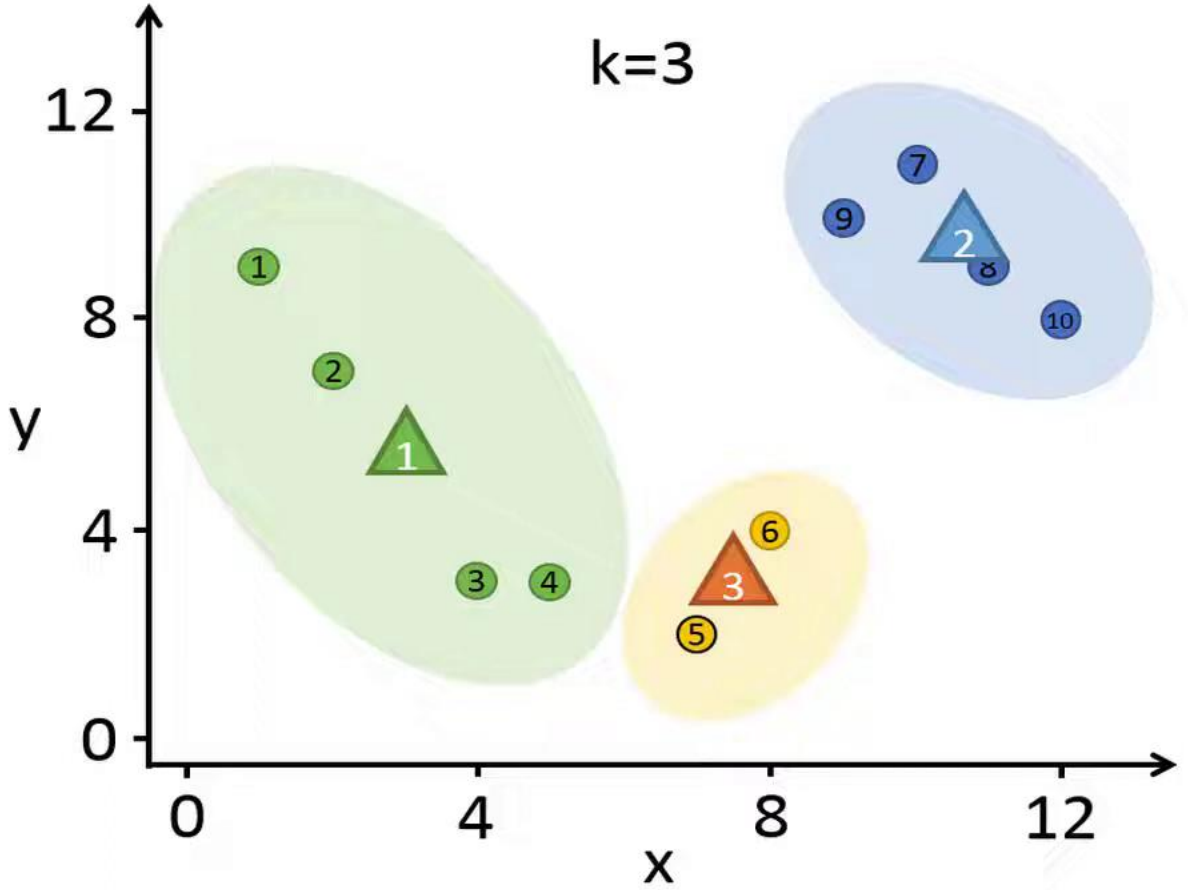
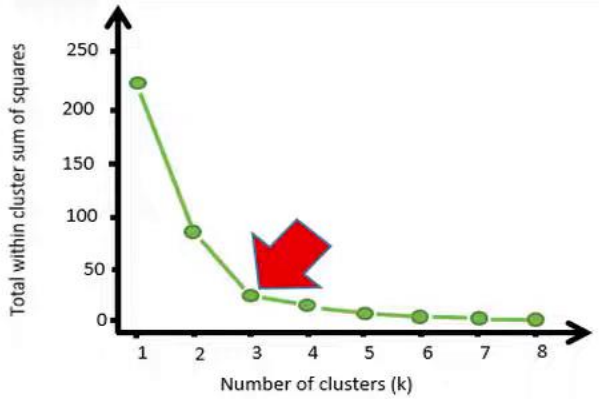
k-means clustering

by Dr. Hafeez Anwar

k-means clustering

	x	y
1	1	9
2	2	7
3	4	3
4	5	3
5	7	2
6	8	4
7	10	11
8	11	9
9	9	10
10	12	8





- How k-means clustering works
- How to calculate the within cluster sum of squares
- Select an appropriate value of *k* based on the elbow method

Clustering

K-means clustering is a method to divide the data into *k* clusters or groups. In comparison to hierarchical clustering, k-means clustering allows the user to determine the number of clusters that should be generated.

k-means clustering









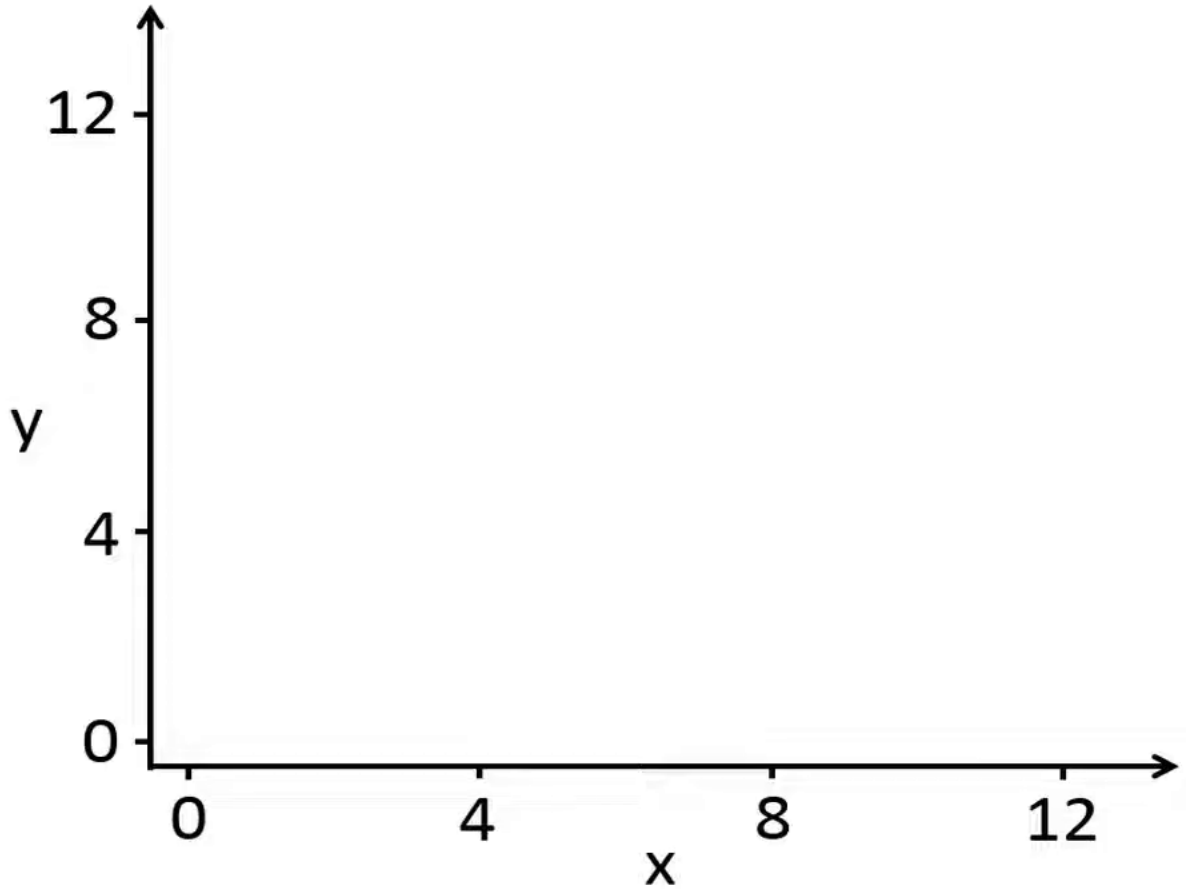


	A	B	C	D	E
Diastolic BP (mmHg)	70	86	78	60	75
Systolic BP (mmHg)	130	147	137	121	134
Weight (kg)	100	90	100	90	106
Height (cm)	180	170	181	171	185
Total cholesterol (mg/dl)	190	215	192	215	200

For example, if we like to divide these five individuals into two groups,

k-means clustering

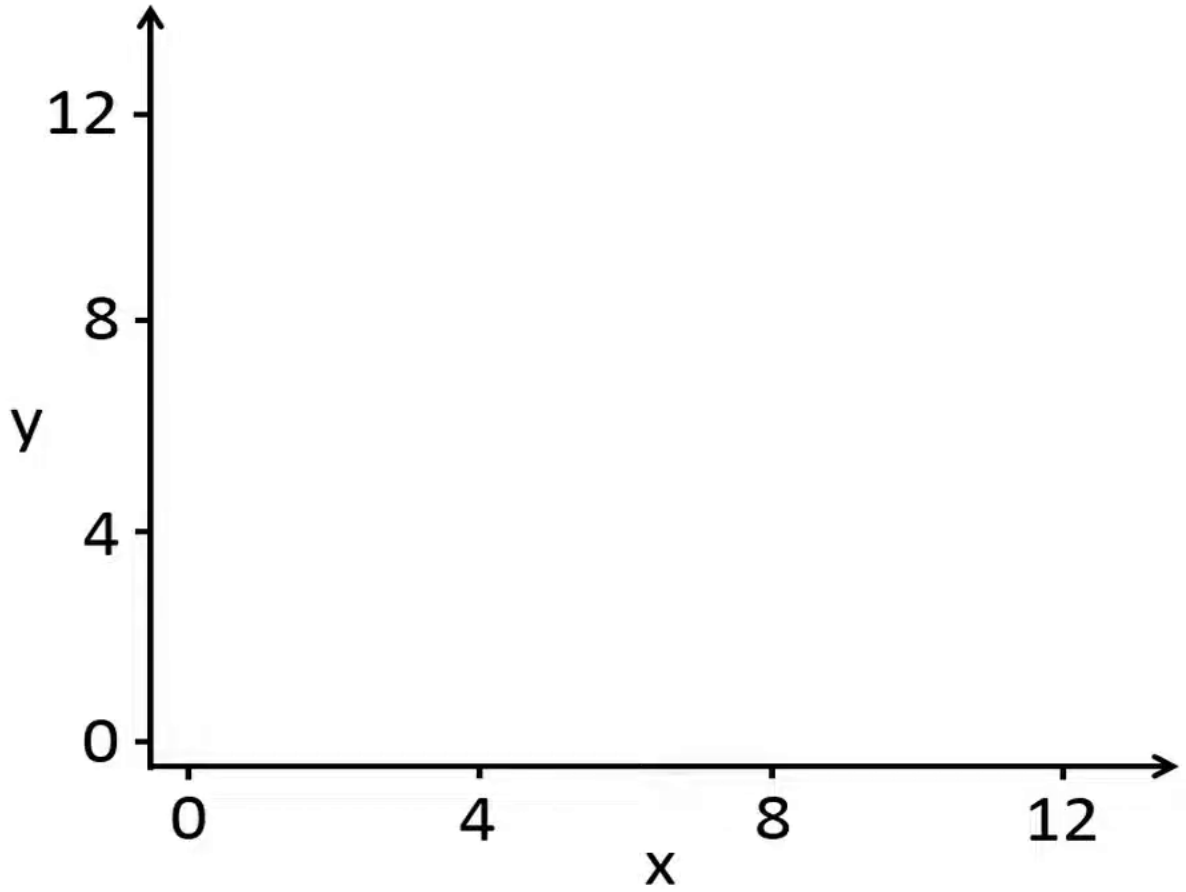
	x	y
1	1	9
2	2	7
3	4	3
4	5	3
5	7	2
6	8	4
7	10	11
8	11	9
9	9	10
10	12	8



To illustrate how k-means clustering works, we will use the following data. We could think of this data as two variables, for example, the body weight and body height, that have been measured on 10 subjects.

k-means clustering

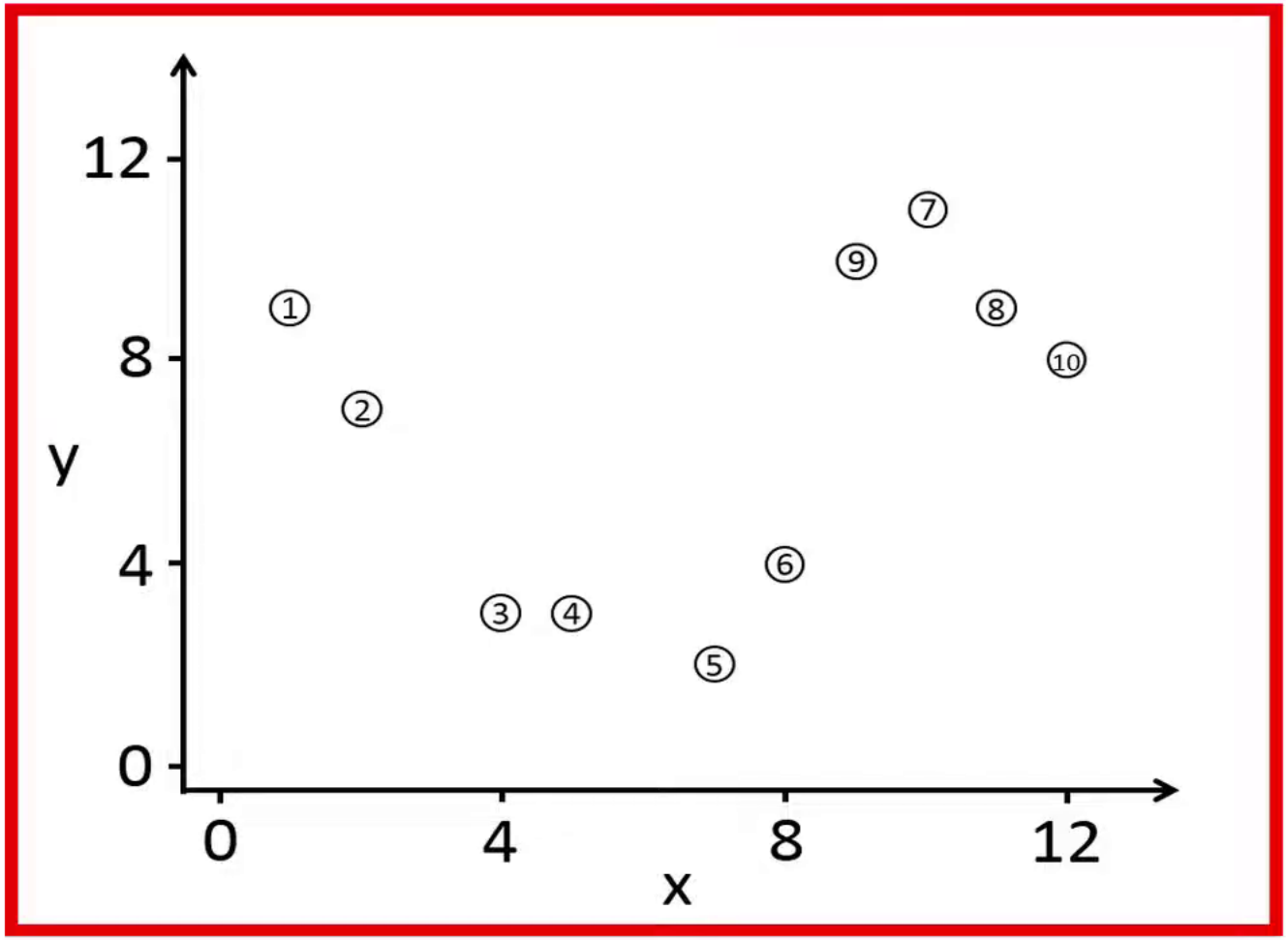
	x	y
1	1	9
2	2	7
3	4	3
4	5	3
5	7	2
6	8	4
7	10	11
8	11	9
9	9	10
10	12	8



Note that k-means clustering can be used with more than two variables. However, it is a lot easier to illustrate how the method works if we only use two dimensions.

k-means clustering

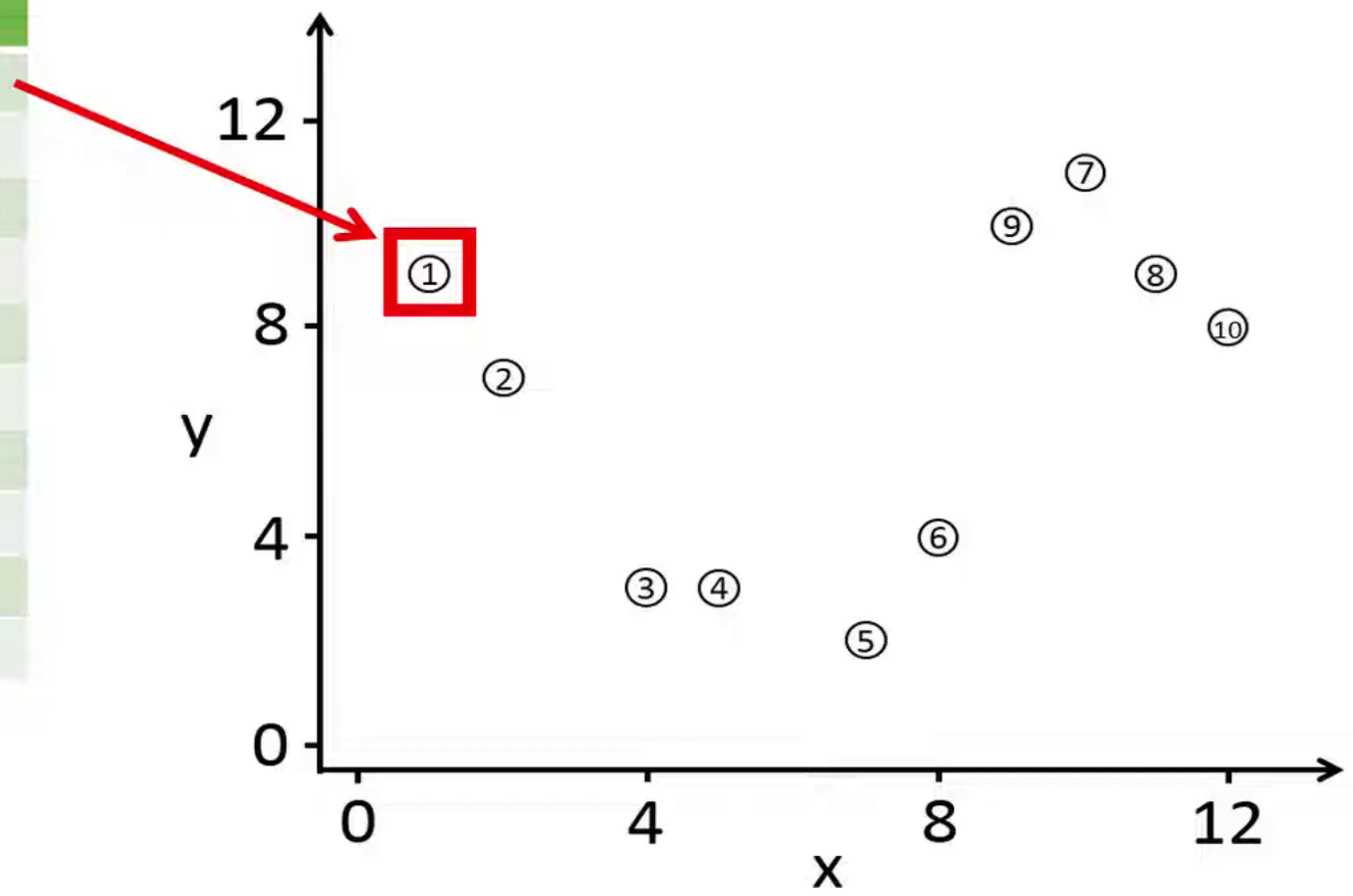
	x	y
1	1	9
2	2	7
3	4	3
4	5	3
5	7	2
6	8	4
7	10	11
8	11	9
9	9	10
10	12	8



Let’s make a scatter plot of the data,

k-means clustering

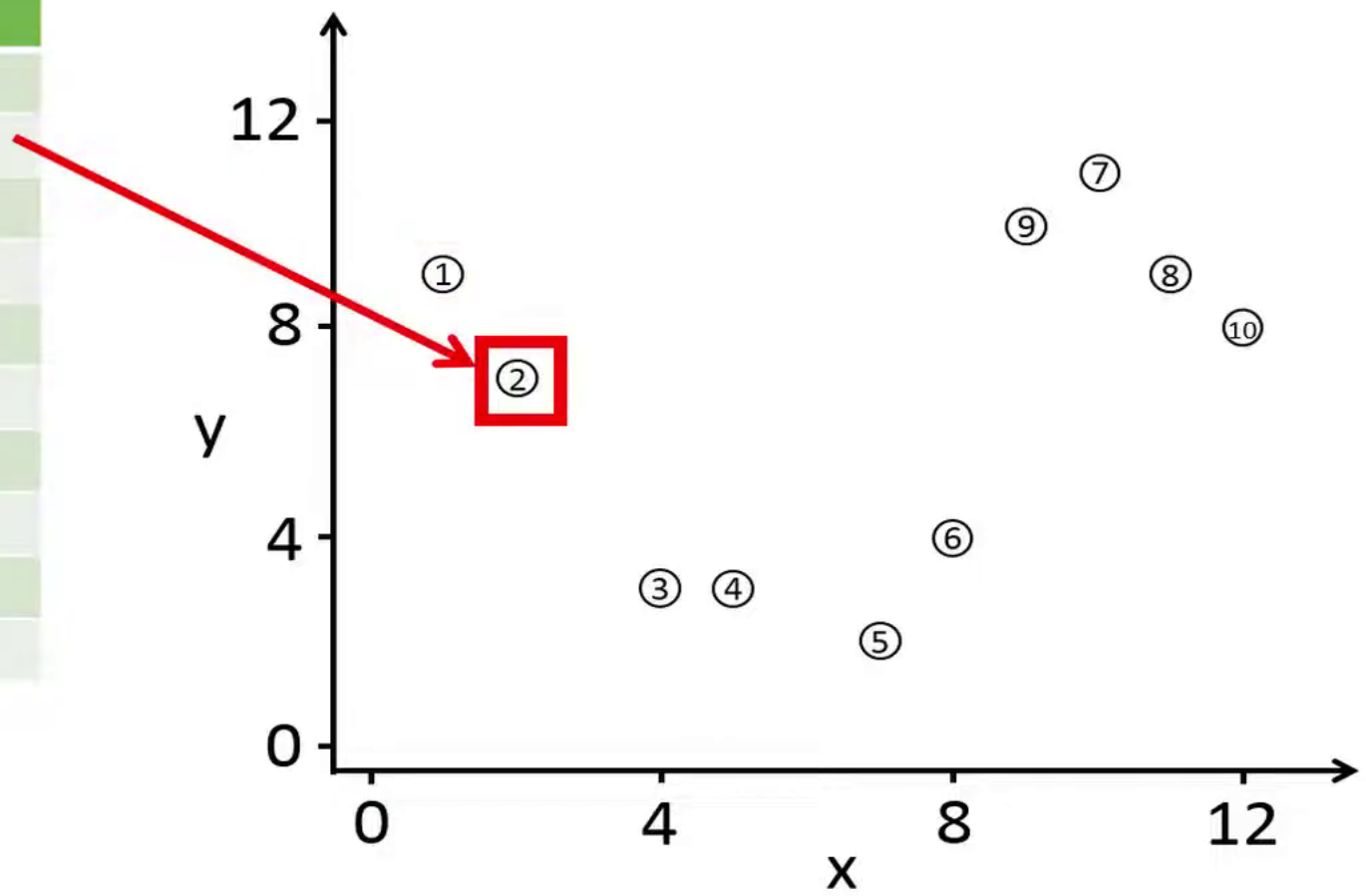
	x	y
1	1	9
2	2	7
3	4	3
4	5	3
5	7	2
6	8	4
7	10	11
8	11	9
9	9	10
10	12	8



where this is data point number one,

k-means clustering

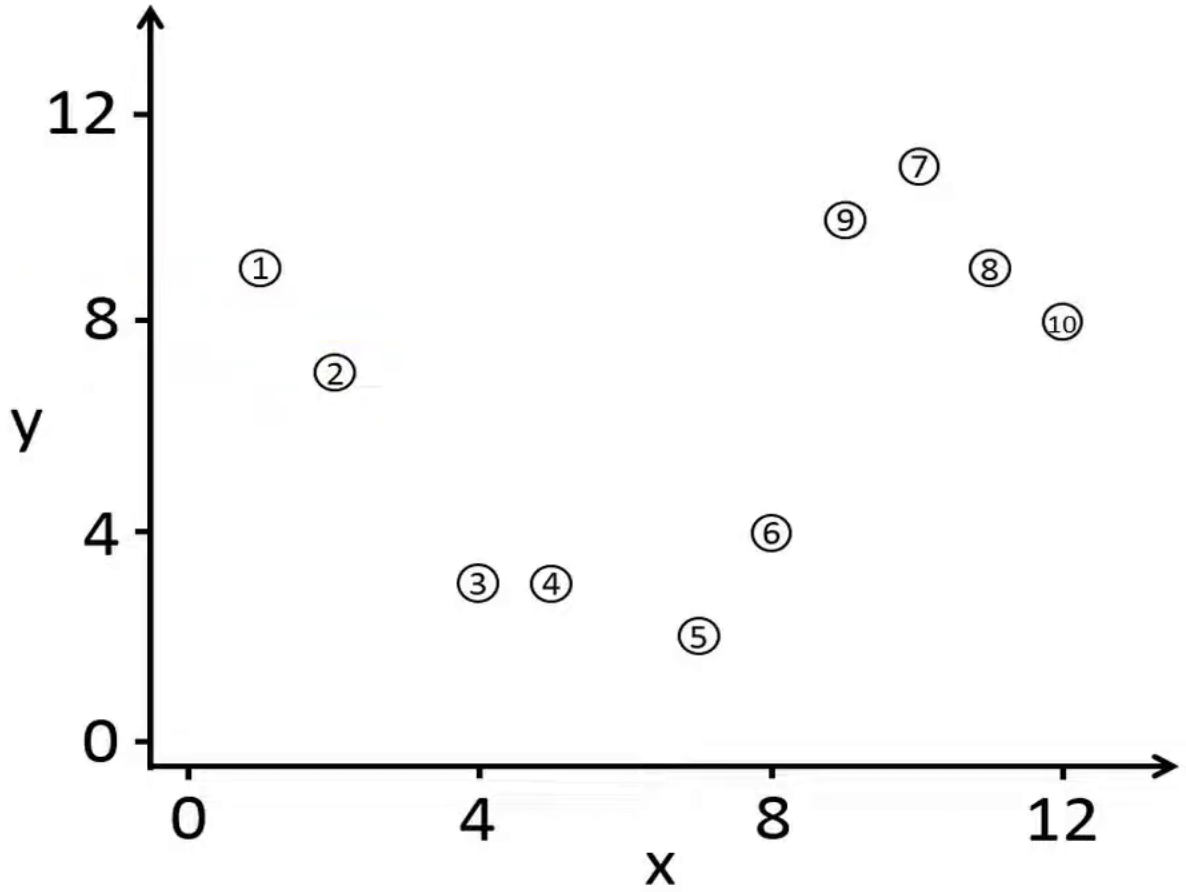
	x	y
1	1	9
2	2	7
3	4	3
4	5	3
5	7	2
6	8	4
7	10	11
8	11	9
9	9	10
10	12	8



and this is data point number 2, and so forth.

k=3

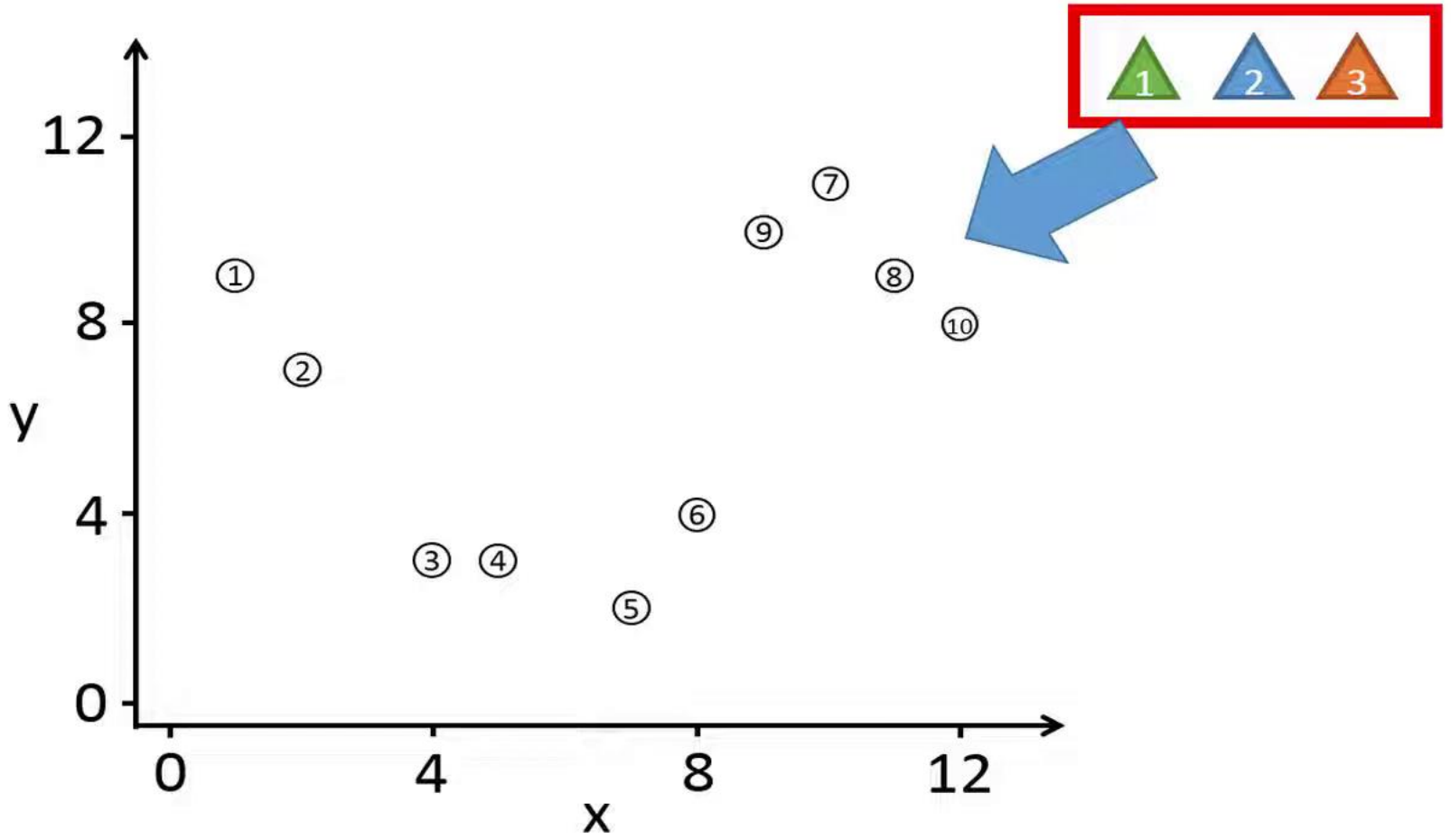
	x	y
1	1	9
2	2	7
3	4	3
4	5	3
5	7	2
6	8	4
7	10	11
8	11	9
9	9	10
10	12	8



The first thing that we should do is to determine the value of k, which is the number of clusters the algorithm should identify. Later, we will try different values of k, but we here begin to set k to three.

k=3

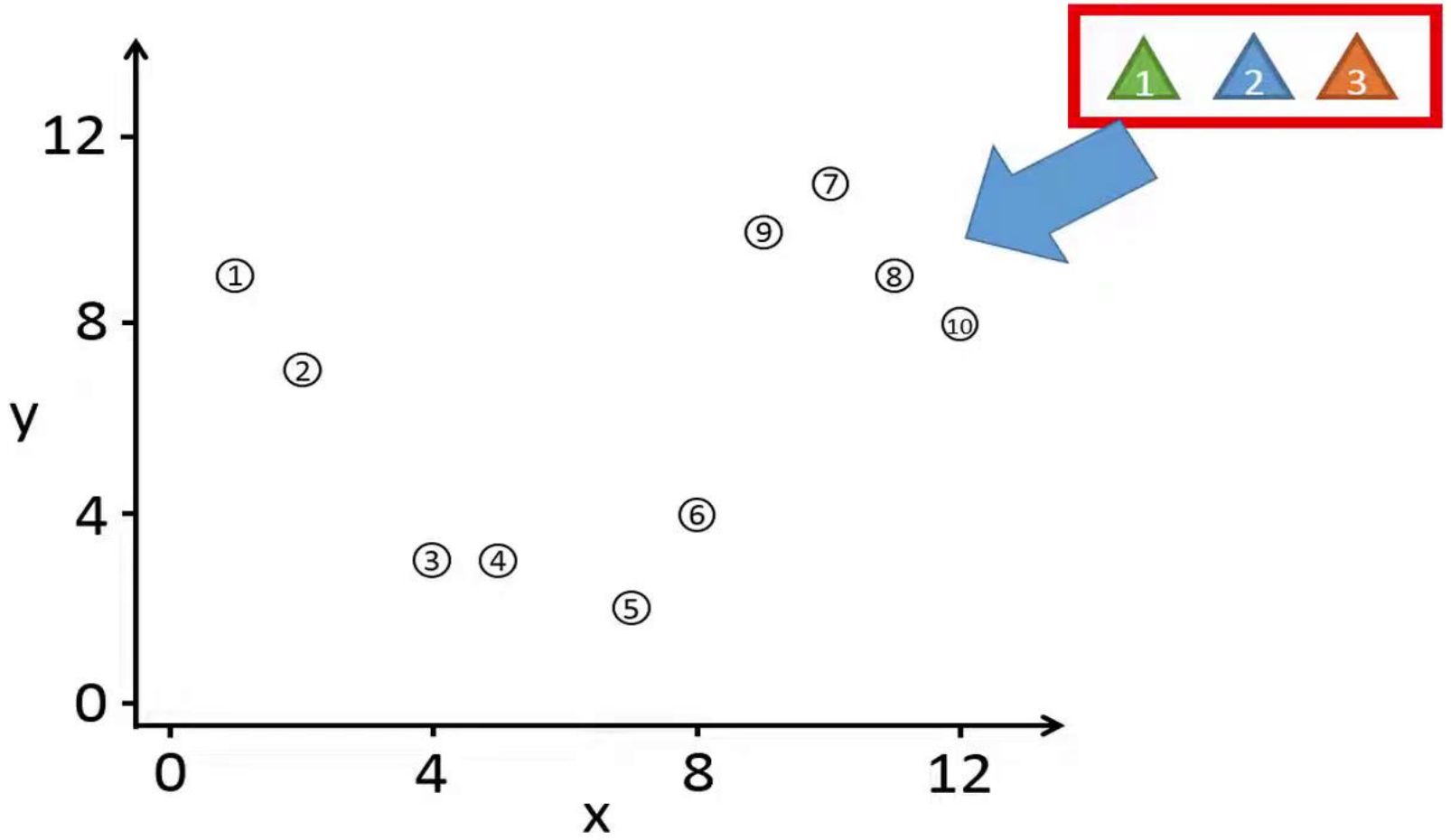
	x	y
1	1	9
2	2	7
3	4	3
4	5	3
5	7	2
6	8	4
7	10	11
8	11	9
9	9	10
10	12	8



Next, we should place the so-called centroids at some initial positions. Since k is equal to three, we need three centroids. Several different methods can be used to randomly select the initial positions of these centroids.

k=3

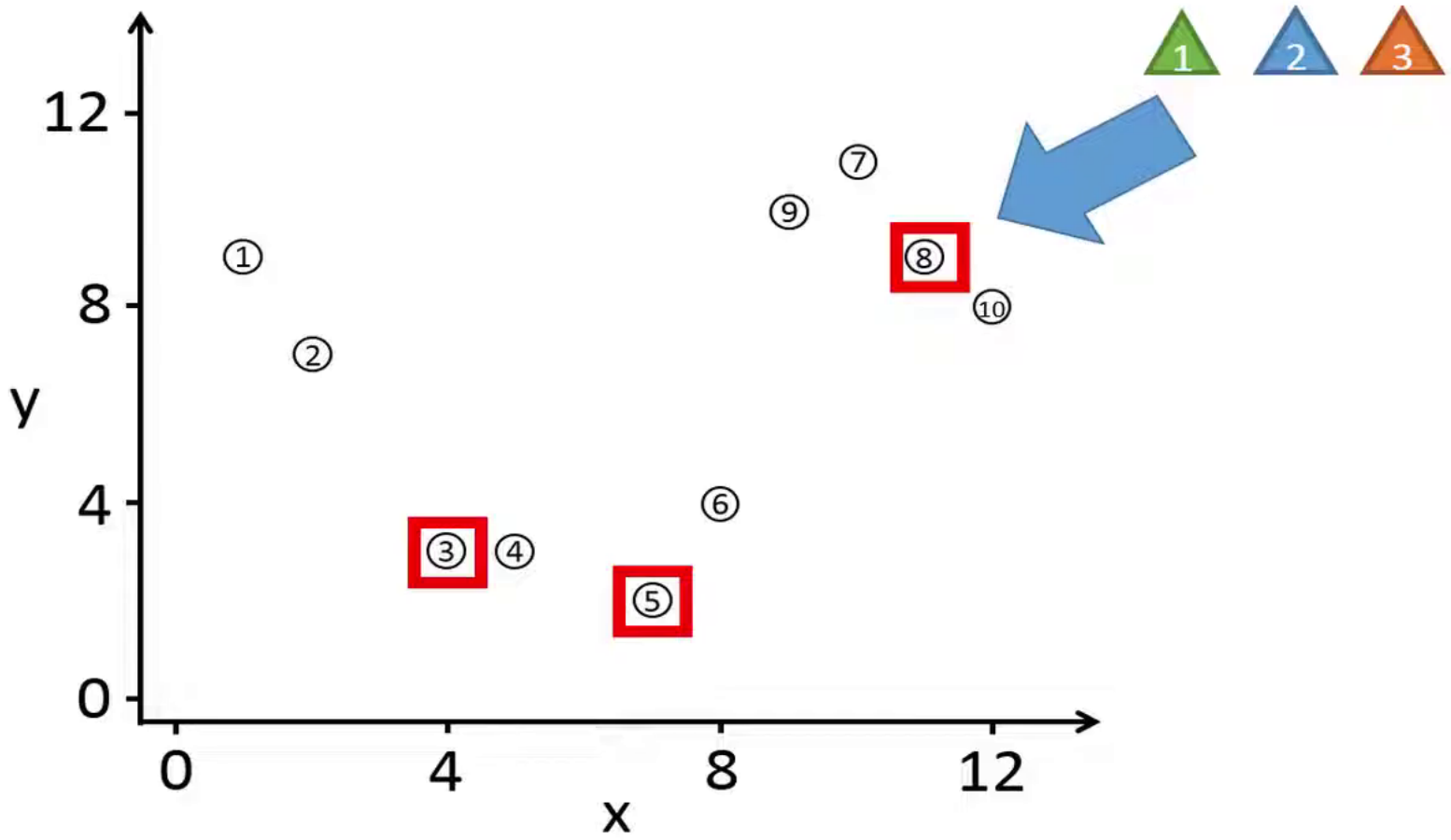
	x	y
1	1	9
2	2	7
3	4	3
4	5	3
5	7	2
6	8	4
7	10	11
8	11	9
9	9	10
10	12	8



One way is to use the Forgy method, which chooses *k* random data points and use the coordinates of these as initial positions for the centroids.

k=3

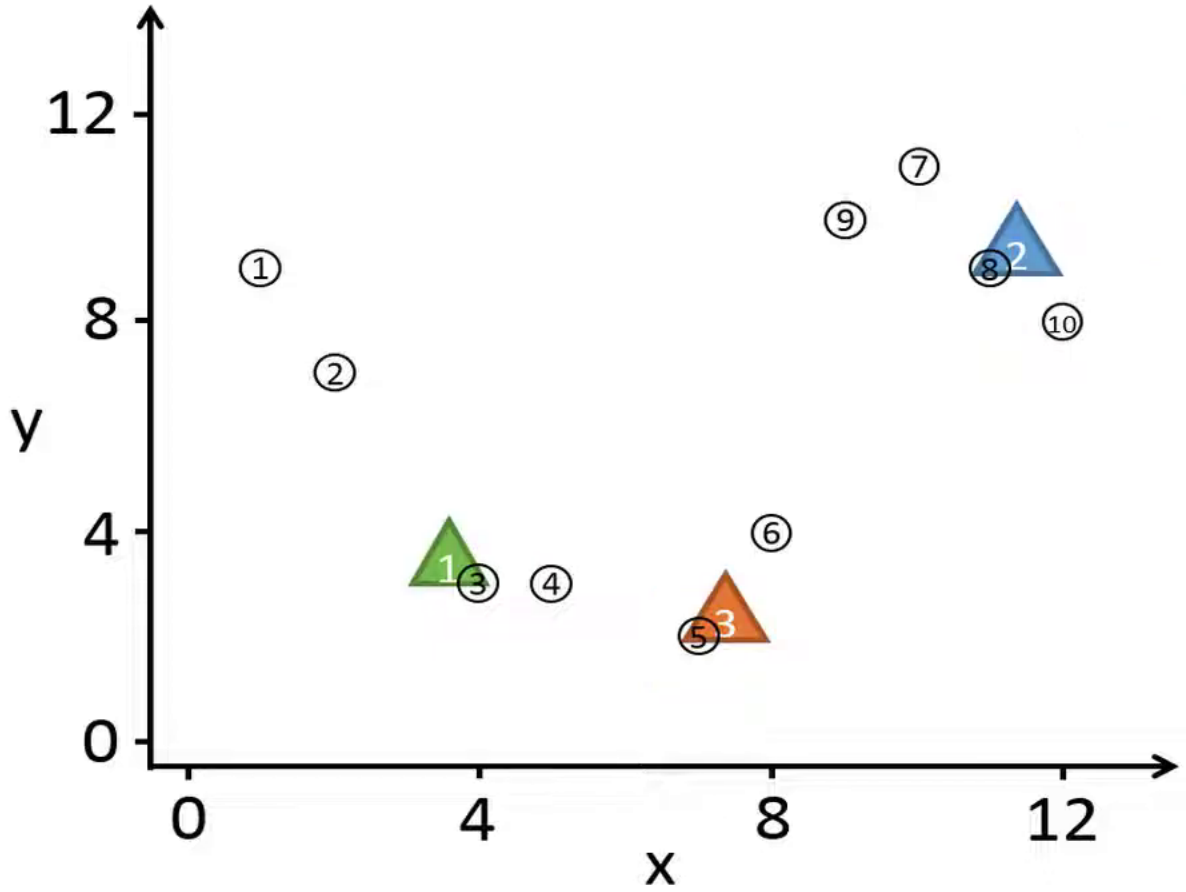
	x	y
1	1	9
2	2	7
3	4	3
4	5	3
5	7	2
6	8	4
7	10	11
8	11	9
9	9	10
10	12	8



For example, let's say that the following three data points were randomly selected.

k=3

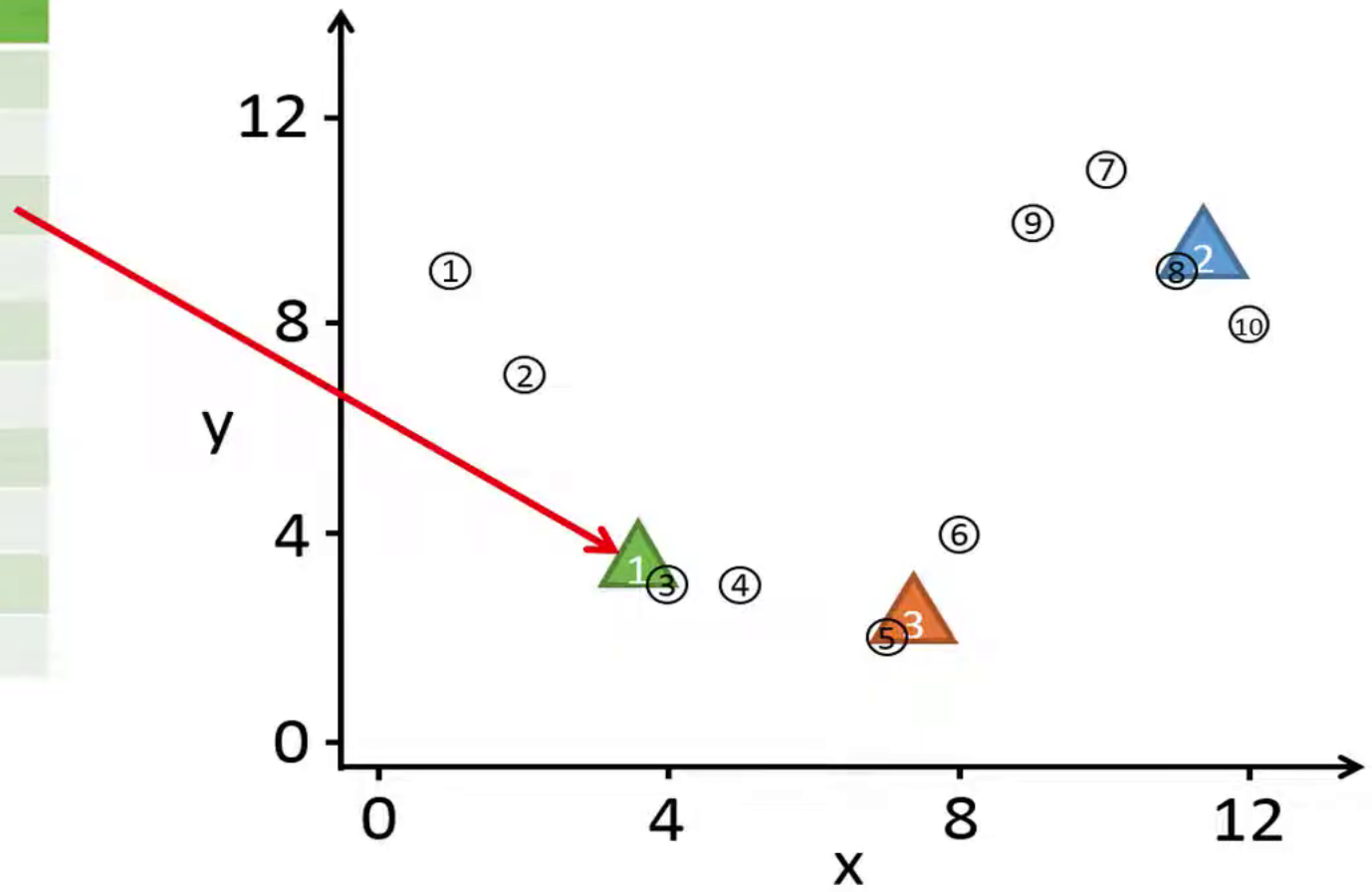
	x	y
1	1	9
2	2	7
3	4	3
4	5	3
5	7	2
6	8	4
7	10	11
8	11	9
9	9	10
10	12	8



We therefore move the centroids to the coordinates of these three data points.

k=3

	x	y
1	1	9
2	2	7
3	4	3
4	5	3
5	7	2
6	8	4
7	10	11
8	11	9
9	9	10
10	12	8

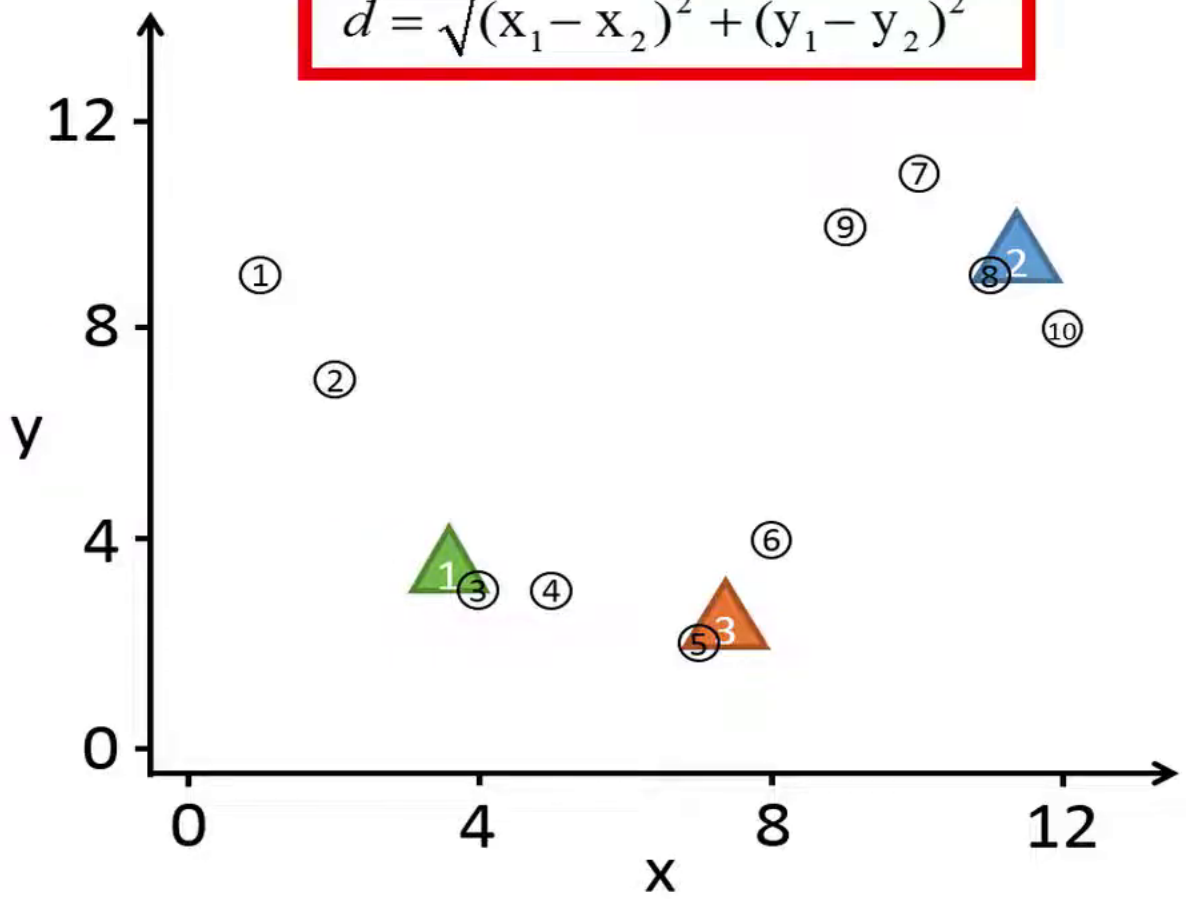


For example, centroid number one has initially the same coordinates as data point number three.

k=3

	x	y
1	1	9
2	2	7
3	4	3
4	5	3
5	7	2
6	8	4
7	10	11
8	11	9
9	9	10
10	12	8

$$d = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$

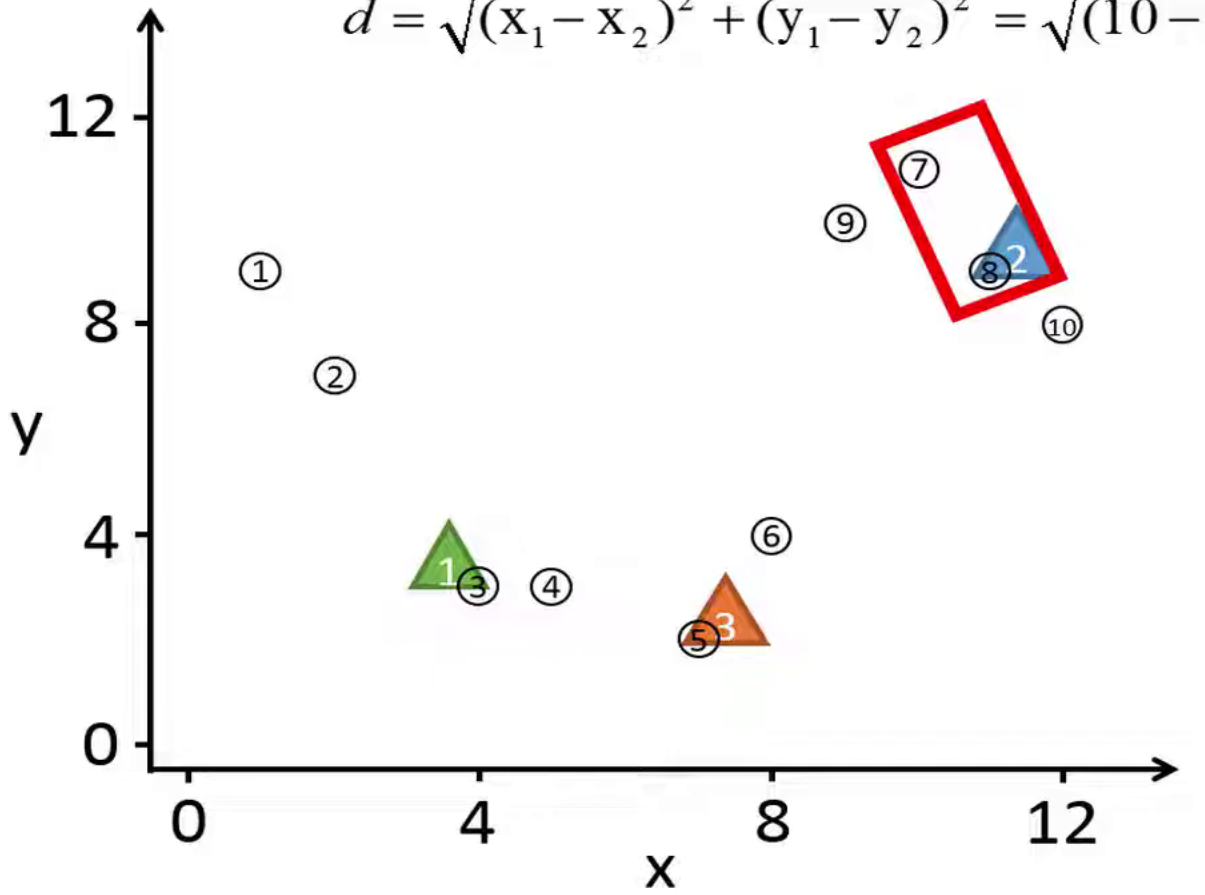


Next, we assign each data point to the closest centroid based on, for example, the Euclidean distance.

k=3

	x	y
1	1	9
2	2	7
3	4	3
4	5	3
5	7	2
6	8	4
7	10	11
8	11	9
9	9	10
10	12	8

$$d = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2} = \sqrt{(10-11)^2 + (11-9)^2} = \sqrt{5}$$

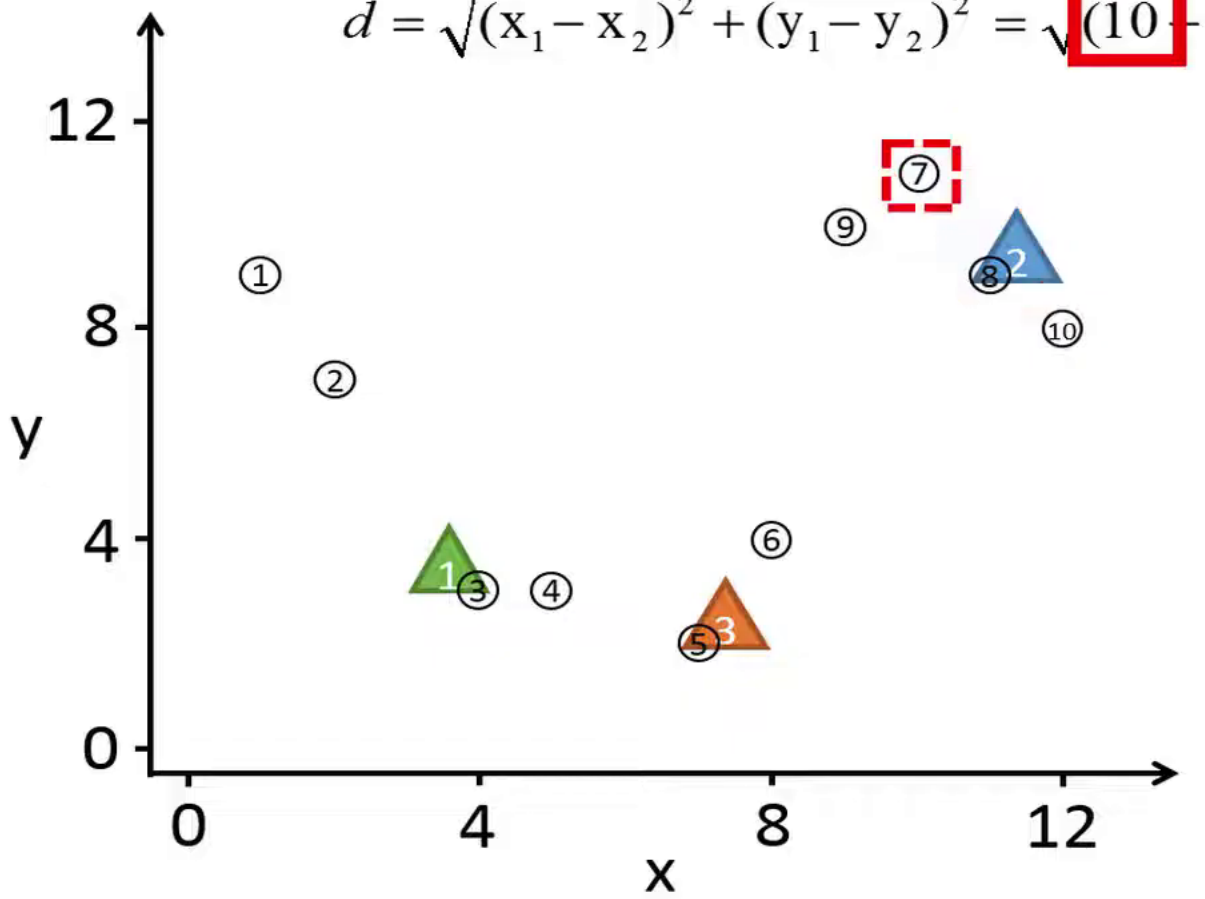


To calculate the Euclidean distance between data point number 7 and the second centroid,

k=3

	x	y
1	1	9
2	2	7
3	4	3
4	5	3
5	7	2
6	8	4
7	10	11
8	11	9
9	9	10
10	12	8

$$d = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2} = \sqrt{(10 - 11)^2 + (11 - 9)^2} = \sqrt{5}$$

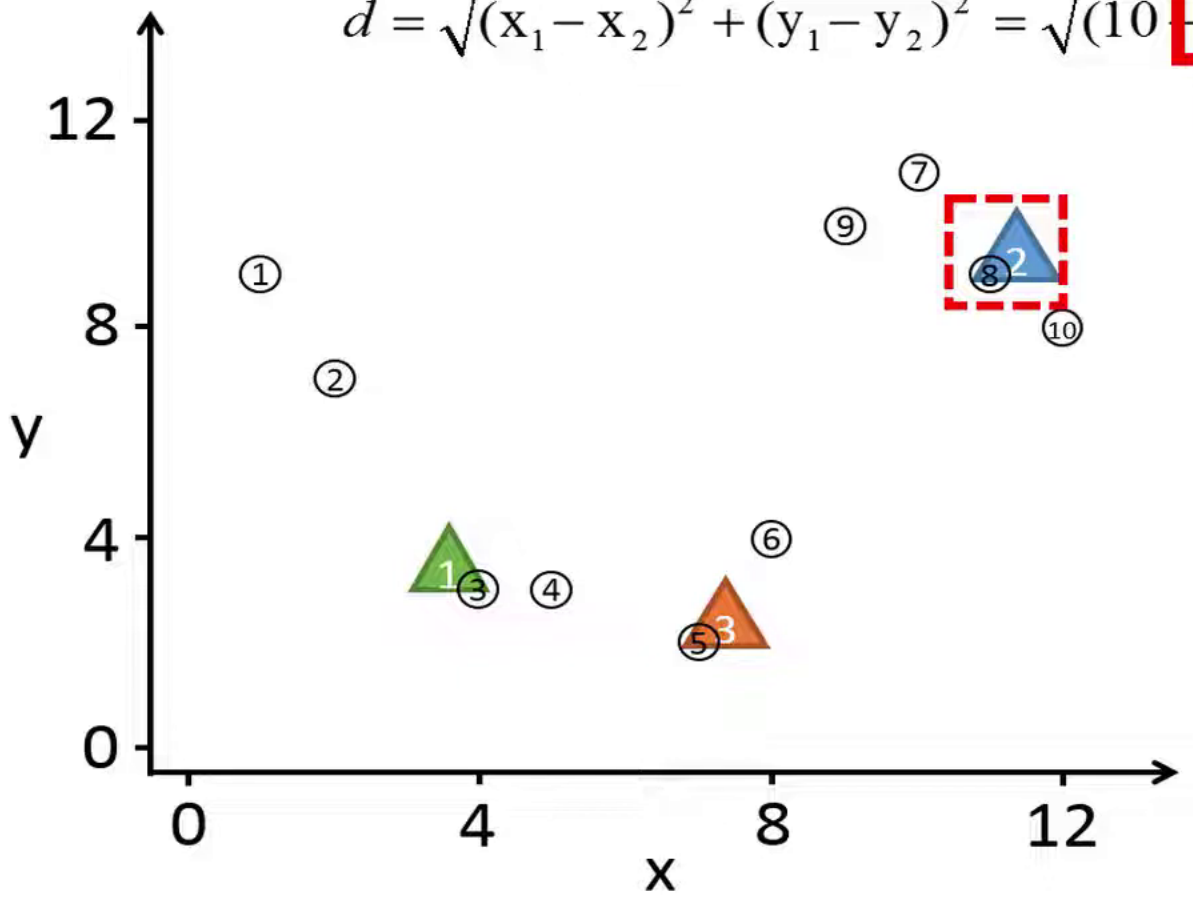


we plug in the x and y coordinates of the data point,

k=3

	x	y
1	1	9
2	2	7
3	4	3
4	5	3
5	7	2
6	8	4
7	10	11
8	11	9
9	9	10
10	12	8

$$d = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2} = \sqrt{(10 - 11)^2 + (11 - 9)^2} = \sqrt{5}$$

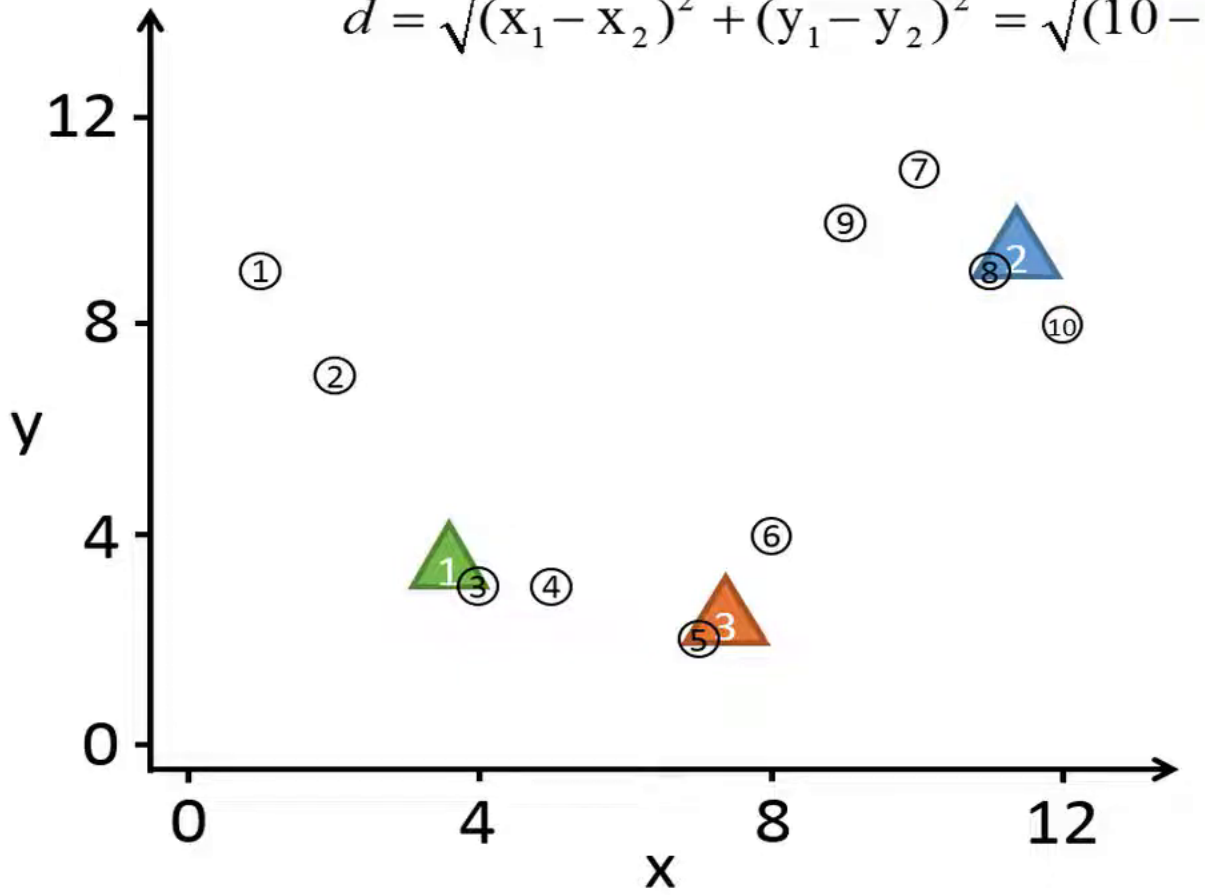


and the centroid,

k=3

	x	y
1	1	9
2	2	7
3	4	3
4	5	3
5	7	2
6	8	4
7	10	11
8	11	9
9	9	10
10	12	8

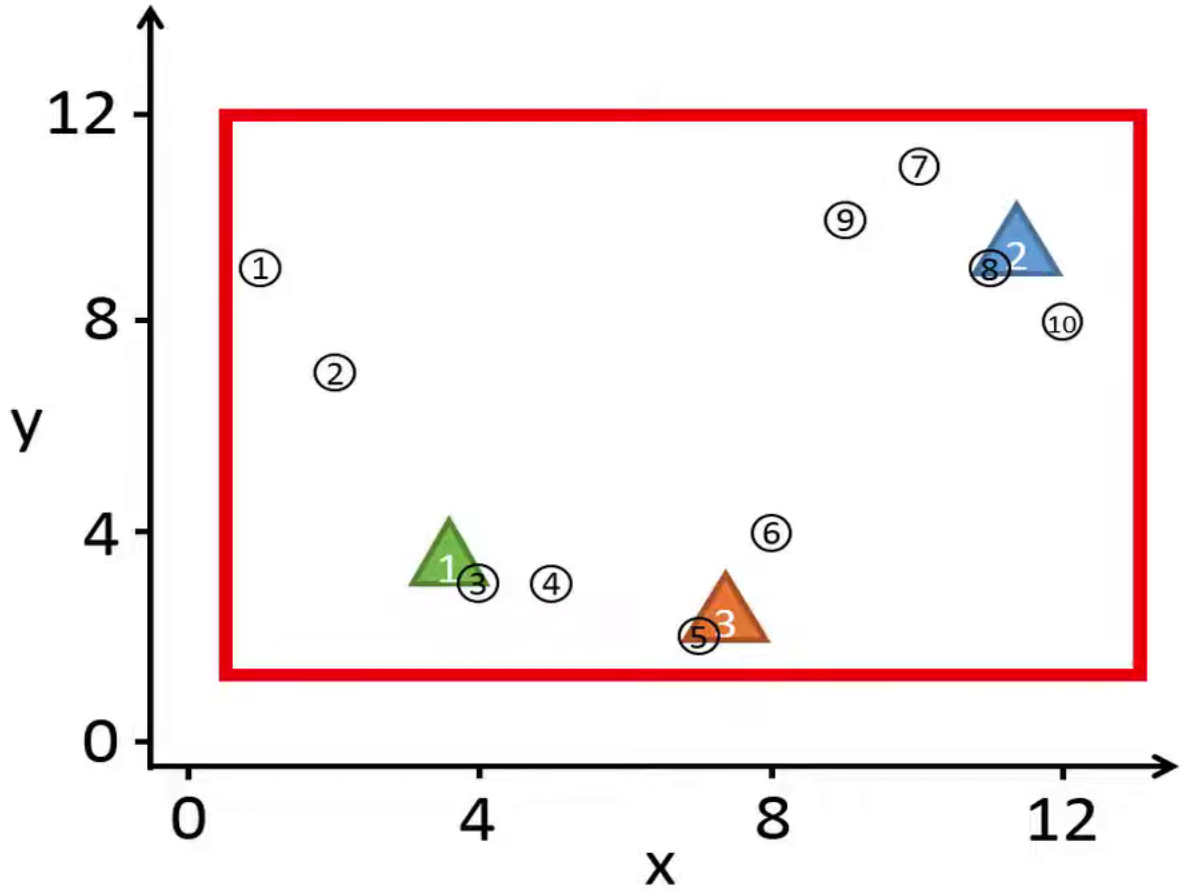
$$d = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2} = \sqrt{(10-11)^2 + (11-9)^2} = \sqrt{5}$$



which results in a distance that is equal to the square root of five.

k=3

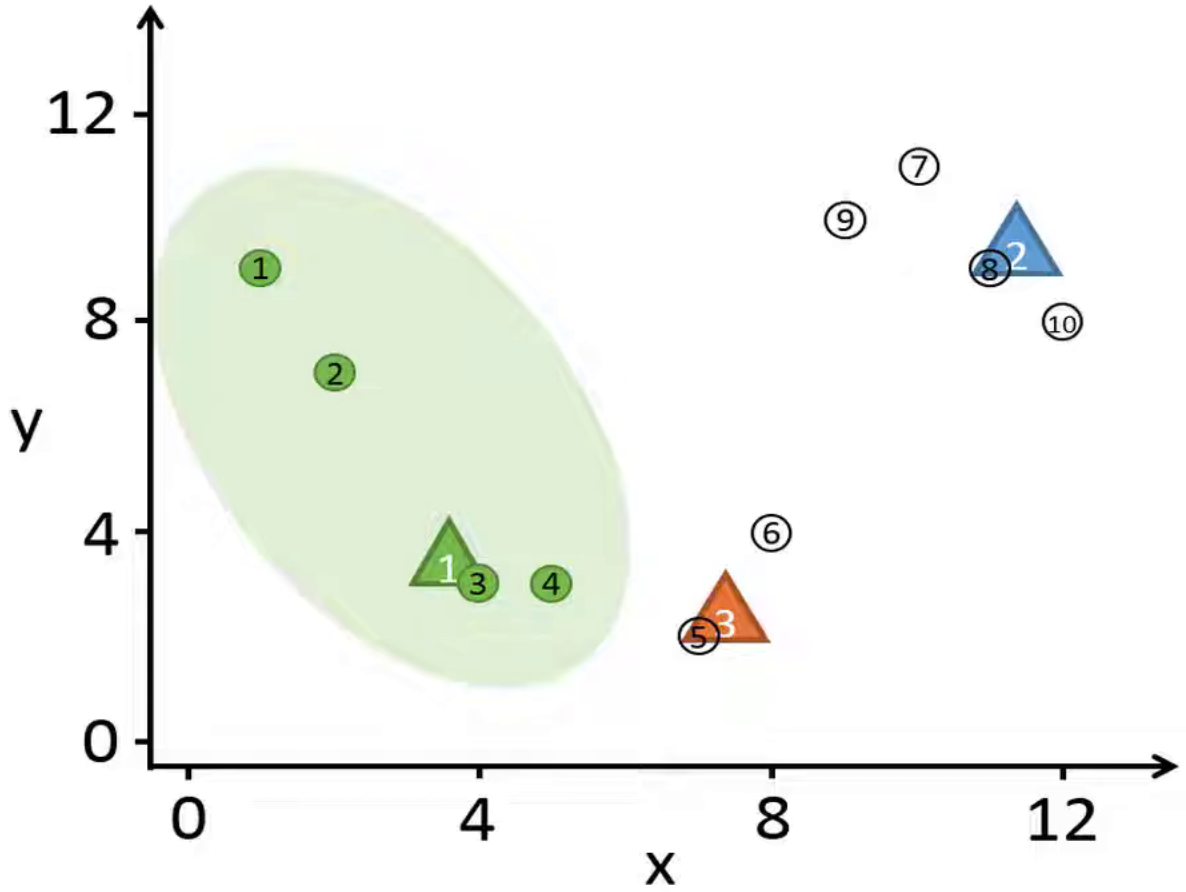
	x	y
1	1	9
2	2	7
3	4	3
4	5	3
5	7	2
6	8	4
7	10	11
8	11	9
9	9	10
10	12	8



We calculate the Euclidean distance between each data point and all the centroids.

k=3

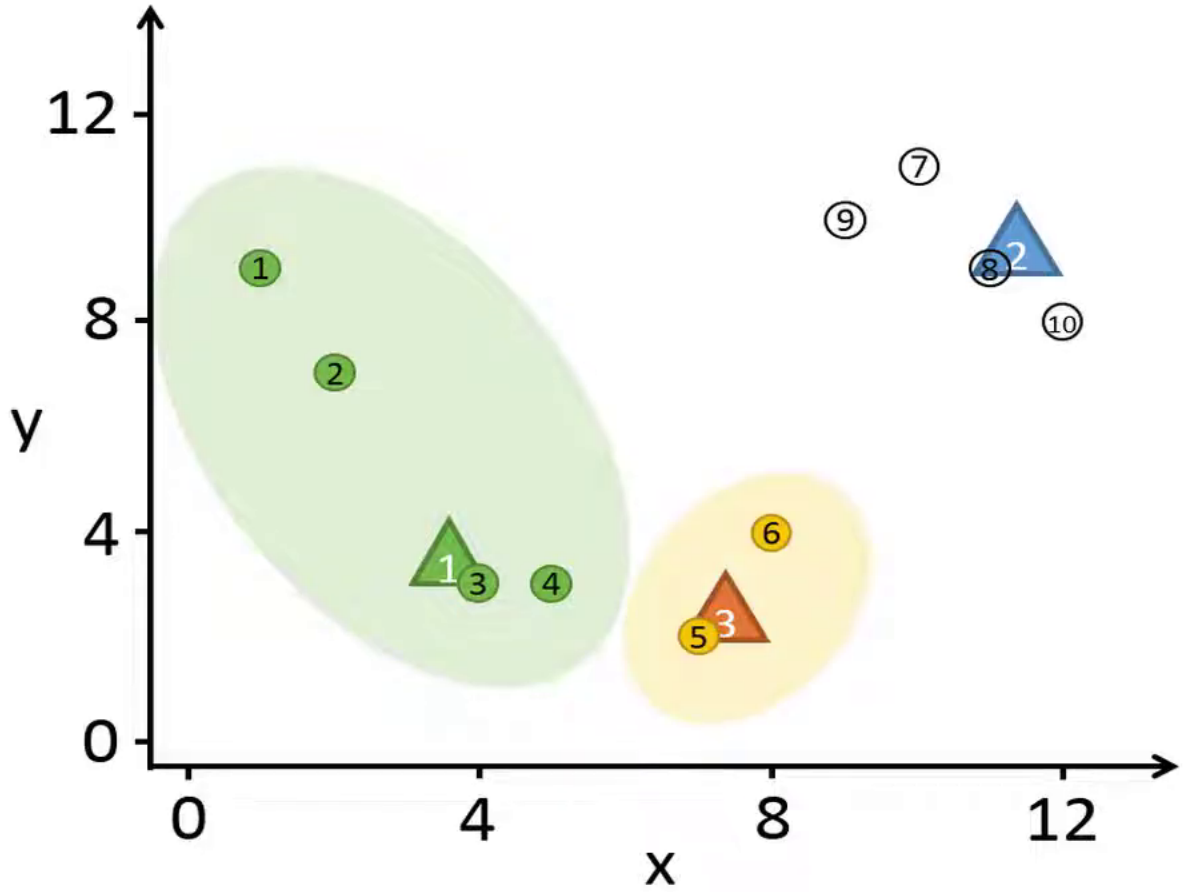
	x	y
1	1	9
2	2	7
3	4	3
4	5	3
5	7	2
6	8	4
7	10	11
8	11	9
9	9	10
10	12	8



After calculating all these distances, we know that these four data points are closest to centroid number one,

k=3

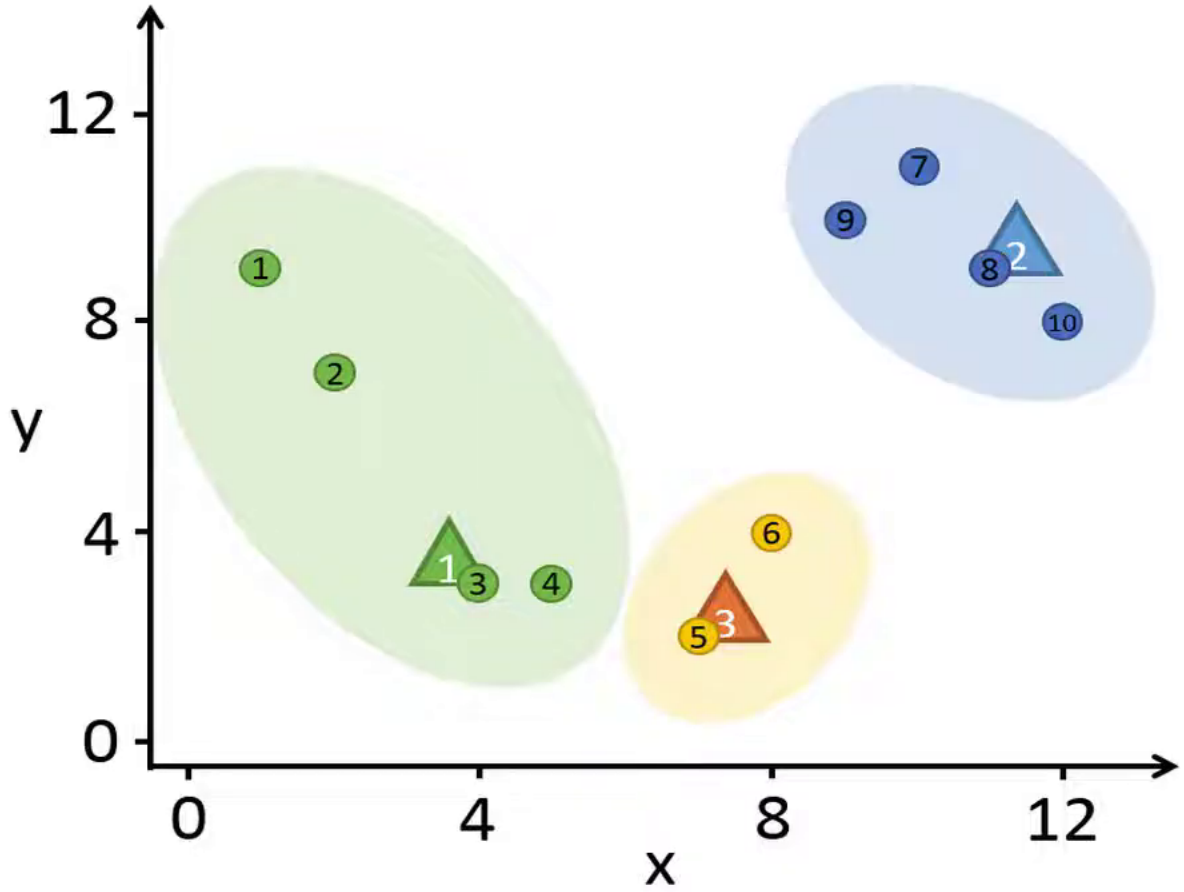
	x	y
1	1	9
2	2	7
3	4	3
4	5	3
5	7	2
6	8	4
7	10	11
8	11	9
9	9	10
10	12	8



whereas these two data points are closest to centroid number three,

k=3

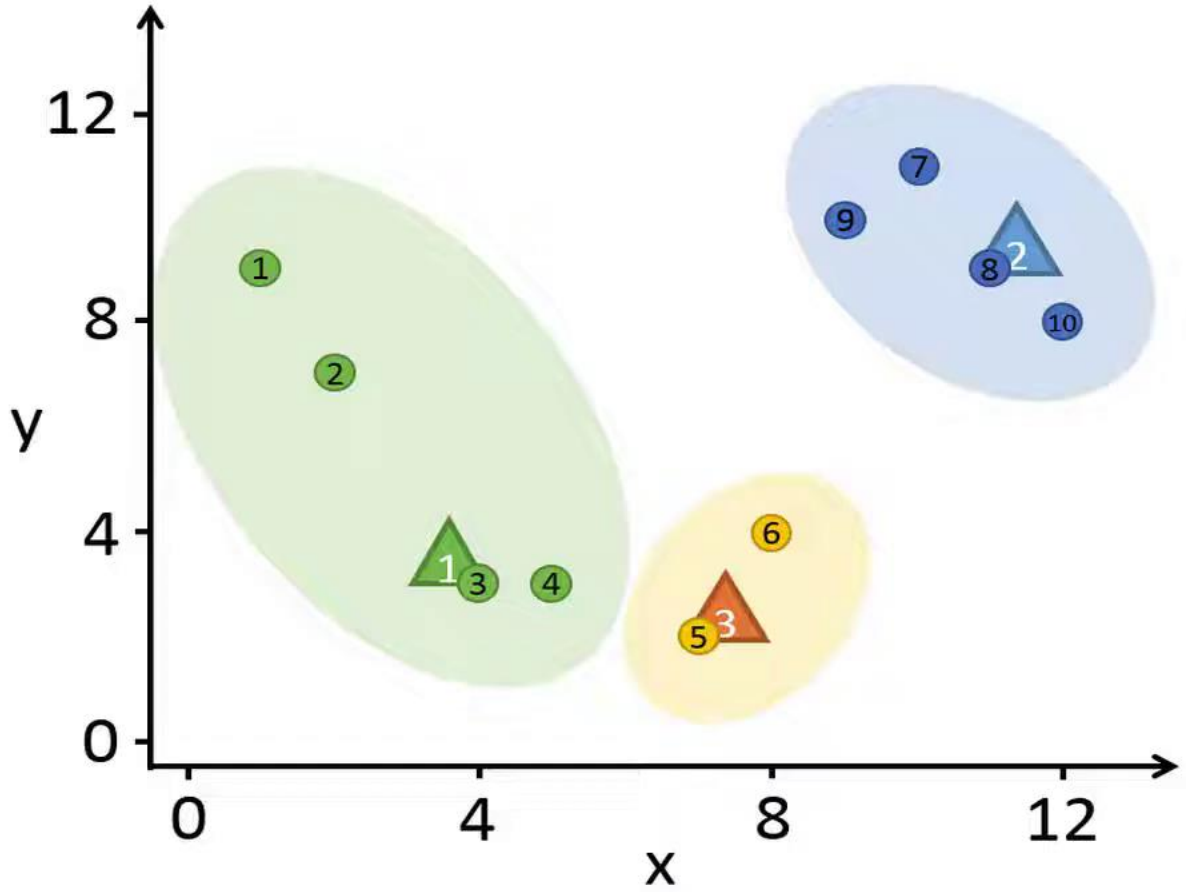
	x	y
1	1	9
2	2	7
3	4	3
4	5	3
5	7	2
6	8	4
7	10	11
8	11	9
9	9	10
10	12	8



and that these data points are closest to centroid number two. We have therefore created three initial clusters.

k=3

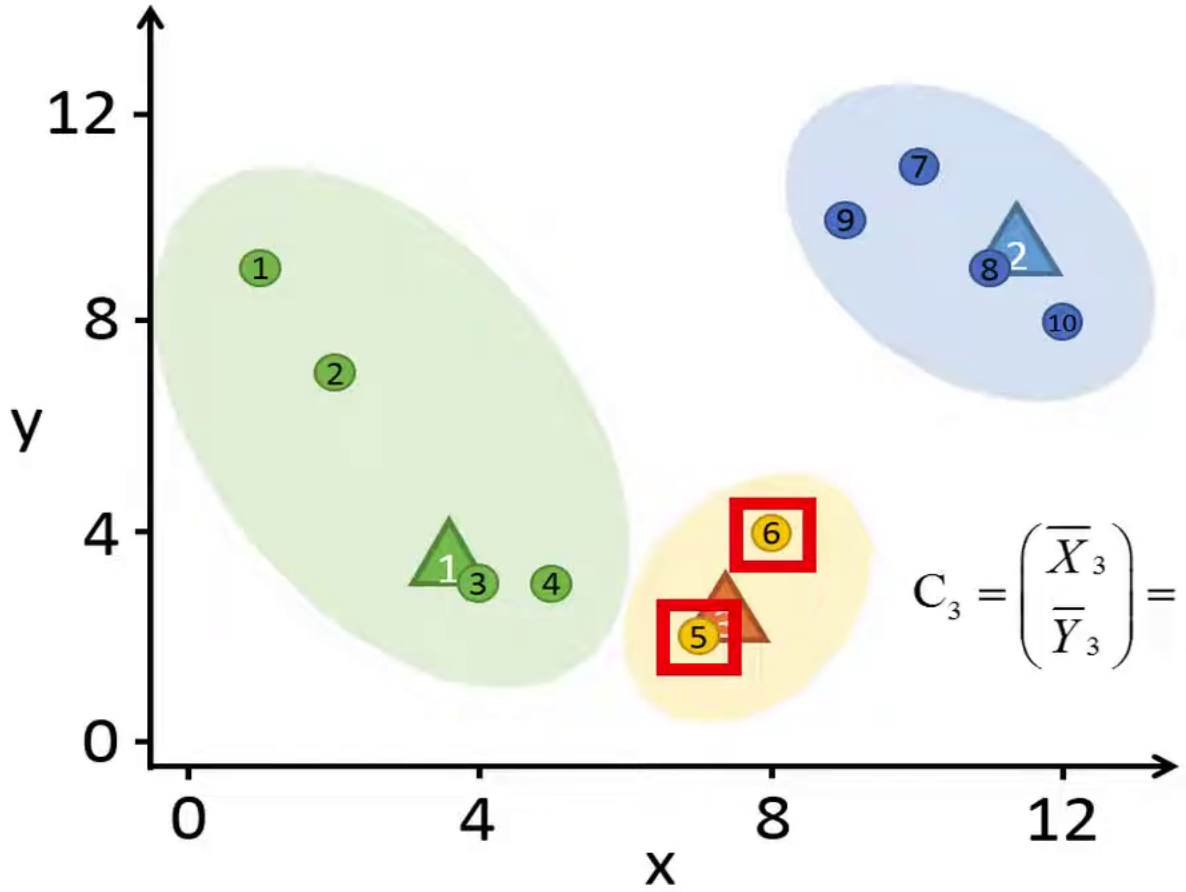
	x	y
1	1	9
2	2	7
3	4	3
4	5	3
5	7	2
6	8	4
7	10	11
8	11	9
9	9	10
10	12	8



Next, we update the positions of the centroids so that they correspond to the mean x and y positions of the data points in the corresponding cluster.

k=3

	x	y
1	1	9
2	2	7
3	4	3
4	5	3
5	7	2
6	8	4
7	10	11
8	11	9
9	9	10
10	12	8

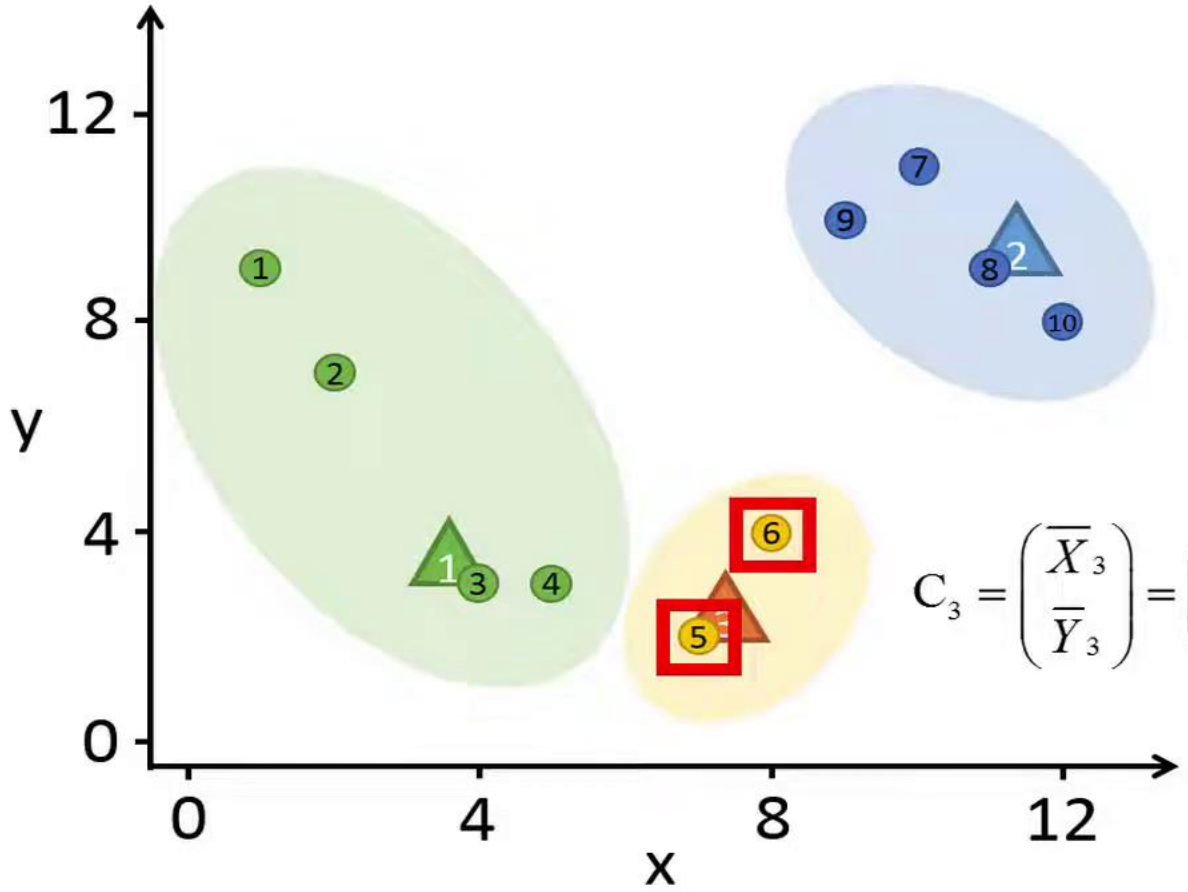


$$C_3 = \begin{pmatrix} \overline{X}_3 \\ \overline{Y}_3 \end{pmatrix} = \begin{pmatrix} (7+8)/2 \\ (2+4)/2 \end{pmatrix} = \begin{pmatrix} 7.5 \\ 3 \end{pmatrix}$$

For example, the mean of the x-coordinates of the two data points in cluster three is 7.5,

k=3

	x	y
1	1	9
2	2	7
3	4	3
4	5	3
5	7	2
6	8	4
7	10	11
8	11	9
9	9	10
10	12	8

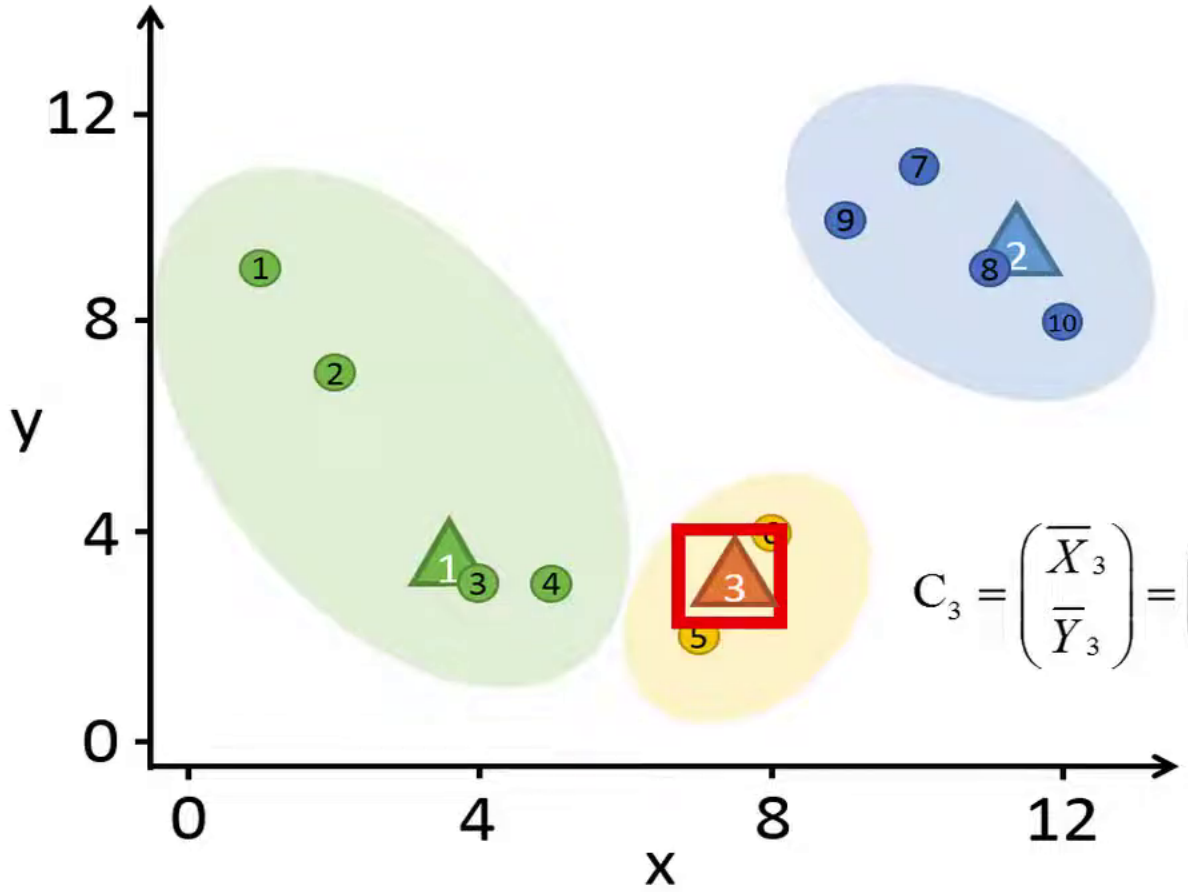


$$C_3 = \begin{pmatrix} \overline{X}_3 \\ \overline{Y}_3 \end{pmatrix} = \begin{pmatrix} (7+8)/2 \\ (2+4)/2 \end{pmatrix} = \begin{pmatrix} 7.5 \\ 3 \end{pmatrix}$$

and the mean of the y-coordinates is three.

k=3

	x	y
1	1	9
2	2	7
3	4	3
4	5	3
5	7	2
6	8	4
7	10	11
8	11	9
9	9	10
10	12	8

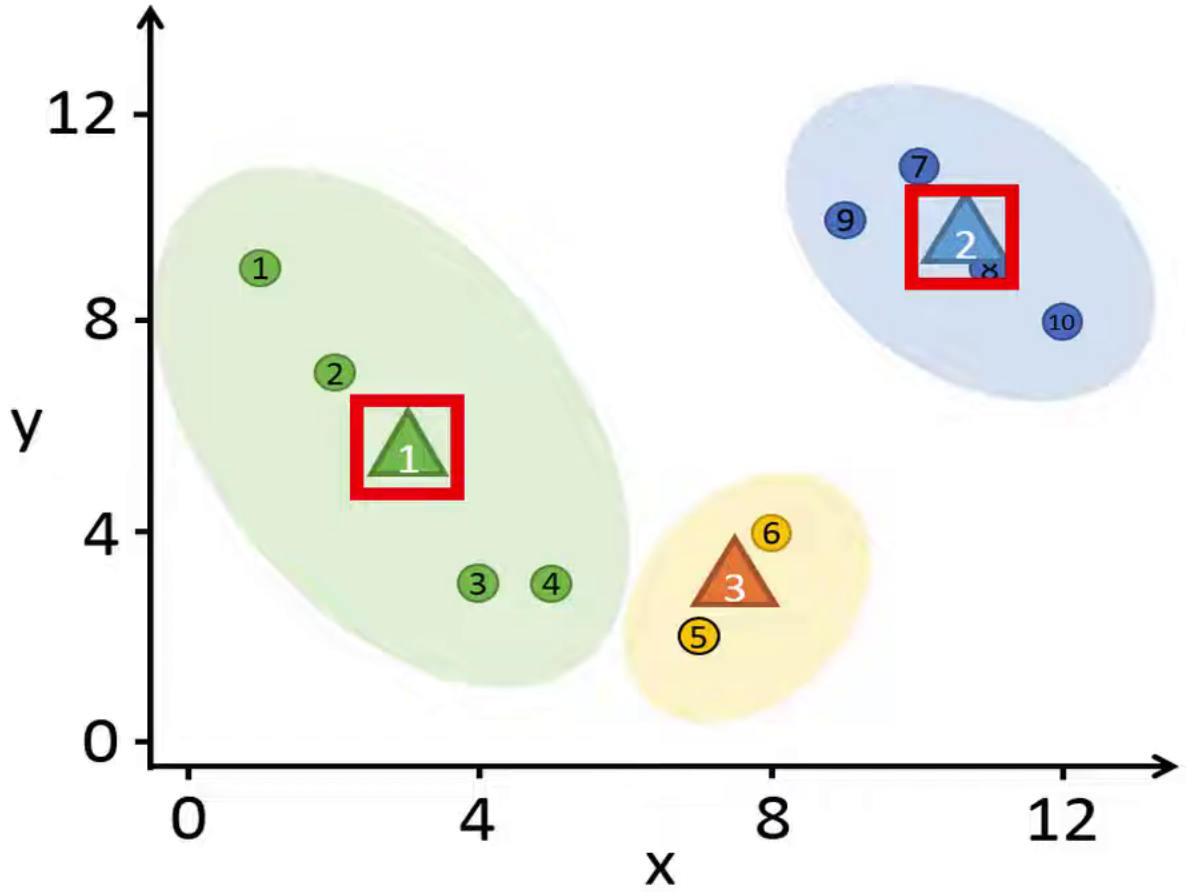


$$C_3 = \begin{pmatrix} \overline{X}_3 \\ \overline{Y}_3 \end{pmatrix} = \begin{pmatrix} (7+8)/2 \\ (2+4)/2 \end{pmatrix} = \begin{pmatrix} 7.5 \\ 3 \end{pmatrix}$$

We therefore move this centroid to the coordinates that represent the mean position of the two data points in the third cluster.

k=3

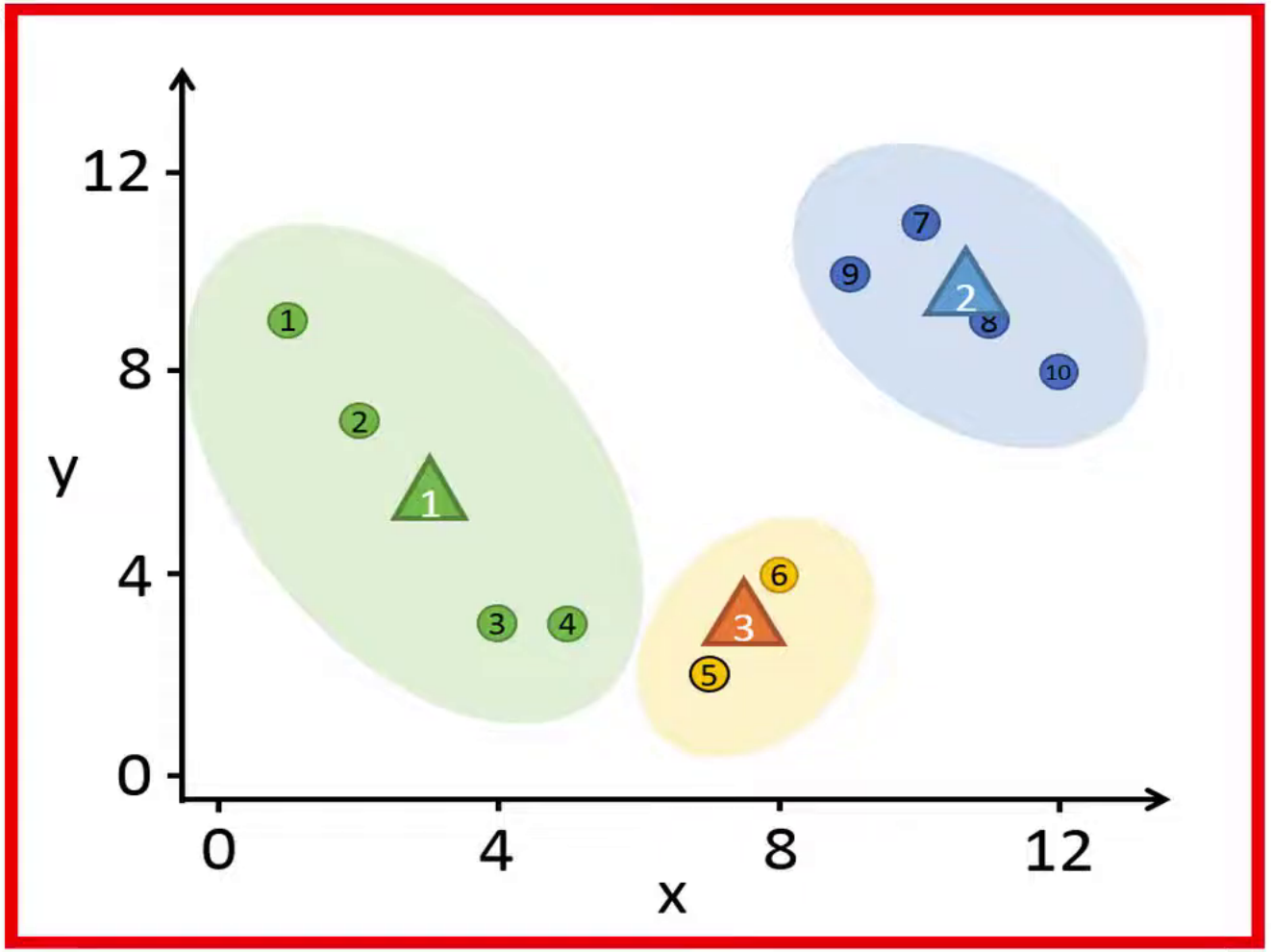
	x	y
1	1	9
2	2	7
3	4	3
4	5	3
5	7	2
6	8	4
7	10	11
8	11	9
9	9	10
10	12	8



Similarly, we update the positions of the centroids of the first and second clusters.

k=3

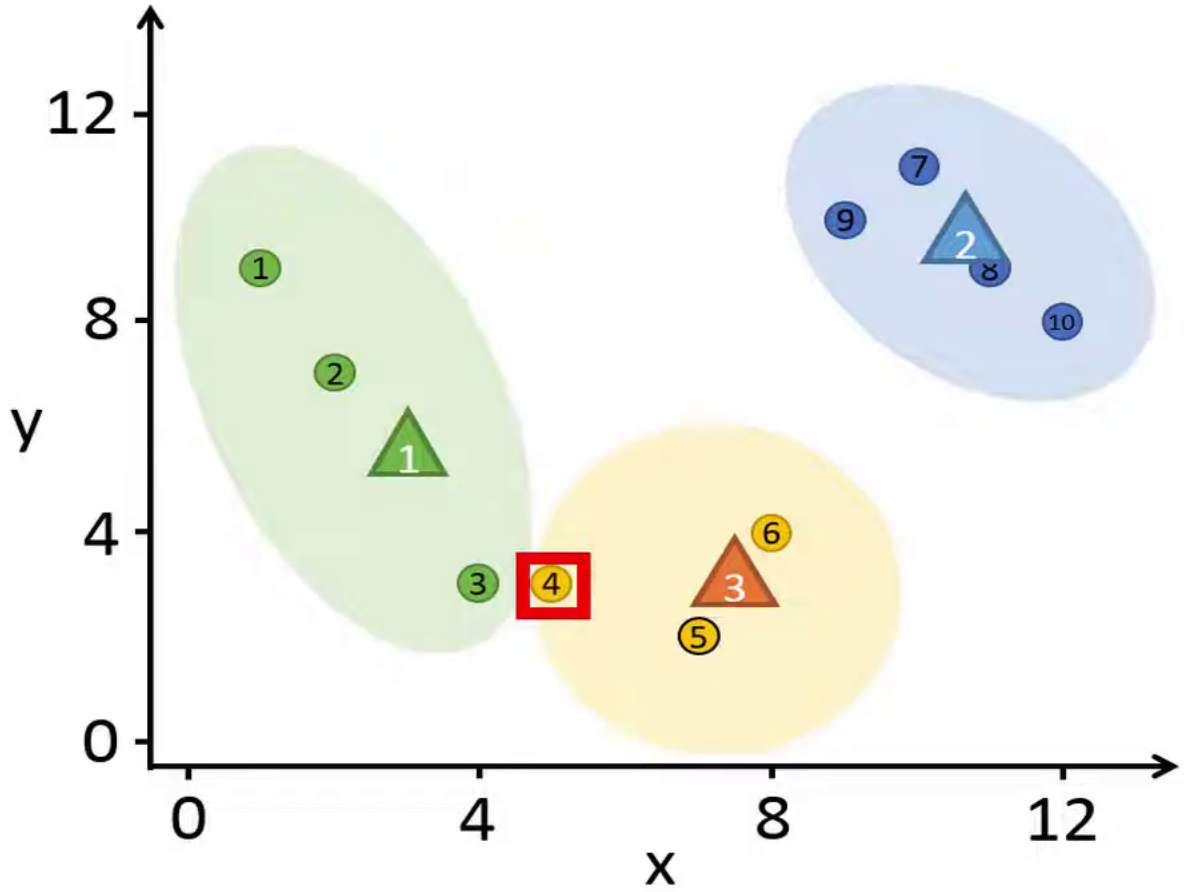
	x	y
1	1	9
2	2	7
3	4	3
4	5	3
5	7	2
6	8	4
7	10	11
8	11	9
9	9	10
10	12	8



We now repeat the previous steps, where we first find the shortest distance between all data points and the centroids.

k=3

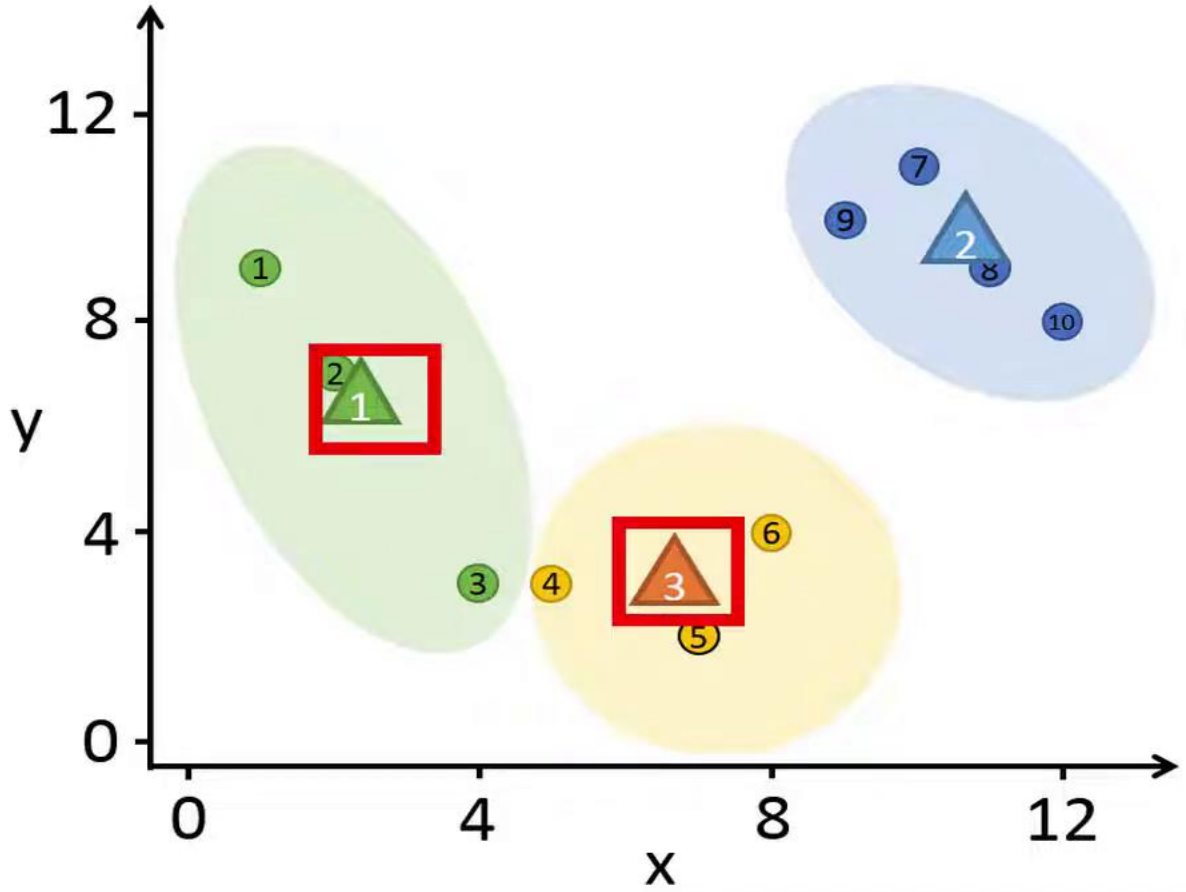
	x	y
1	1	9
2	2	7
3	4	3
4	5	3
5	7	2
6	8	4
7	10	11
8	11	9
9	9	10
10	12	8



it will therefore switch cluster from cluster one to cluster three.

k=3

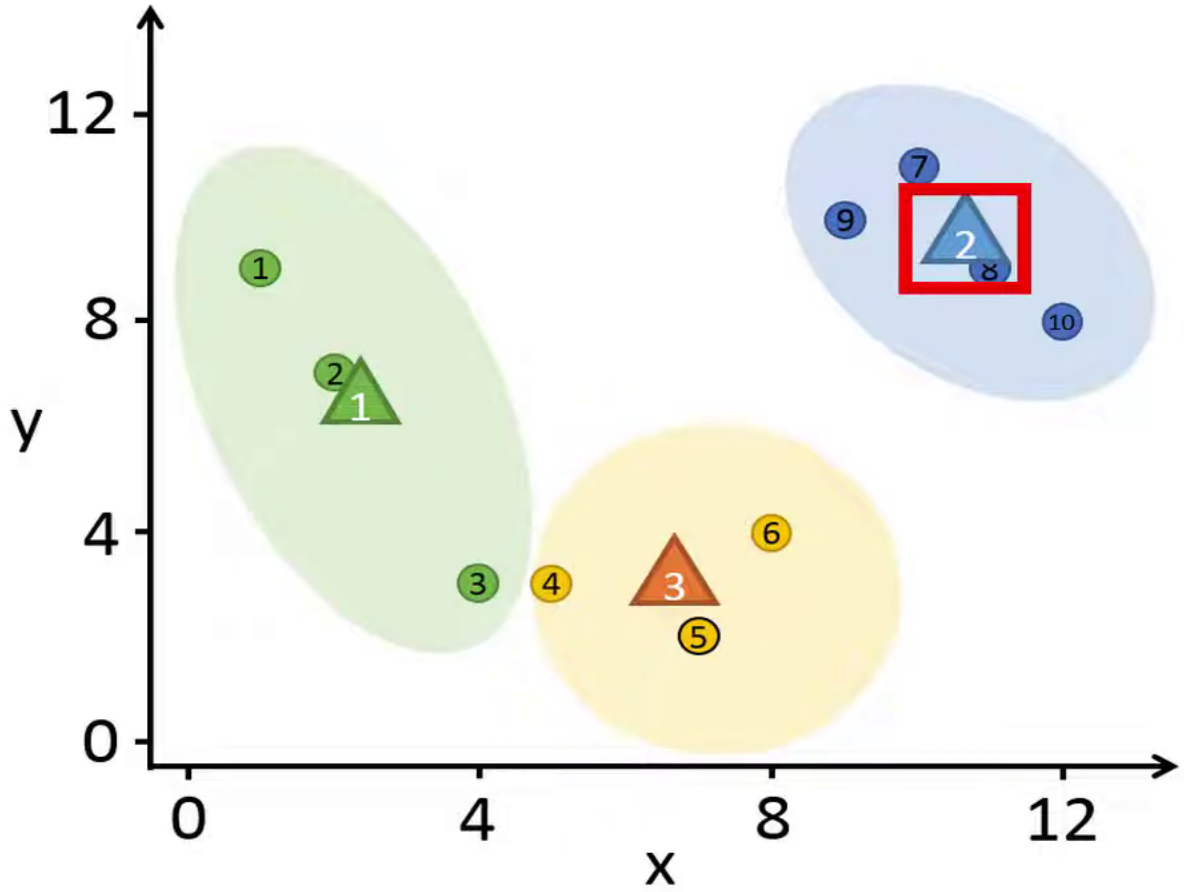
	x	y
1	1	9
2	2	7
3	4	3
4	5	3
5	7	2
6	8	4
7	10	11
8	11	9
9	9	10
10	12	8



Next, we update the positions of the centroids based on their associated data points.

k=3

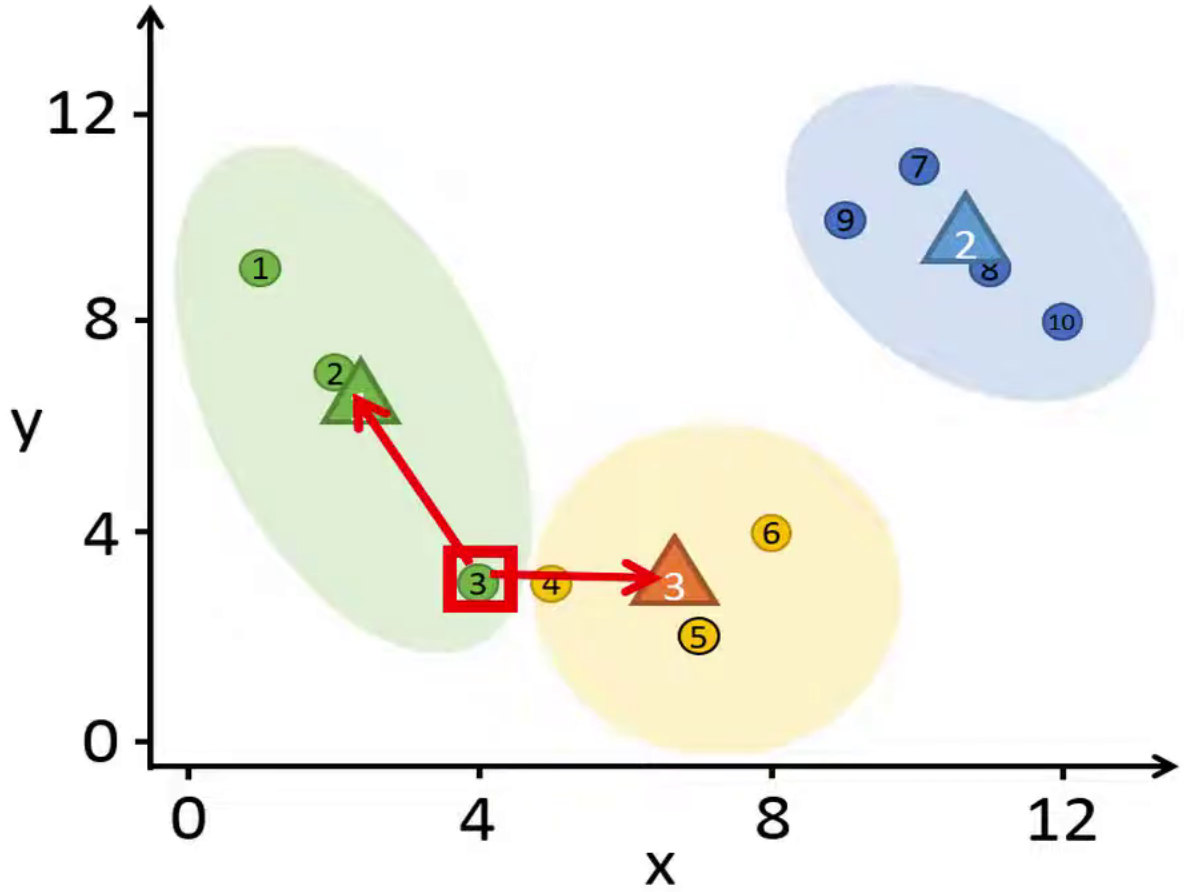
	x	y
1	1	9
2	2	7
3	4	3
4	5	3
5	7	2
6	8	4
7	10	11
8	11	9
9	9	10
10	12	8



Note that this centroid does not change position because the data points in this cluster have not been changed.

k=3

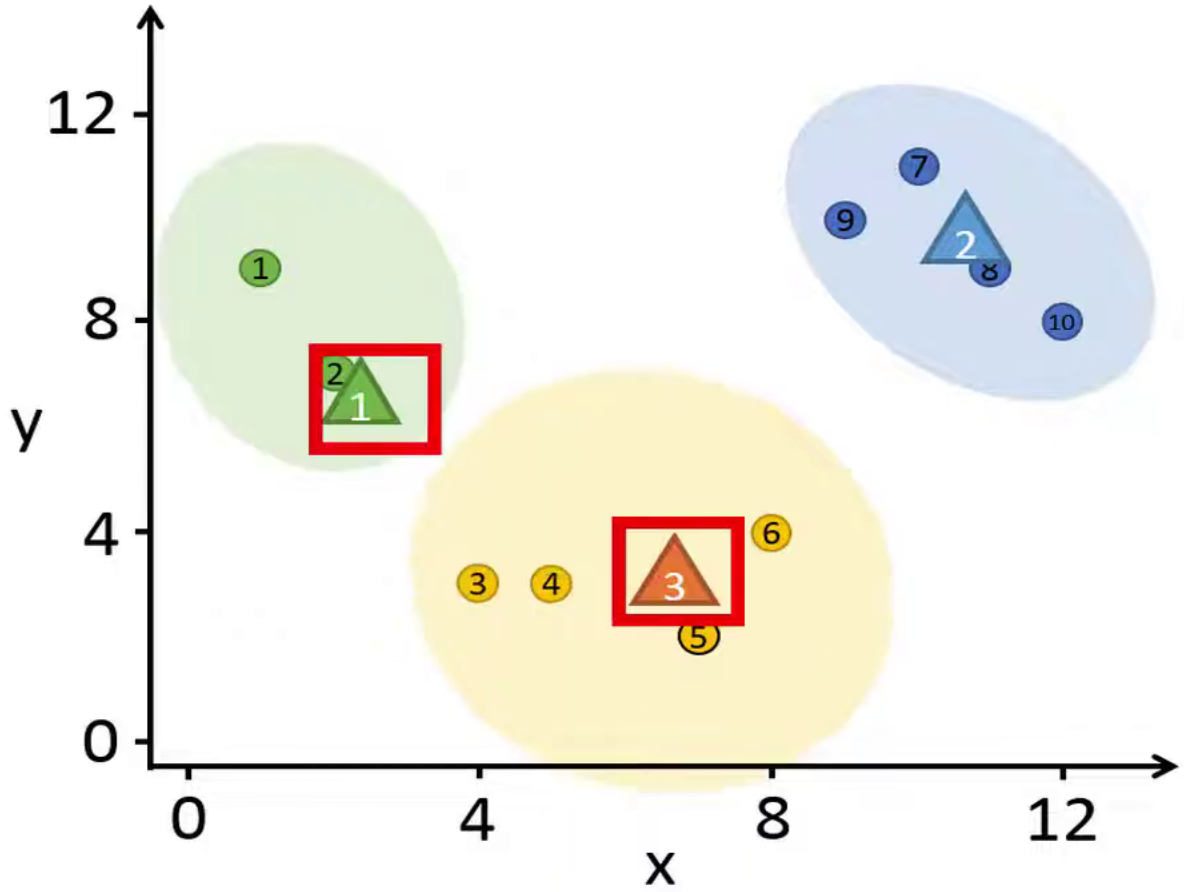
	x	y
1	1	9
2	2	7
3	4	3
4	5	3
5	7	2
6	8	4
7	10	11
8	11	9
9	9	10
10	12	8



Since data point number 3 is now closer to the centroid of the third cluster it will also switch cluster.

k=3

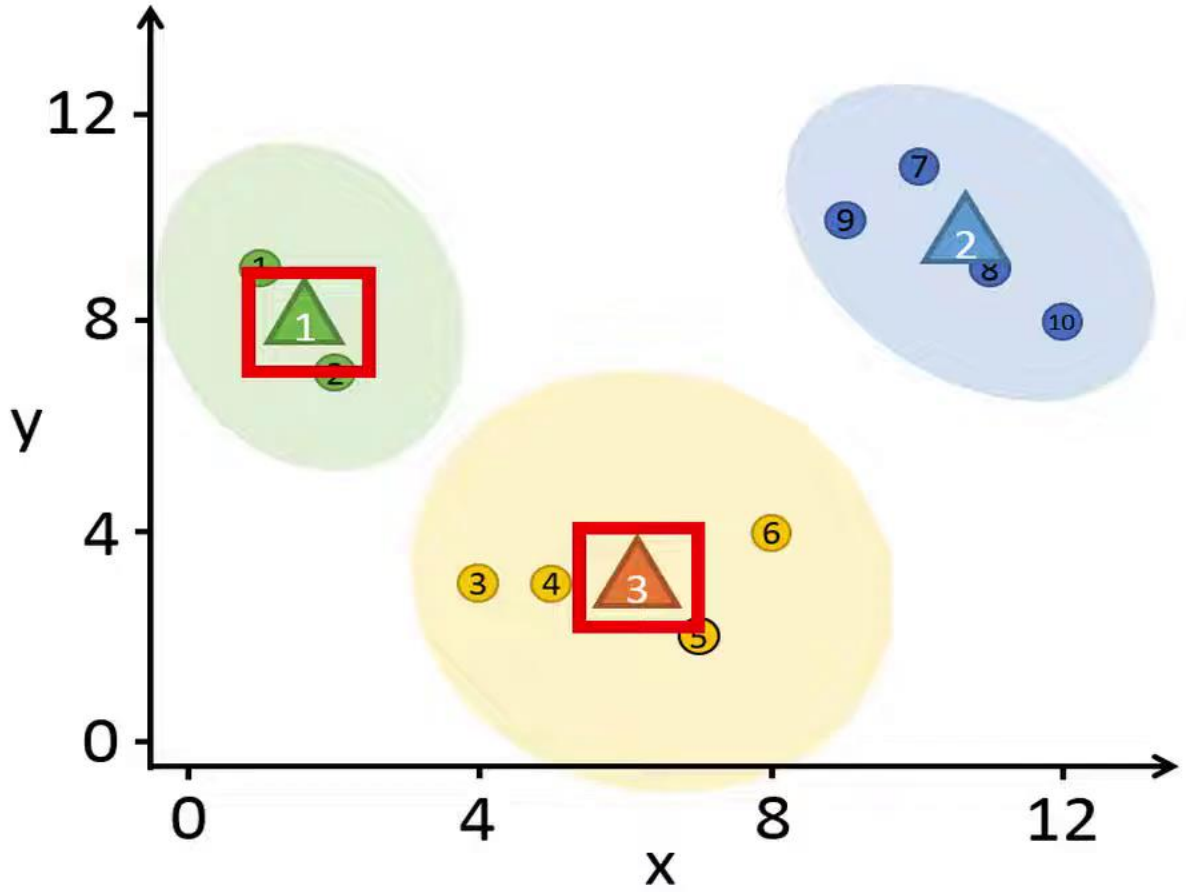
	x	y
1	1	9
2	2	7
3	4	3
4	5	3
5	7	2
6	8	4
7	10	11
8	11	9
9	9	10
10	12	8



Next, we update the positions of the centroids again,

k=3

	x	y
1	1	9
2	2	7
3	4	3
4	5	3
5	7	2
6	8	4
7	10	11
8	11	9
9	9	10
10	12	8



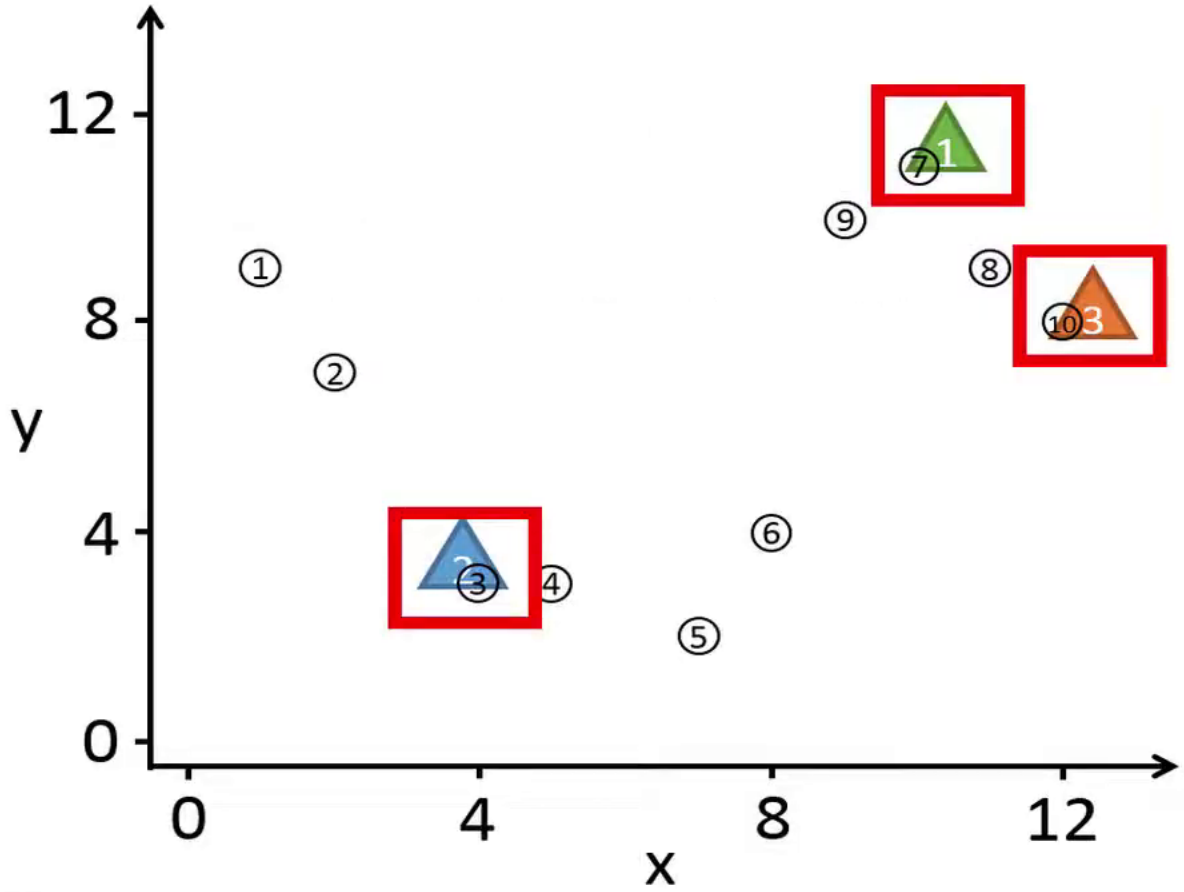
so that they have the following positions. Since all data points are now closest to the centroid of the cluster they belong to, no data point will switch cluster. When this happens, the algorithm will stop.

Initial positions of the centroids

So, do the initial positions of the centroids affect the clustering?

Initial positions of the centroids

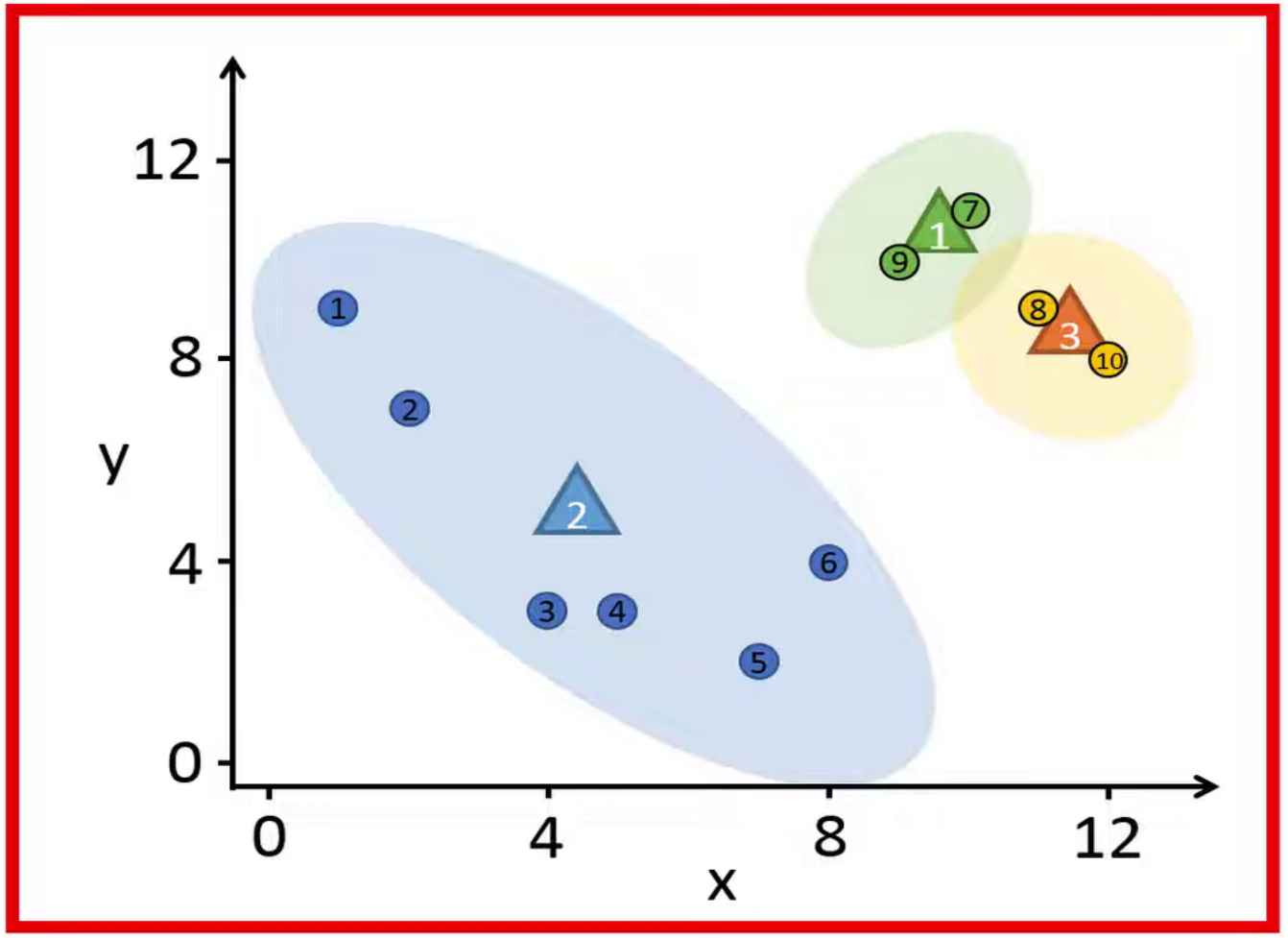
	x	y
1	1	9
2	2	7
3	4	3
4	5	3
5	7	2
6	8	4
7	10	11
8	11	9
9	9	10
10	12	8



For example, what would happen if the centroids are initially placed at the following positions?

Initial positions of the centroids

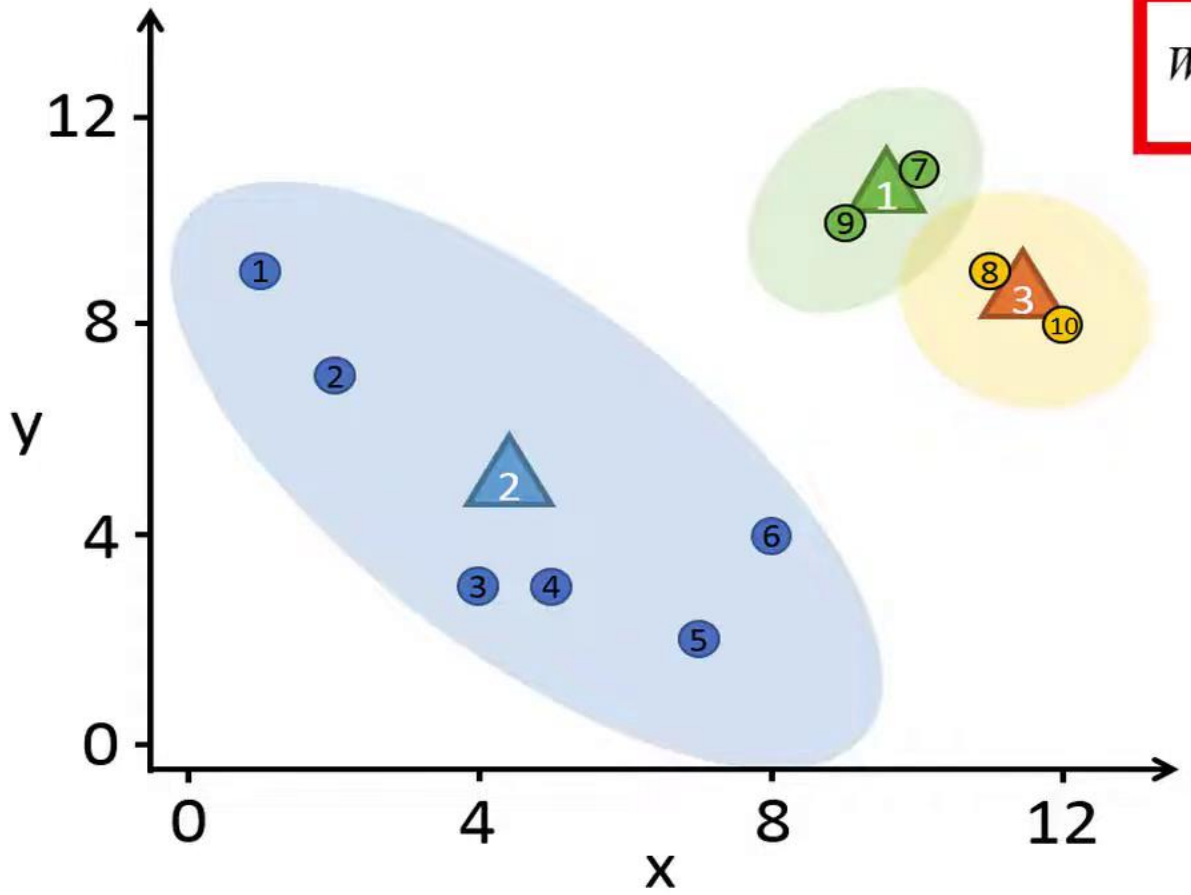
	x	y
1	1	9
2	2	7
3	4	3
4	5	3
5	7	2
6	8	4
7	10	11
8	11	9
9	9	10
10	12	8



The algorithm will then end up with the following three clusters. So, are these clusters better than the ones we generated earlier? To compare different types of outputs, we need some sort of measure for how well the clustering has performed.

Initial positions of the centroids

	x	y
1	1	9
2	2	7
3	4	3
4	5	3
5	7	2
6	8	4
7	10	11
8	11	9
9	9	10
10	12	8

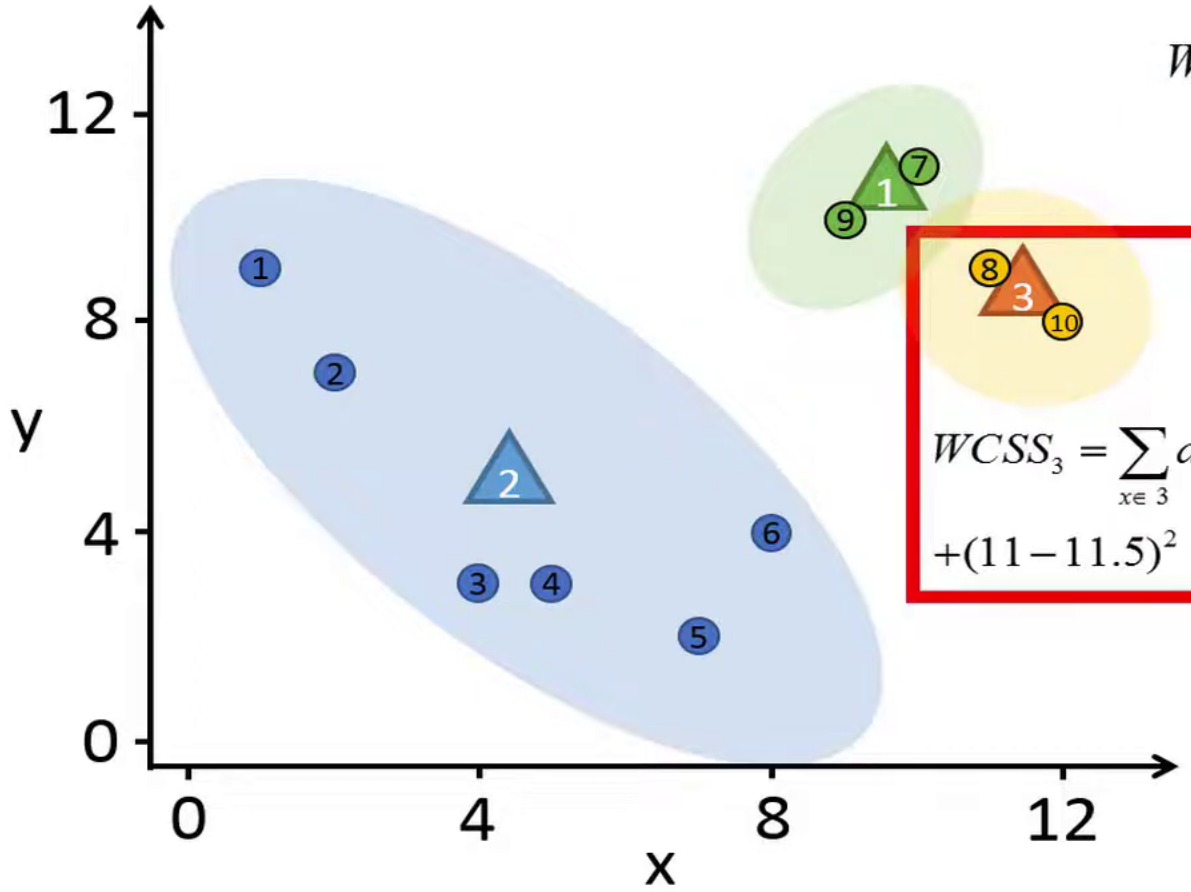


$$WCSS = \sum_{i=1}^{n_c} \sum_{x \in i} d(x, \bar{x}_i)^2$$

One measure is the so-called within cluster sum of squares (WCSS), which is the sum of the squared Euclidean distance between each data point and the centroid of the corresponding cluster. This measure tells how close the data points are to the centroids.

Initial positions of the centroids

	x	y
1	1	9
2	2	7
3	4	3
4	5	3
5	7	2
6	8	4
7	10	11
8	11	9
9	9	10
10	12	8



$$WCSS = \sum_{i=1}^{n_c} \sum_{x \in i} d(x, \bar{x}_i)^2$$

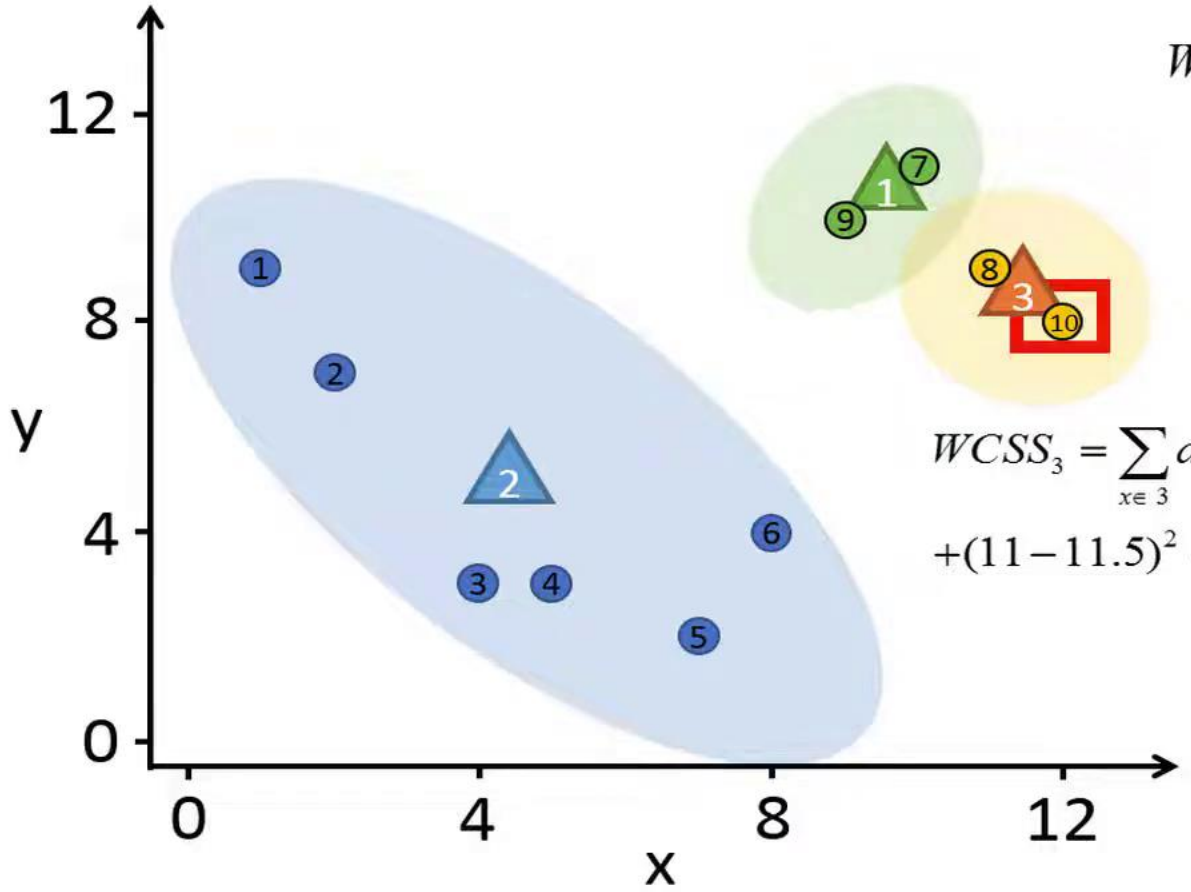
$$WCSS_3 = \sum_{x \in 3} d(x, \bar{x}_3)^2 = (12 - 11.5)^2 + (8 - 8.5)^2$$

$$+(11 - 11.5)^2 + (9 - 8.5)^2 = 1$$

For example, the within cluster sum of squares of the third cluster is calculated like this,

Initial positions of the centroids

	x	y
1	1	9
2	2	7
3	4	3
4	5	3
5	7	2
6	8	4
7	10	11
8	11	9
9	9	10
10	12	8



$$WCSS = \sum_{i=1}^{n_c} \sum_{x \in i} d(x, \bar{x}_i)^2$$

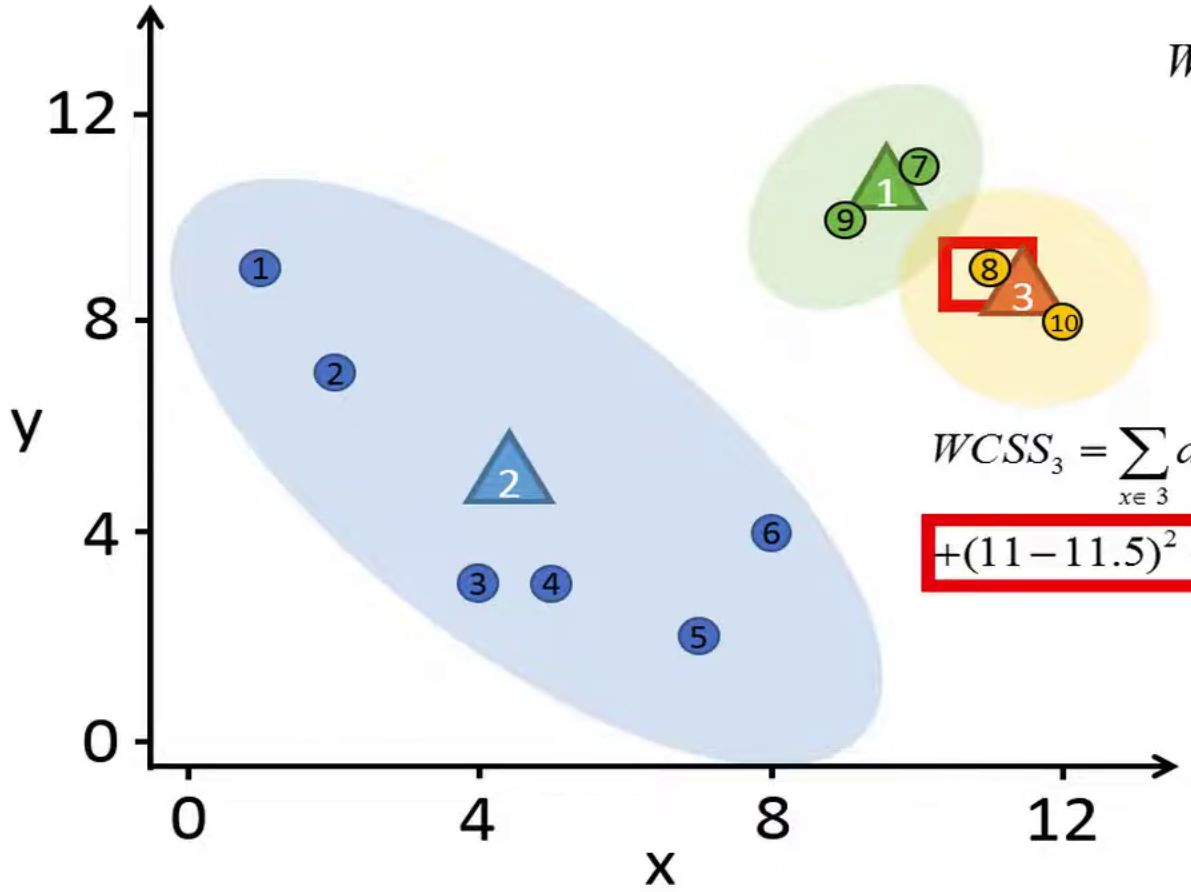
$$WCSS_3 = \sum_{x \in 3} d(x, \bar{x}_3)^2 = (12-11.5)^2 + (8-8.5)^2$$

$$+(11-11.5)^2 + (9-8.5)^2 = 1$$

where these terms represent the squared Euclidean distance between data point number 10 and the centroid of the third cluster,

Initial positions of the centroids

	x	y
1	1	9
2	2	7
3	4	3
4	5	3
5	7	2
6	8	4
7	10	11
8	11	9
9	9	10
10	12	8



$$WCSS = \sum_{i=1}^{n_c} \sum_{x \in i} d(x, \bar{x}_i)^2$$

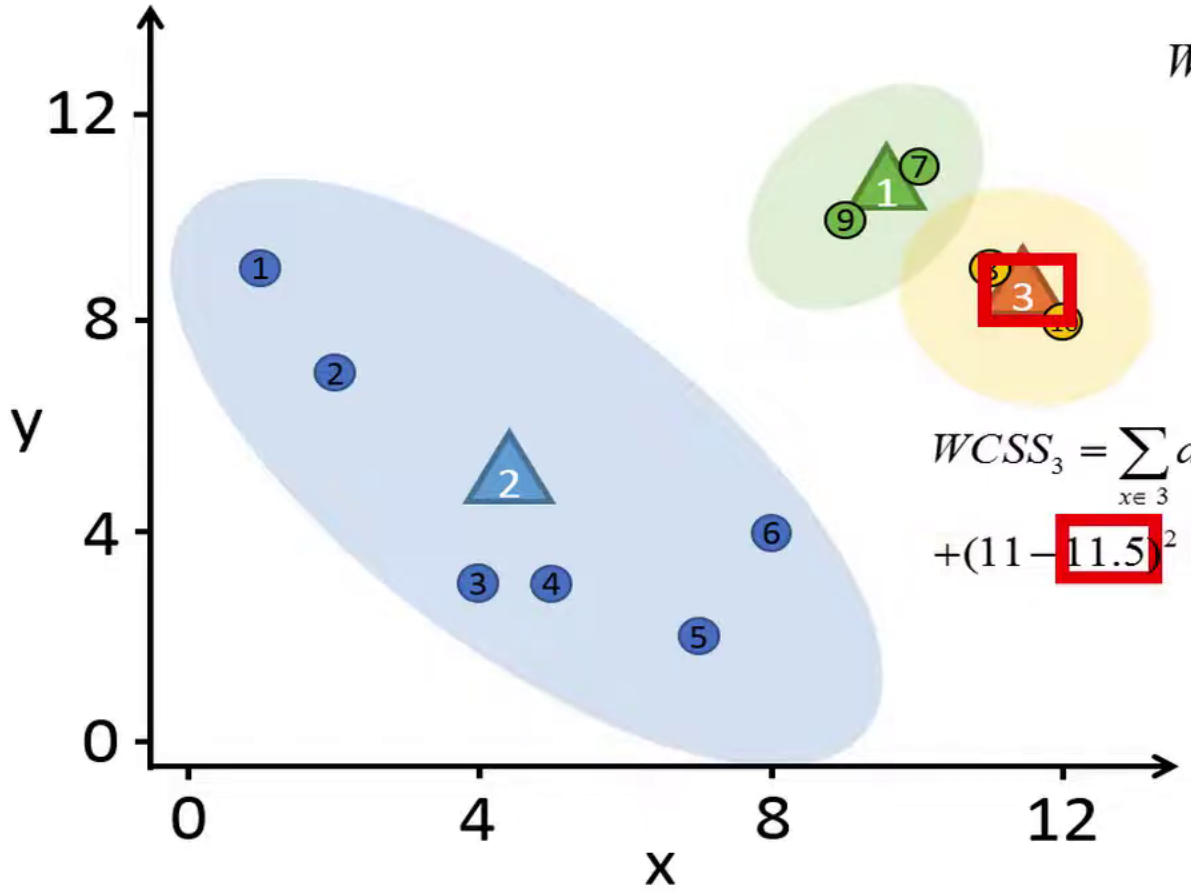
$$WCSS_3 = \sum_{x \in 3} d(x, \bar{x}_3)^2 = (12-11.5)^2 + (8-8.5)^2$$

$$+(11-11.5)^2 + (9-8.5)^2 = 1$$

and this is the distance between data point number 8 and the centroid.

Initial positions of the centroids

	x	y
1	1	9
2	2	7
3	4	3
4	5	3
5	7	2
6	8	4
7	10	11
8	11	9
9	9	10
10	12	8



$$WCSS = \sum_{i=1}^{n_c} \sum_{x \in i} d(x, \bar{x}_i)^2$$

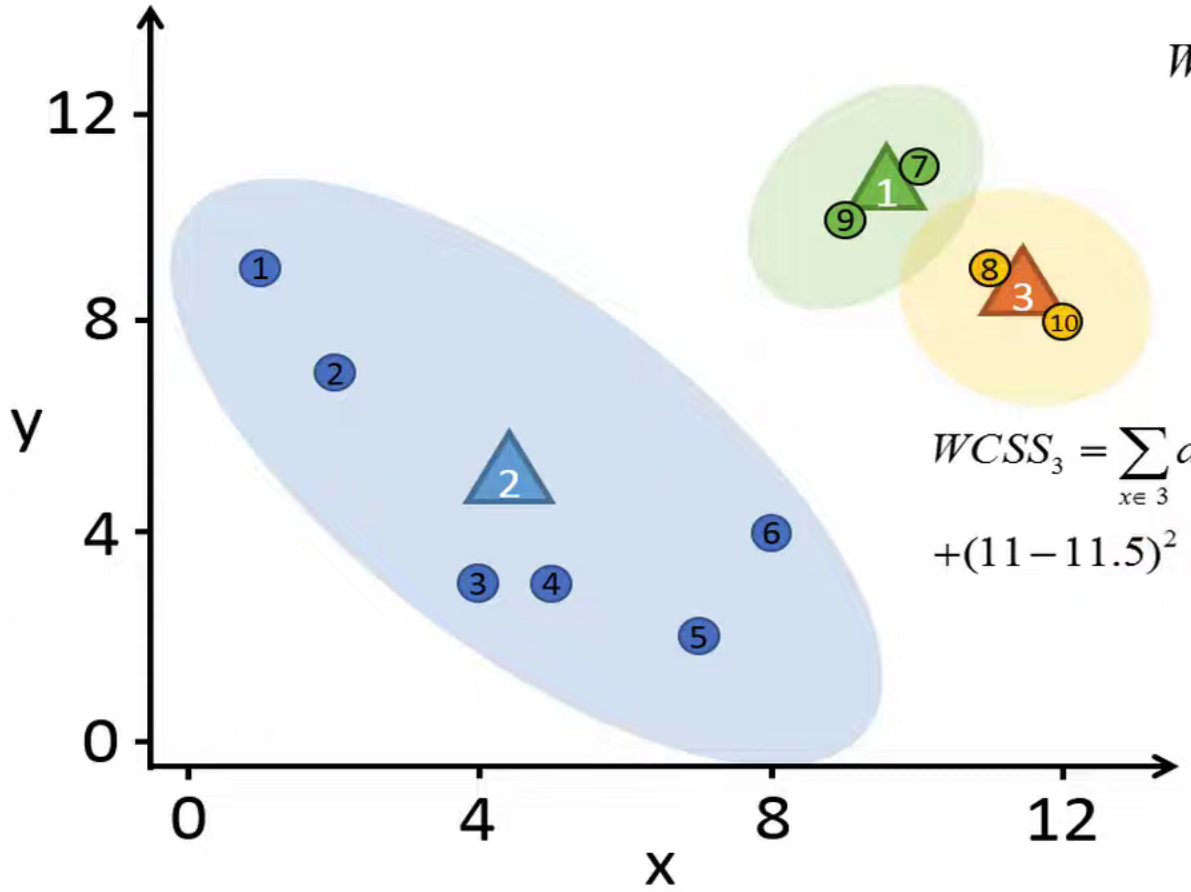
$$WCSS_3 = \sum_{x \in 3} d(x, \bar{x}_3)^2 = (12 - 11.5)^2 + (8 - 8.5)^2$$

$$+(11 - 11.5)^2 + (9 - 8.5)^2 = 1$$

The x- and y-coordinates of the third centroid are here 11.5 and 8.5.

Initial positions of the centroids

	x	y
1	1	9
2	2	7
3	4	3
4	5	3
5	7	2
6	8	4
7	10	11
8	11	9
9	9	10
10	12	8



$$WCSS = \sum_{i=1}^{n_c} \sum_{x \in i} d(x, \bar{x}_i)^2$$

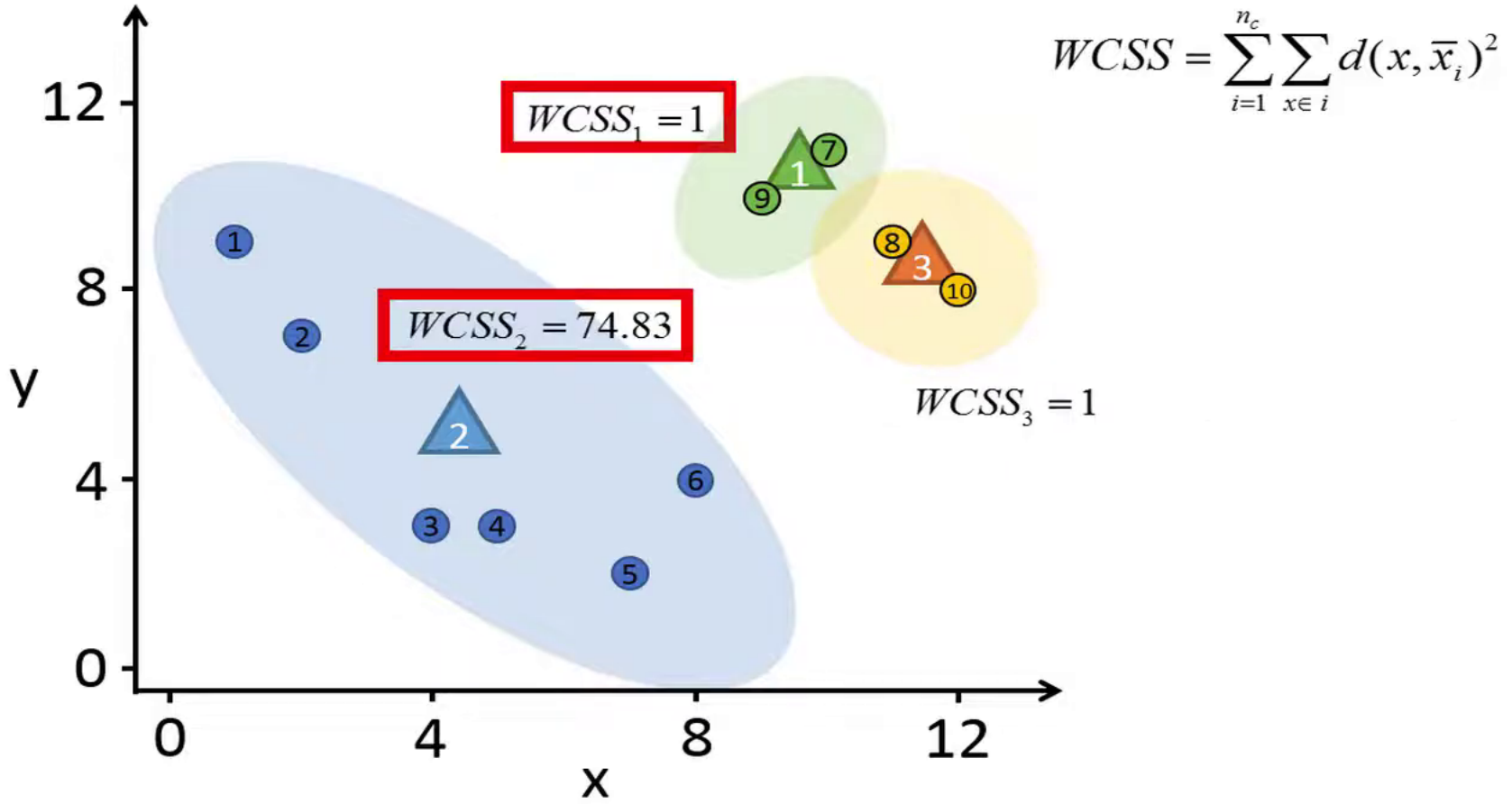
$$WCSS_3 = \sum_{x \in 3} d(x, \bar{x}_3)^2 = (12-11.5)^2 + (8-8.5)^2$$

$$+(11-11.5)^2 + (9-8.5)^2 = 1$$

The sum of the squared Euclidean distances in cluster three is here equal to one.

Initial positions of the centroids

	x	y
1	1	9
2	2	7
3	4	3
4	5	3
5	7	2
6	8	4
7	10	11
8	11	9
9	9	10
10	12	8

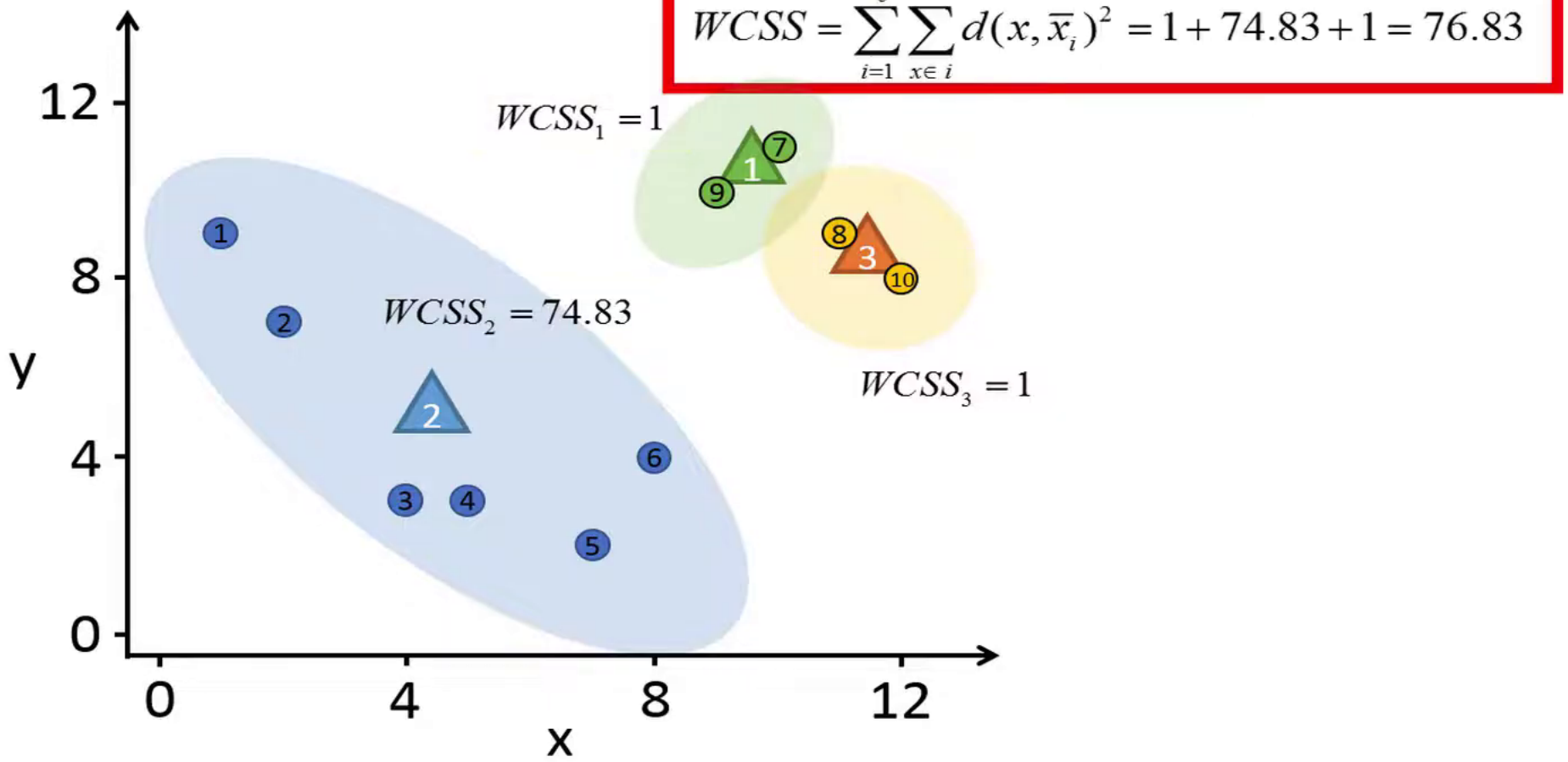


We calculate the within cluster sum of squares for the other two clusters in a similar way.

Initial positions of the centroids

	x	y
1	1	9
2	2	7
3	4	3
4	5	3
5	7	2
6	8	4
7	10	11
8	11	9
9	9	10
10	12	8

$$WCSS = \sum_{i=1}^{n_c} \sum_{x \in i} d(x, \bar{x}_i)^2 = 1 + 74.83 + 1 = 76.83$$

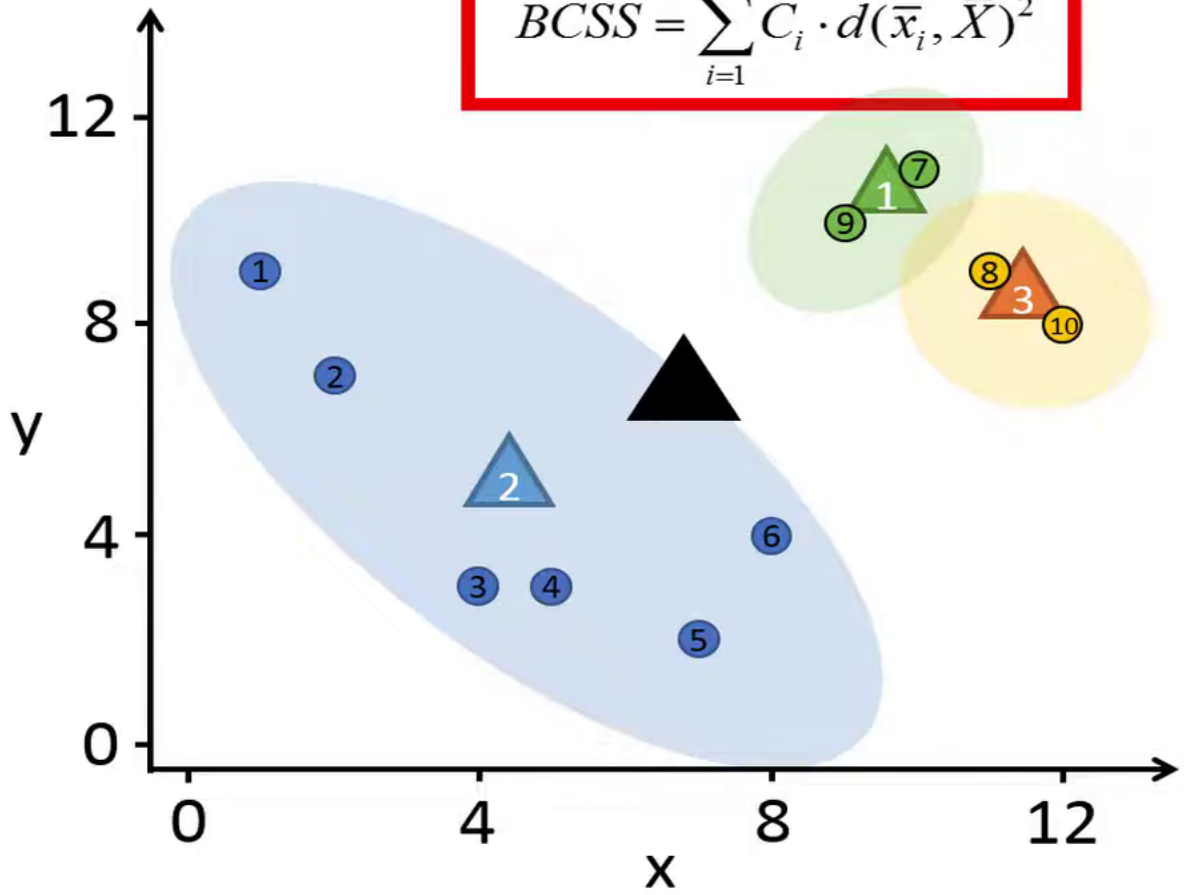


Then we sum these to get the total within cluster sum of squares, which is here equal to 76.83.

Initial positions of the centroids

	x	y
1	1	9
2	2	7
3	4	3
4	5	3
5	7	2
6	8	4
7	10	11
8	11	9
9	9	10
10	12	8

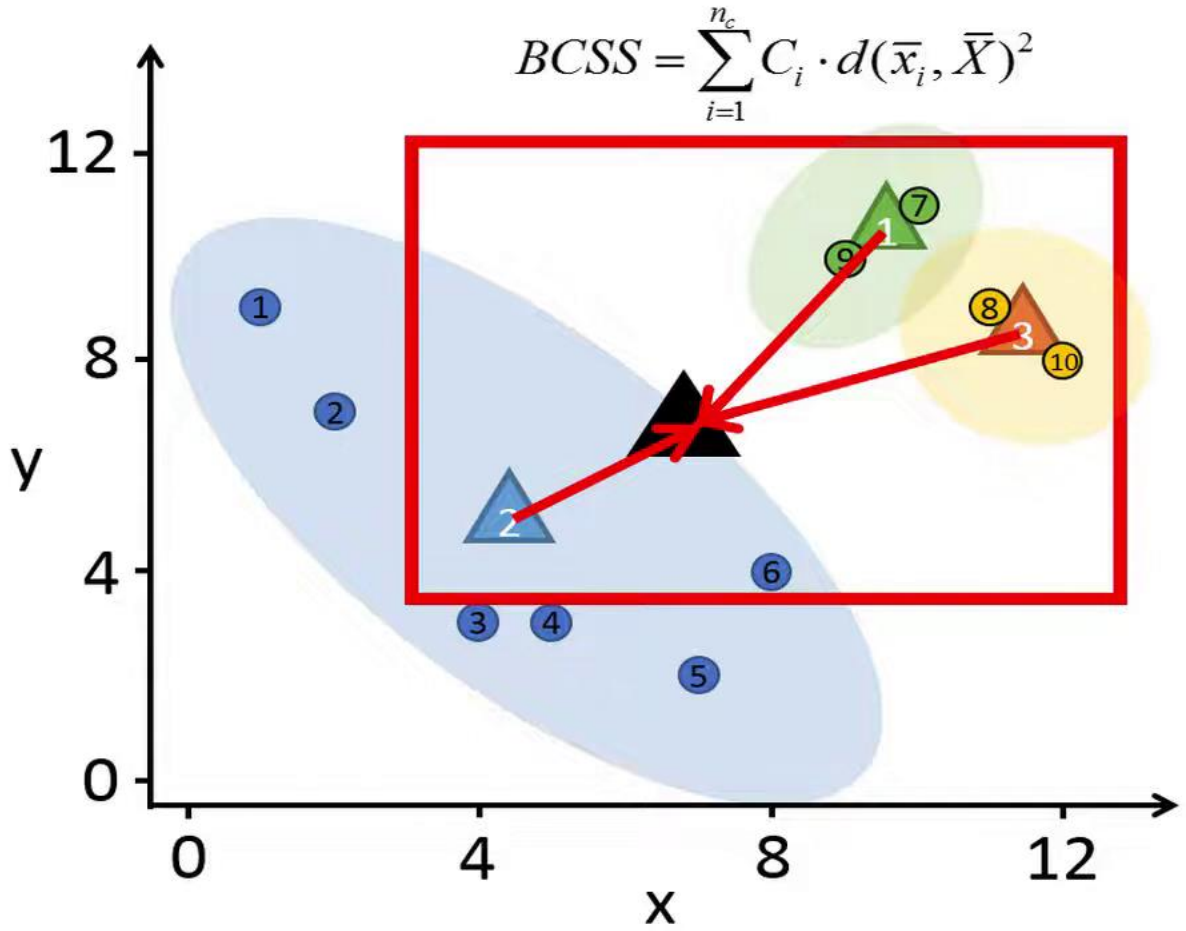
$$BCSS = \sum_{i=1}^{n_c} C_i \cdot d(\bar{x}_i, \bar{X})^2$$



An additional measure that is reported by many software tools is the between cluster sum of squares.

Initial positions of the centroids

	x	y
1	1	9
2	2	7
3	4	3
4	5	3
5	7	2
6	8	4
7	10	11
8	11	9
9	9	10
10	12	8

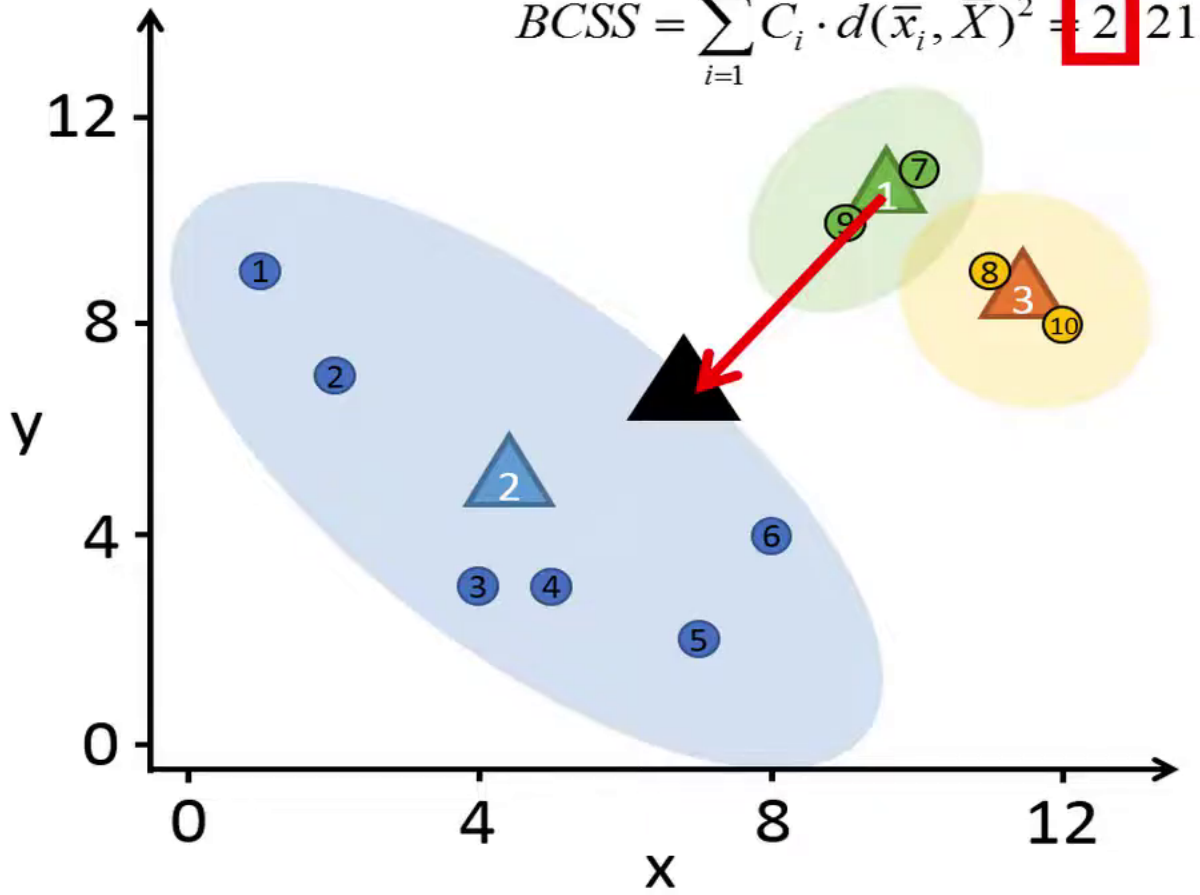


This is a measure of how far away the centroids are from the midpoint in the data set.

Initial positions of the centroids

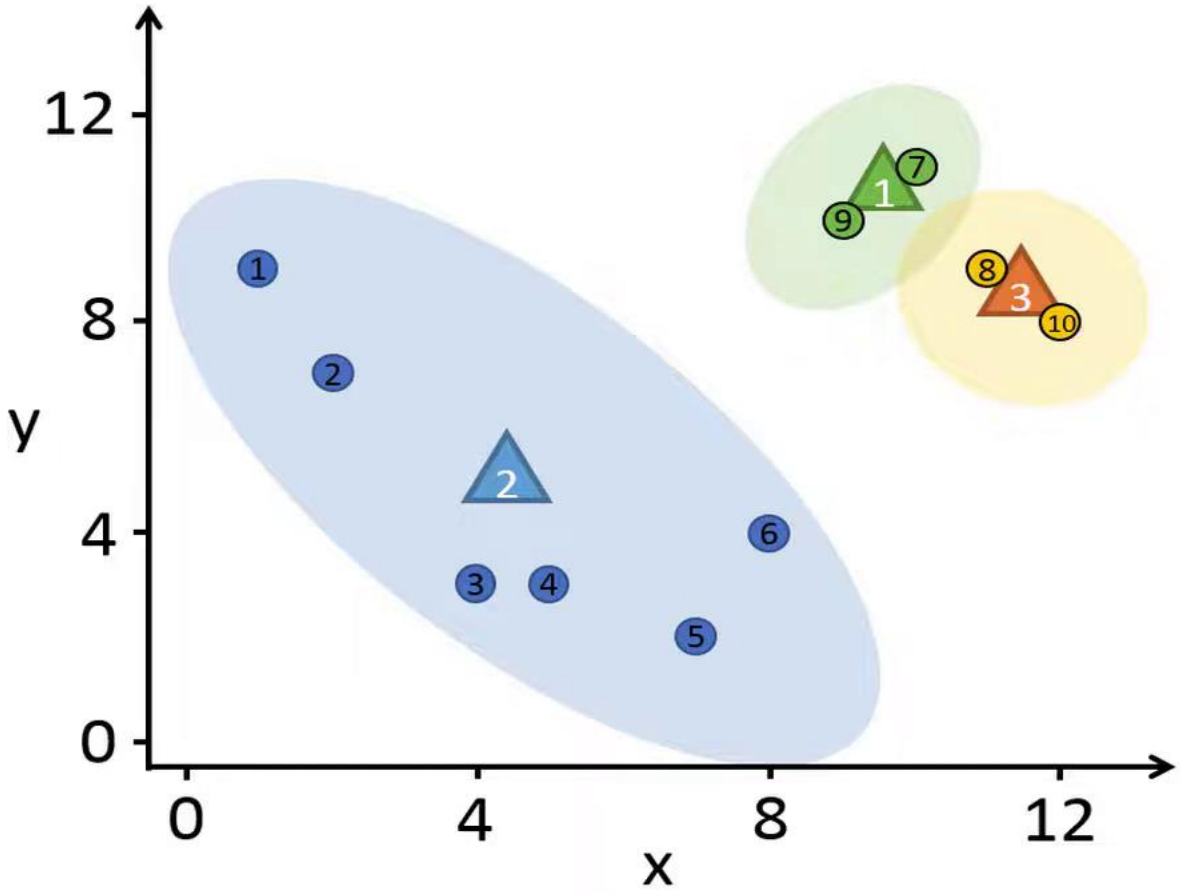
	x	y
1	1	9
2	2	7
3	4	3
4	5	3
5	7	2
6	8	4
7	10	11
8	11	9
9	9	10
10	12	8

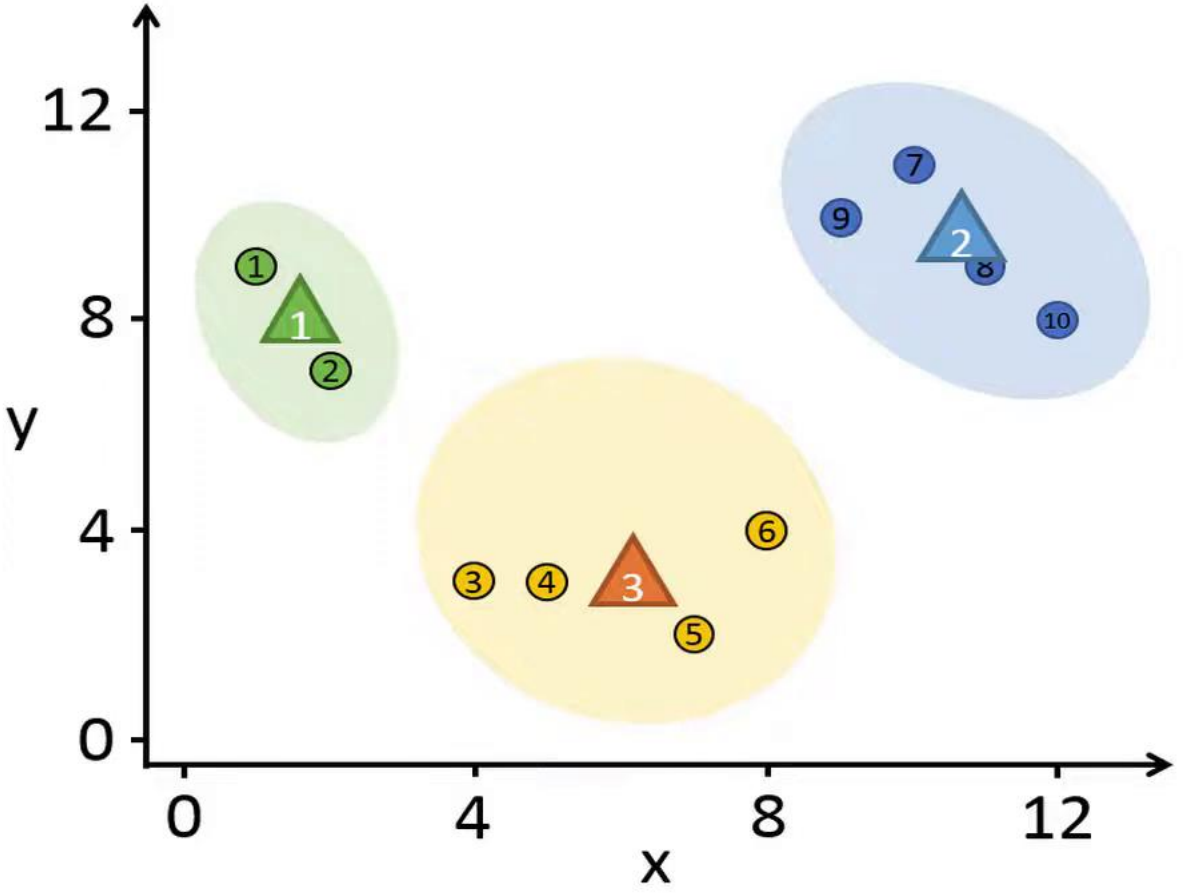
$$BCSS = \sum_{i=1}^{n_c} C_i \cdot d(\bar{x}_i, \bar{X})^2 = 2\,21.97 + 6 \cdot 9.50 + 2 \cdot 24.77 = 150.47$$



Since this centroid is associated to a cluster with two data points, the distance is multiplied by two.

Initial positions of the centroids

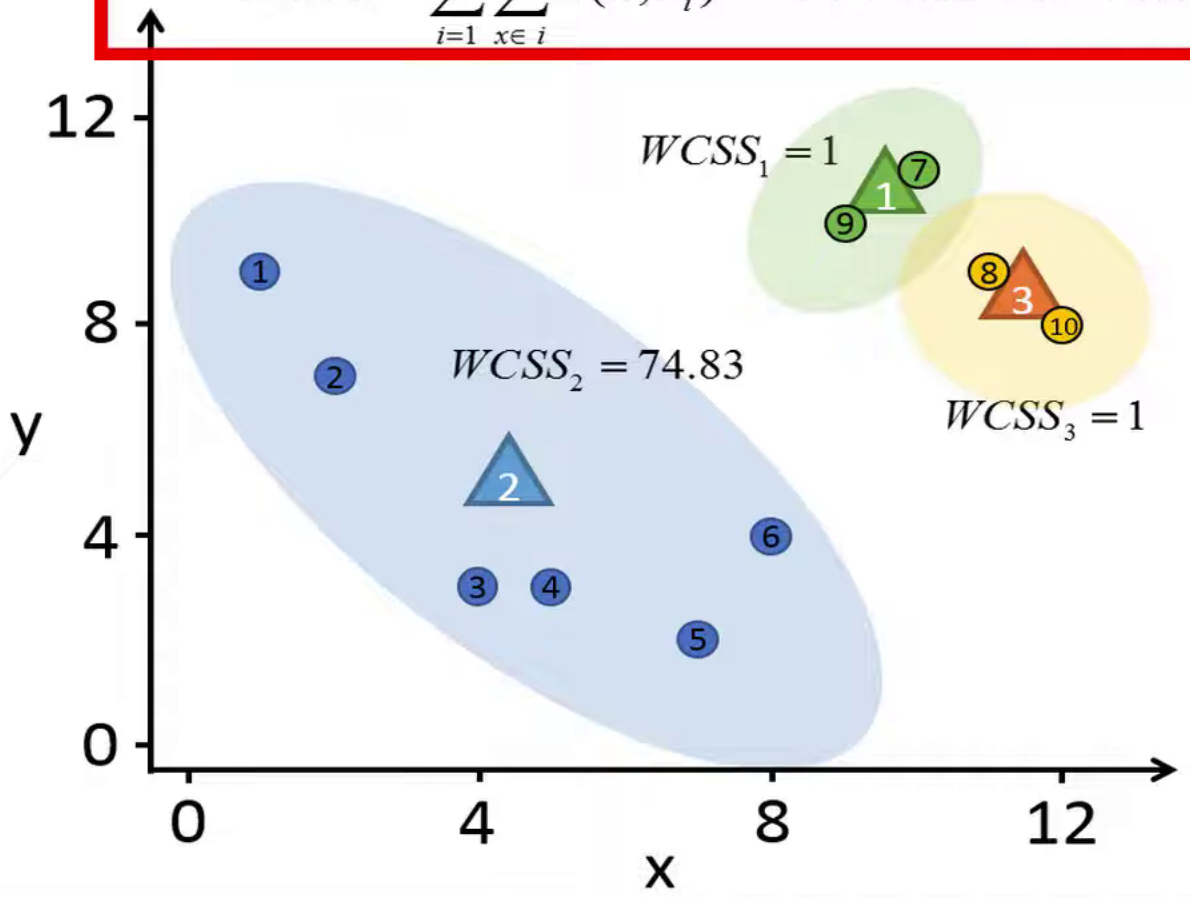




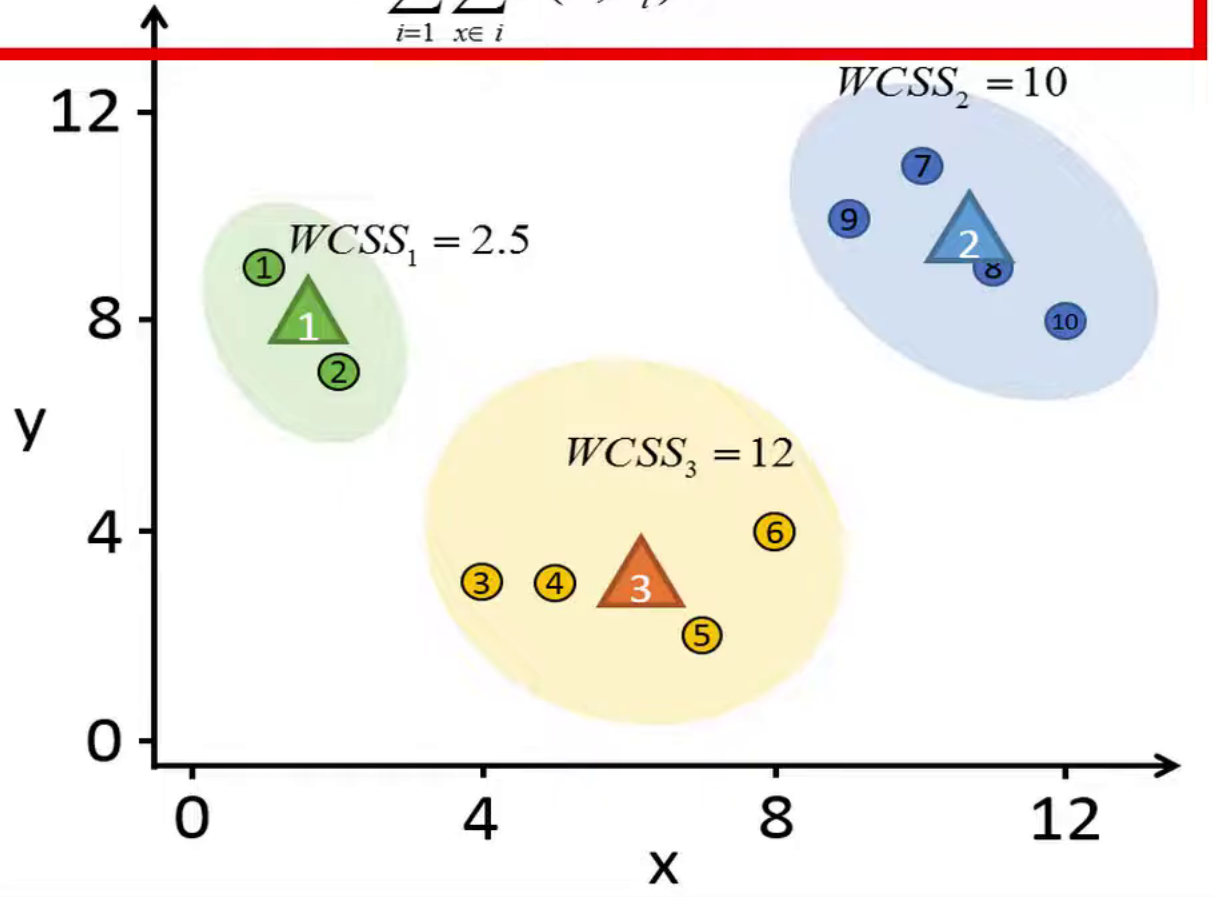
Remember that k-means clustering may result in different outputs depending on the initial location of the centroids. We will here use the total within cluster sum of squares to determine which of these two outputs that generate the "best" clusters.

Initial positions of the centroids

$$WCSS = \sum_{i=1}^{n_c} \sum_{x \in i} d(x, \bar{x}_i)^2 = 1 + 74.83 + 1 = 76.83$$



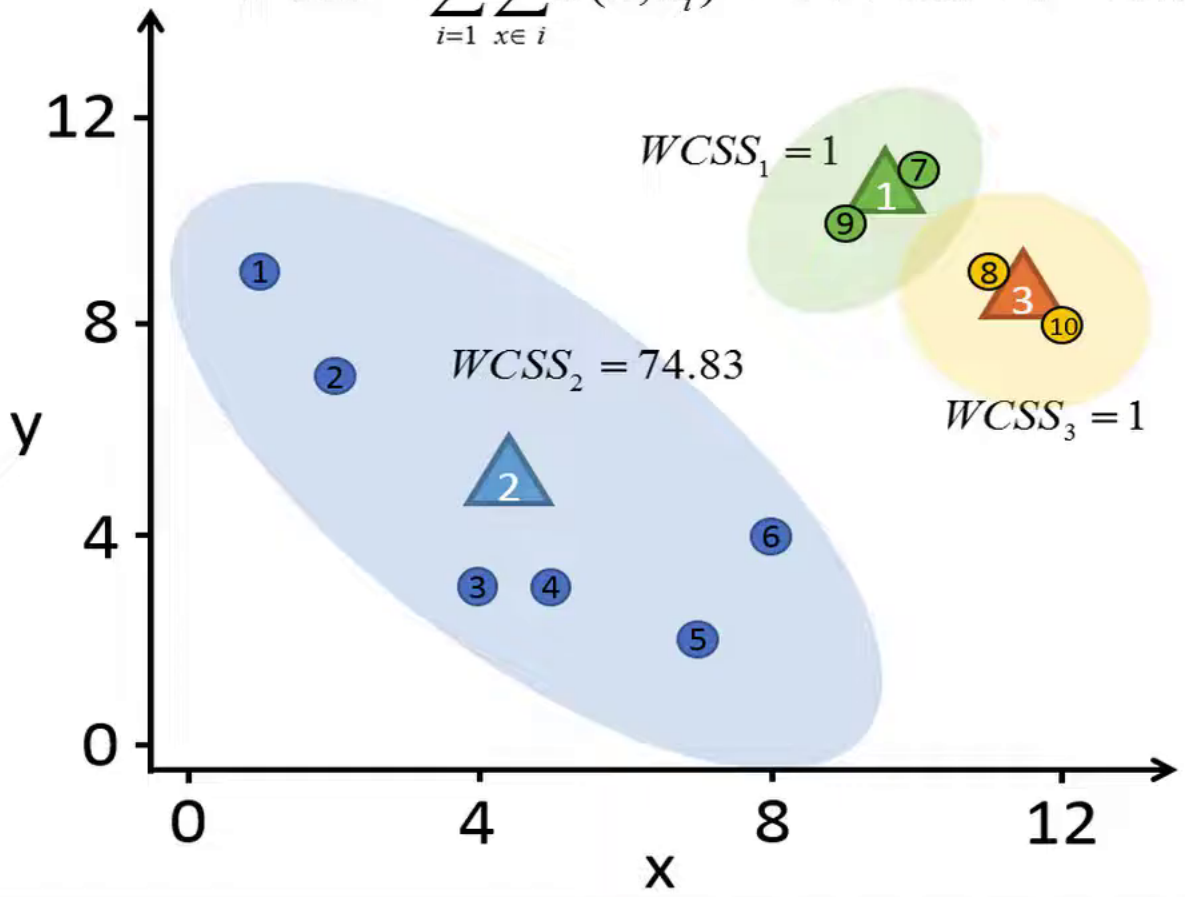
$$WCSS = \sum_{i=1}^{n_c} \sum_{x \in i} d(x, \bar{x}_i)^2 = 2.5 + 10 + 12 = 24.5$$



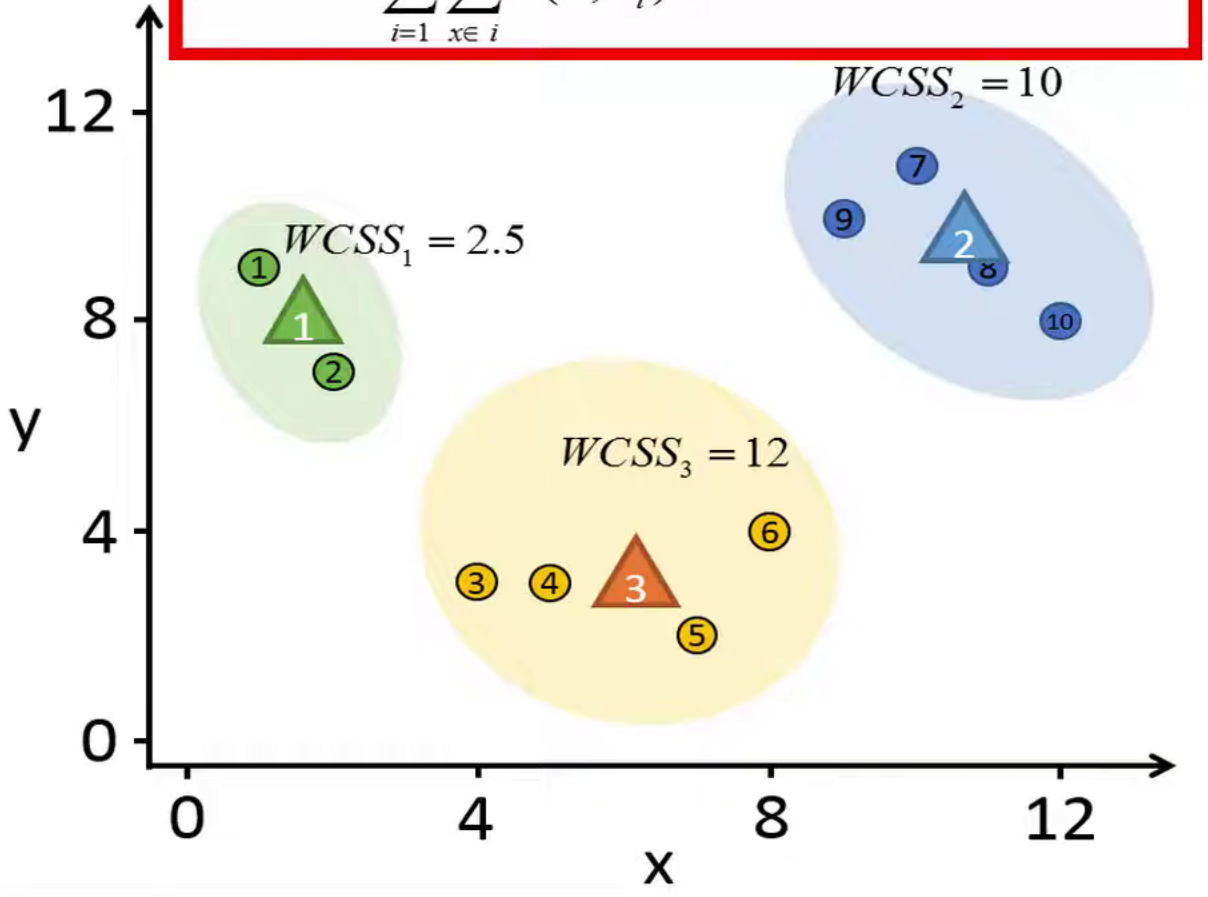
We therefore calculate the total within cluster sum of squares of the two different outputs.

Initial positions of the centroids

$$WCSS = \sum_{i=1}^{n_c} \sum_{x \in i} d(x, \bar{x}_i)^2 = 1 + 74.83 + 1 = 76.83$$



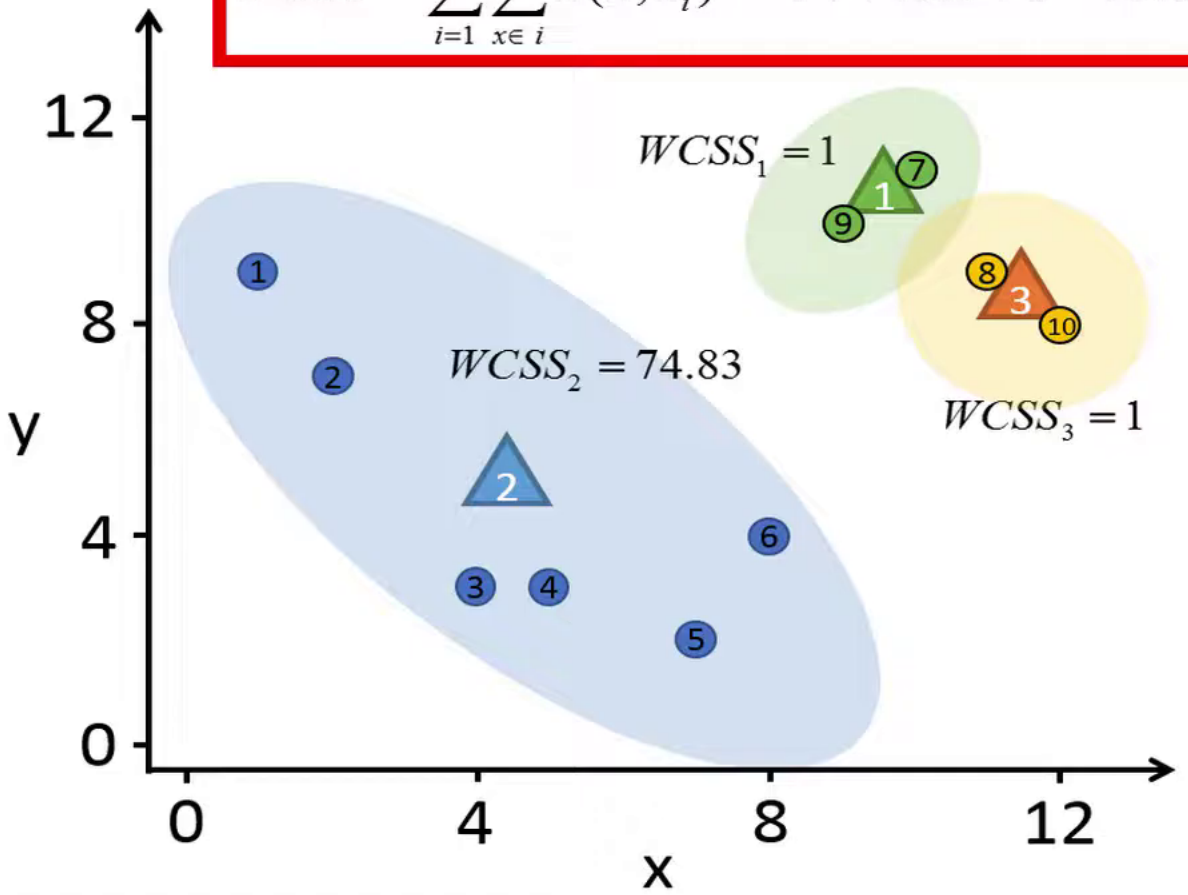
$$WCSS = \sum_{i=1}^{n_c} \sum_{x \in i} d(x, \bar{x}_i)^2 = 2.5 + 10 + 12 = 24.5$$



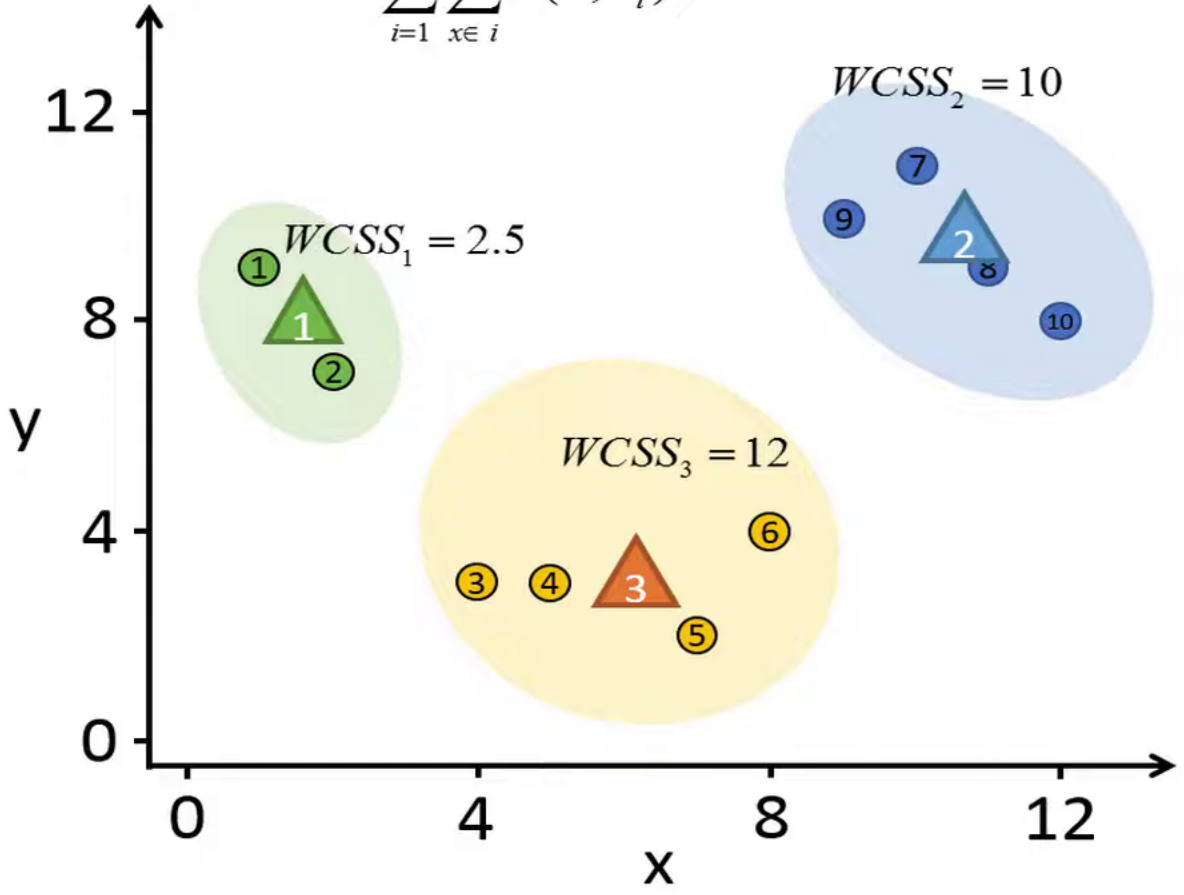
Since the within cluster sum of squares is much smaller in the case to the right,

Initial positions of the centroids

$$WCSS = \sum_{i=1}^{n_c} \sum_{x \in i} d(x, \bar{x}_i)^2 = 1 + 74.83 + 1 = 76.83$$



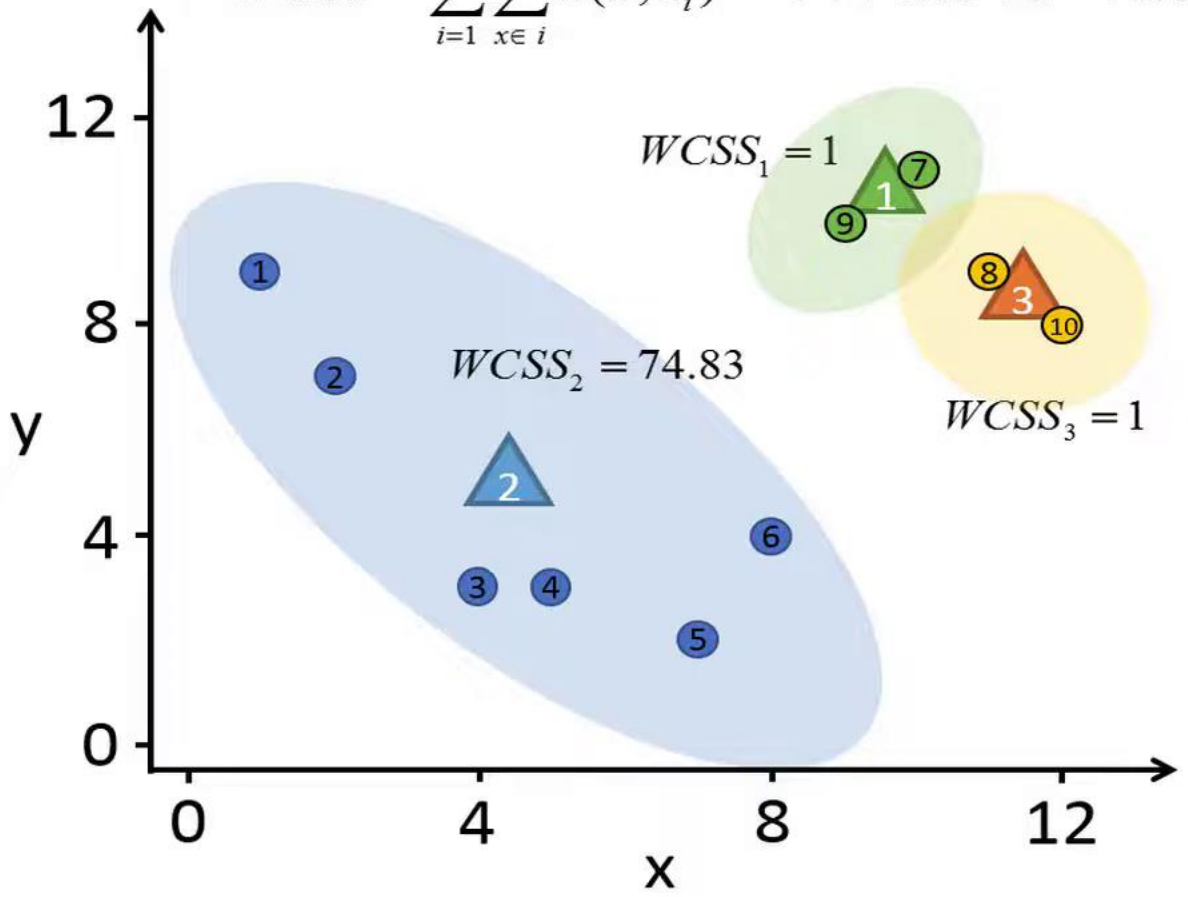
$$WCSS = \sum_{i=1}^{n_c} \sum_{x \in i} d(x, \bar{x}_i)^2 = 2.5 + 10 + 12 = 24.5$$



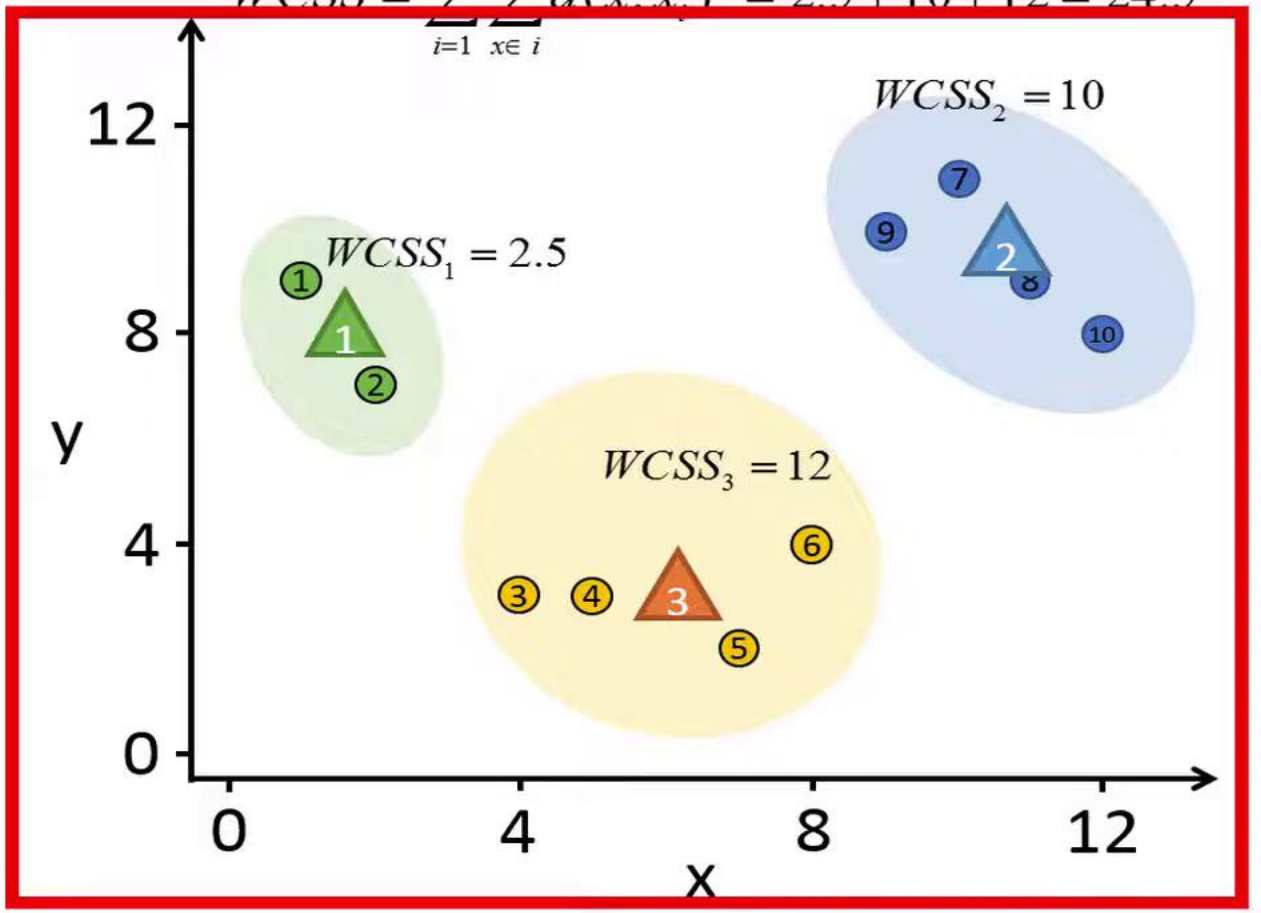
compared to the one to the left,

Initial positions of the centroids

$$WCSS = \sum_{i=1}^{n_c} \sum_{x \in i} d(x, \bar{x}_i)^2 = 1 + 74.83 + 1 = 76.83$$



$$WCSS = \sum_{i=1}^{n_c} \sum_{x \in i} d(x, \bar{x}_i)^2 = 2.5 + 10 + 12 = 24.5$$



we would select this clustering, which seems reasonable because the data points are generally closer to the centroids, and the clusters are well separated.

Try different values of *k*

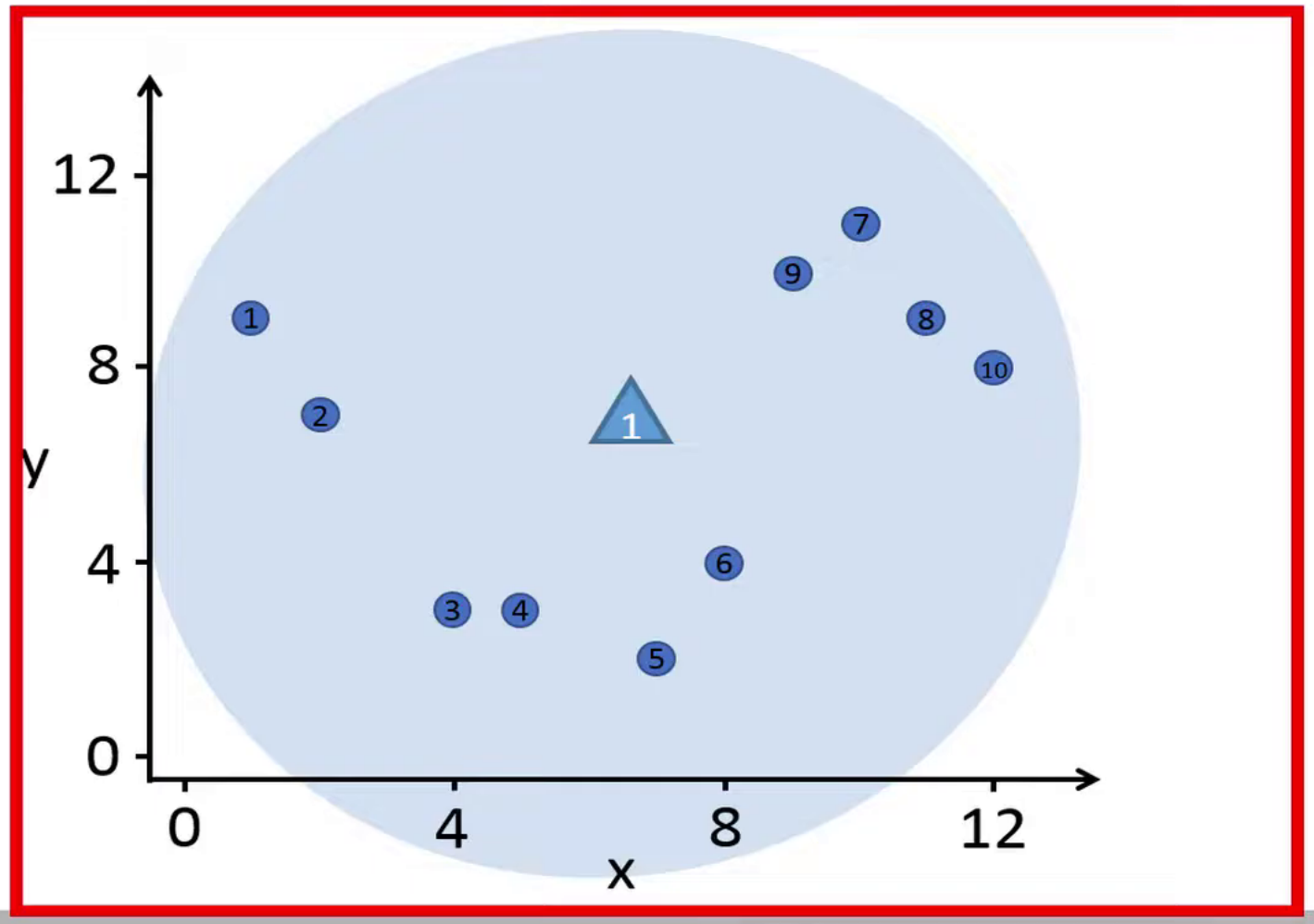
It is therefore important that we try many different initial positions of the centroids and that we then select the clustering that results in the smallest within cluster sum of squares.

Try different values of *k*

It is therefore important that we try many different initial positions of the centroids and that we then select the clustering that results in the smallest within cluster sum of squares.

The same method can also be used when we evaluate different values of *k*.

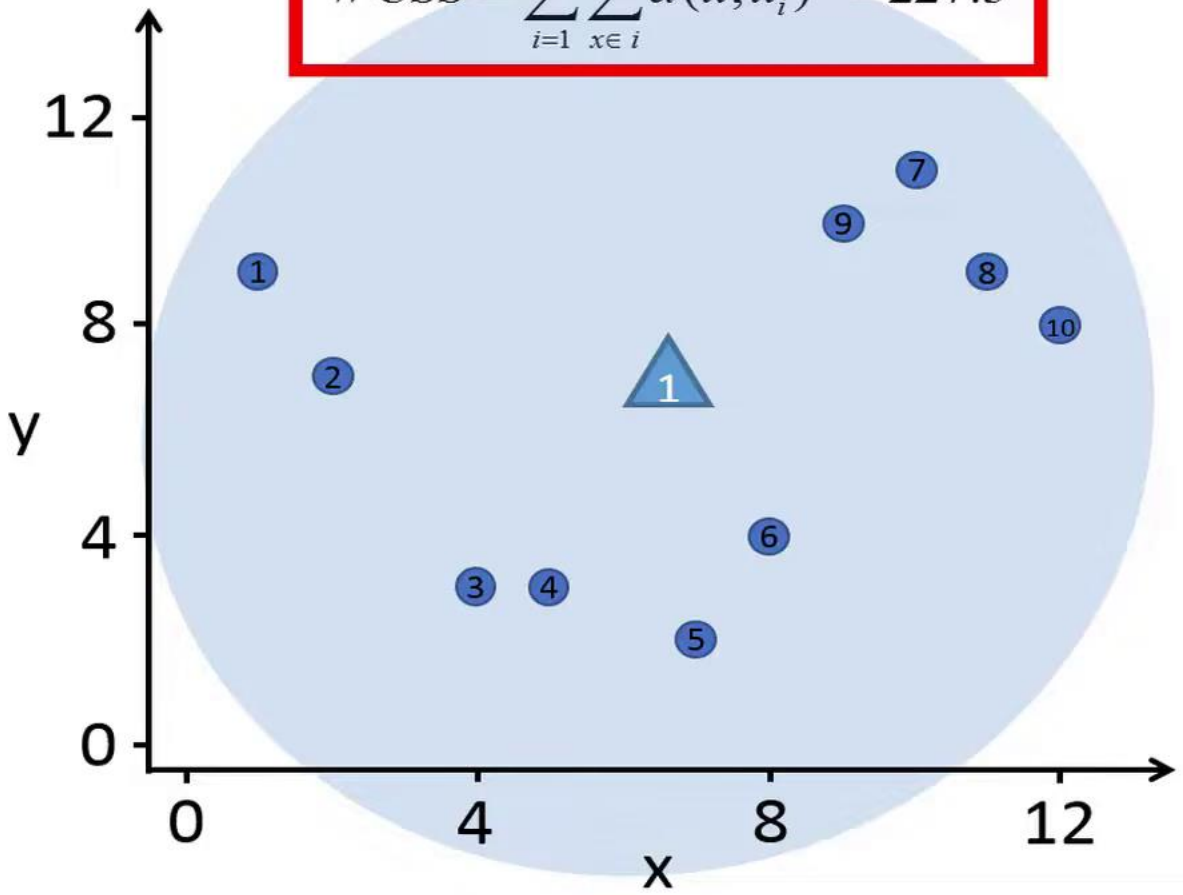
k=1



If we would set k to one, which means that we only have one cluster,

k=1

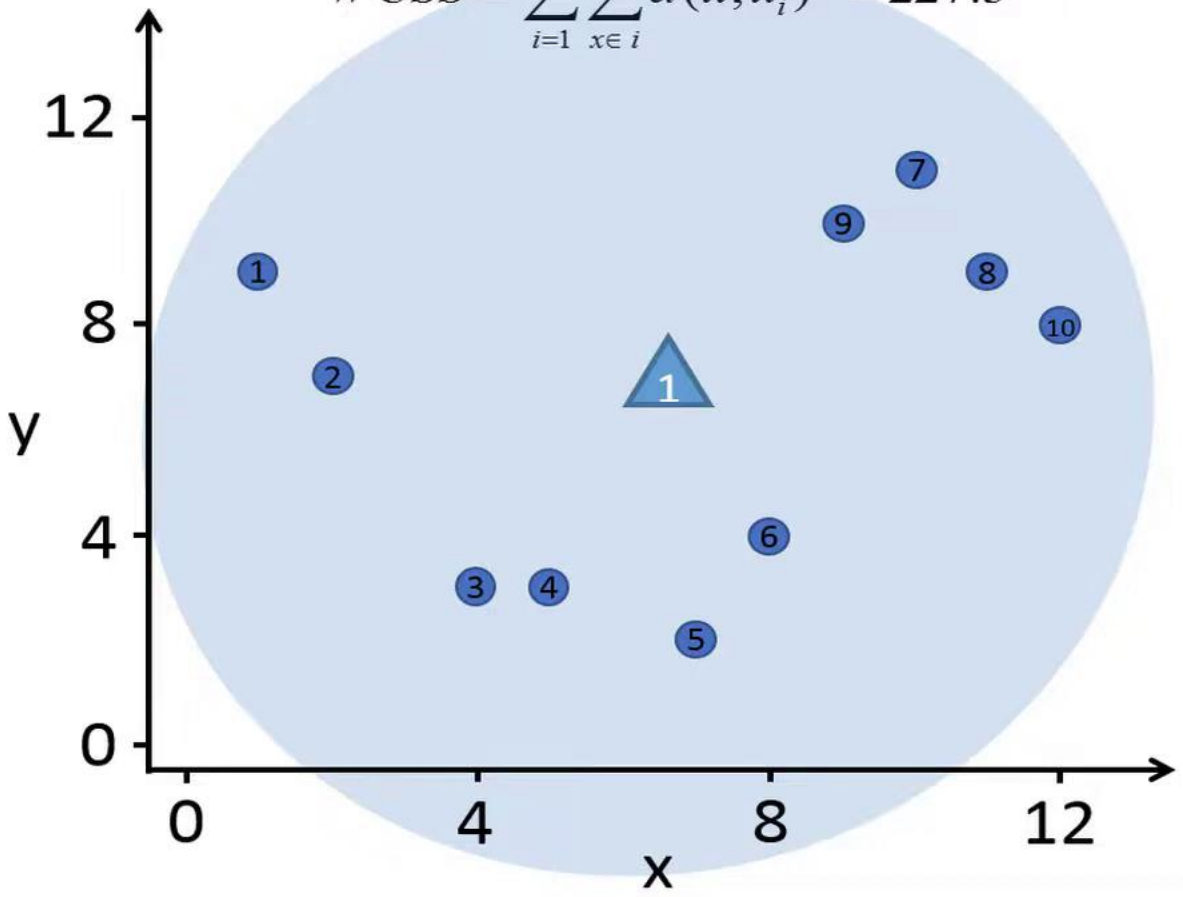
$$WCSS = \sum_{i=1}^{n_c} \sum_{x \in i} d(x, \bar{x}_i)^2 = 227.3$$

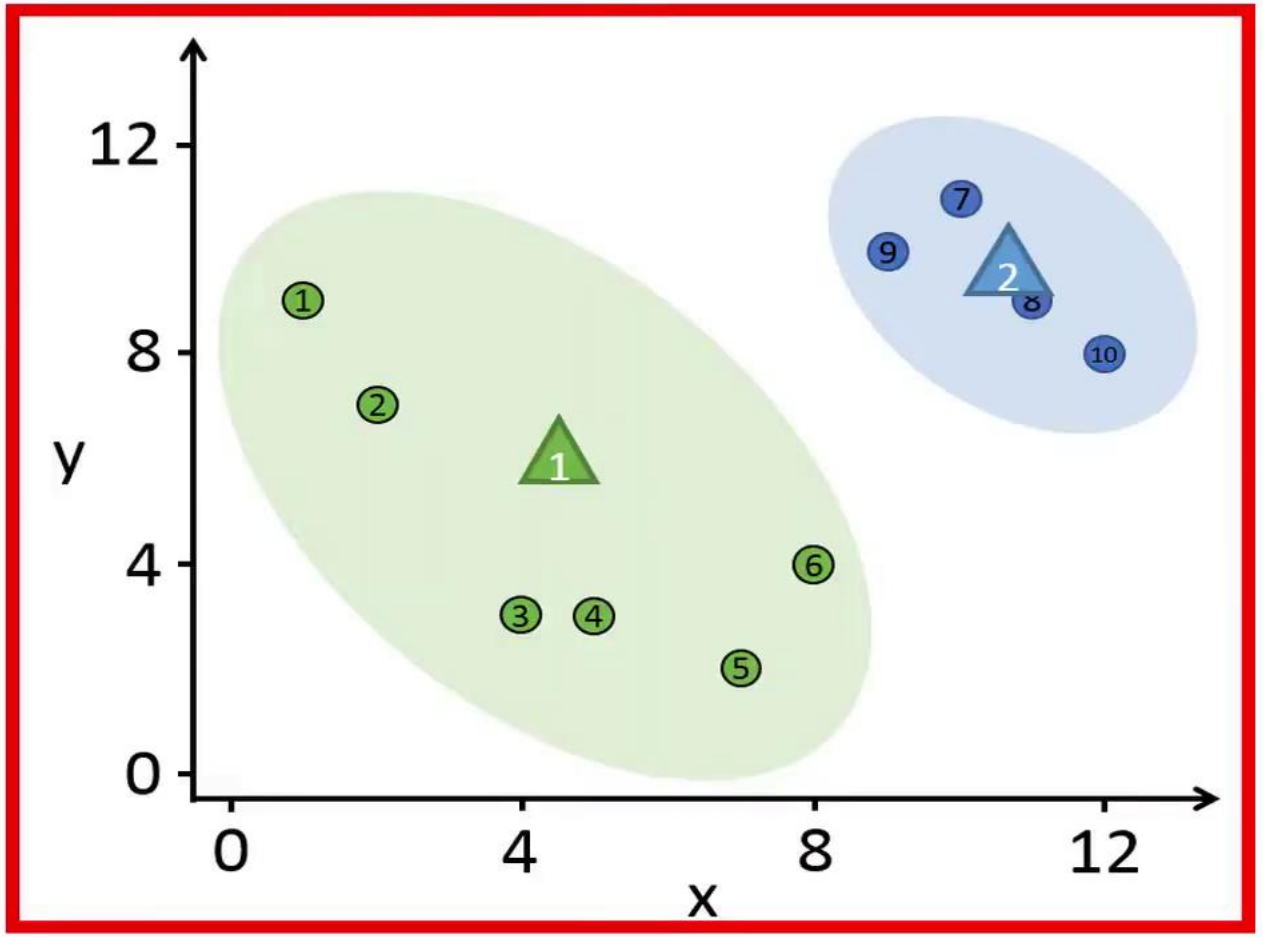


the within cluster sum of squares is equal to 227.3. Since we only have one cluster in this case, the between cluster sum of squares is equal to zero.

k=2

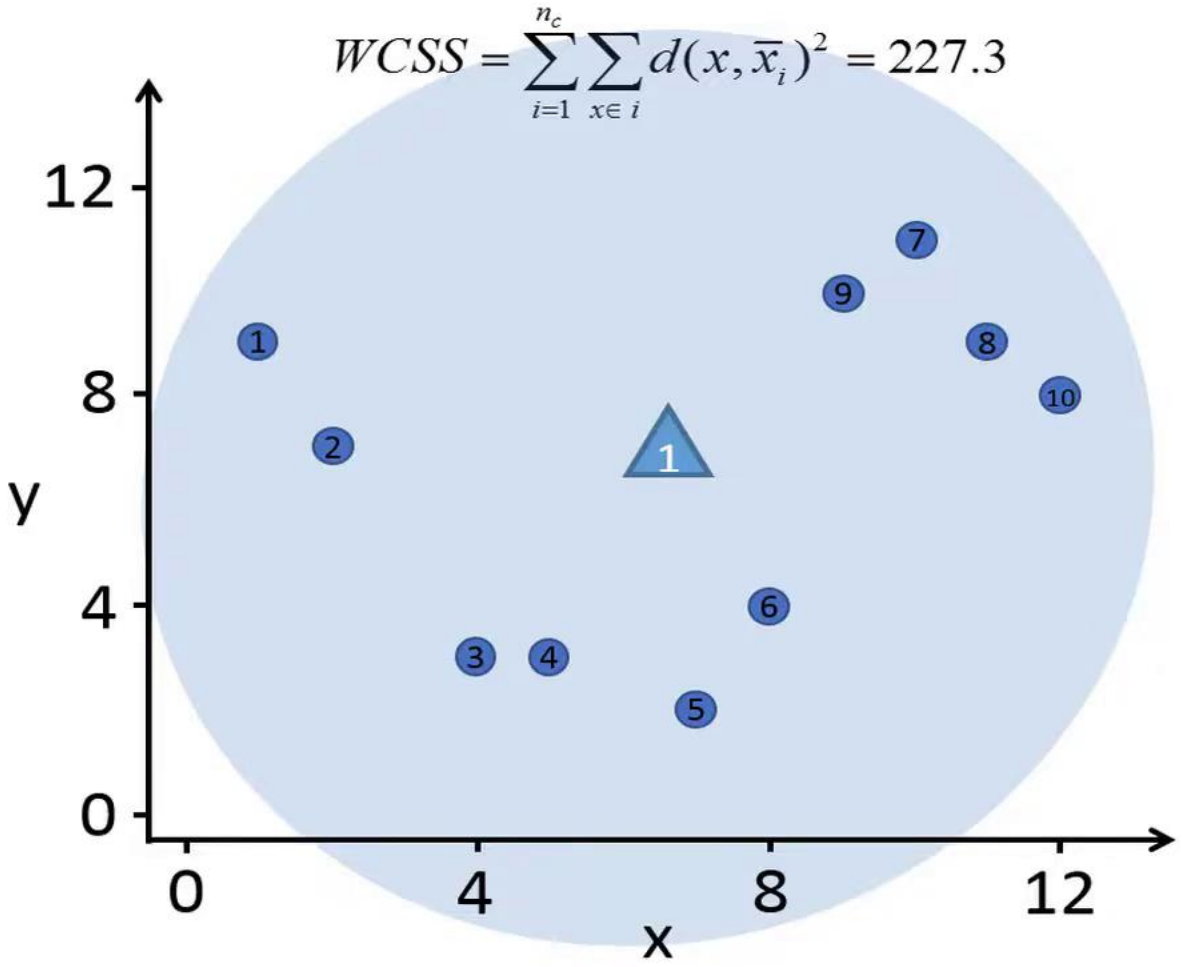
$$WCSS = \sum_{i=1}^{n_c} \sum_{x \in i} d(x, \bar{x}_i)^2 = 227.3$$

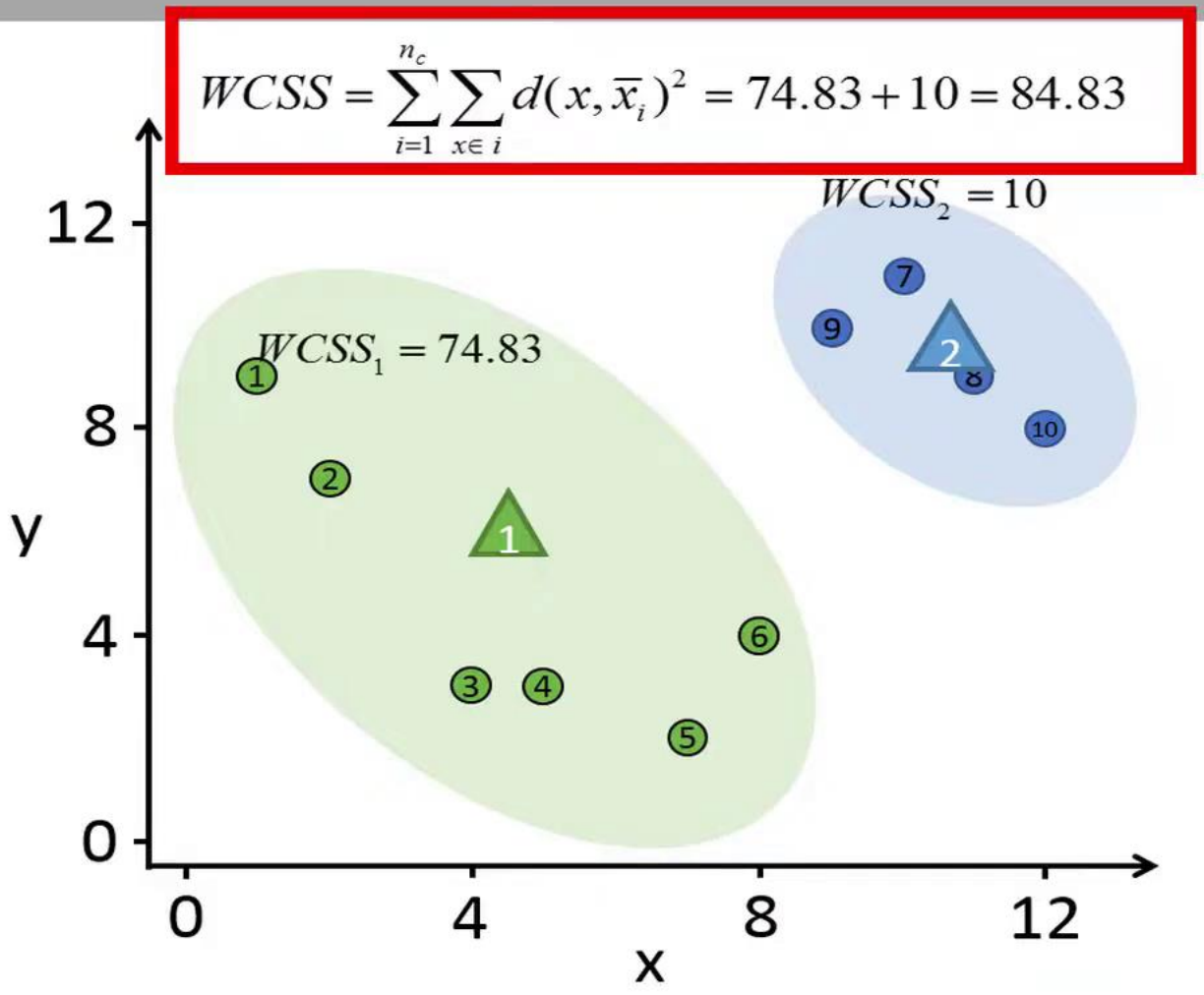




If we set *k* to 2, the following two clusters will be generated,

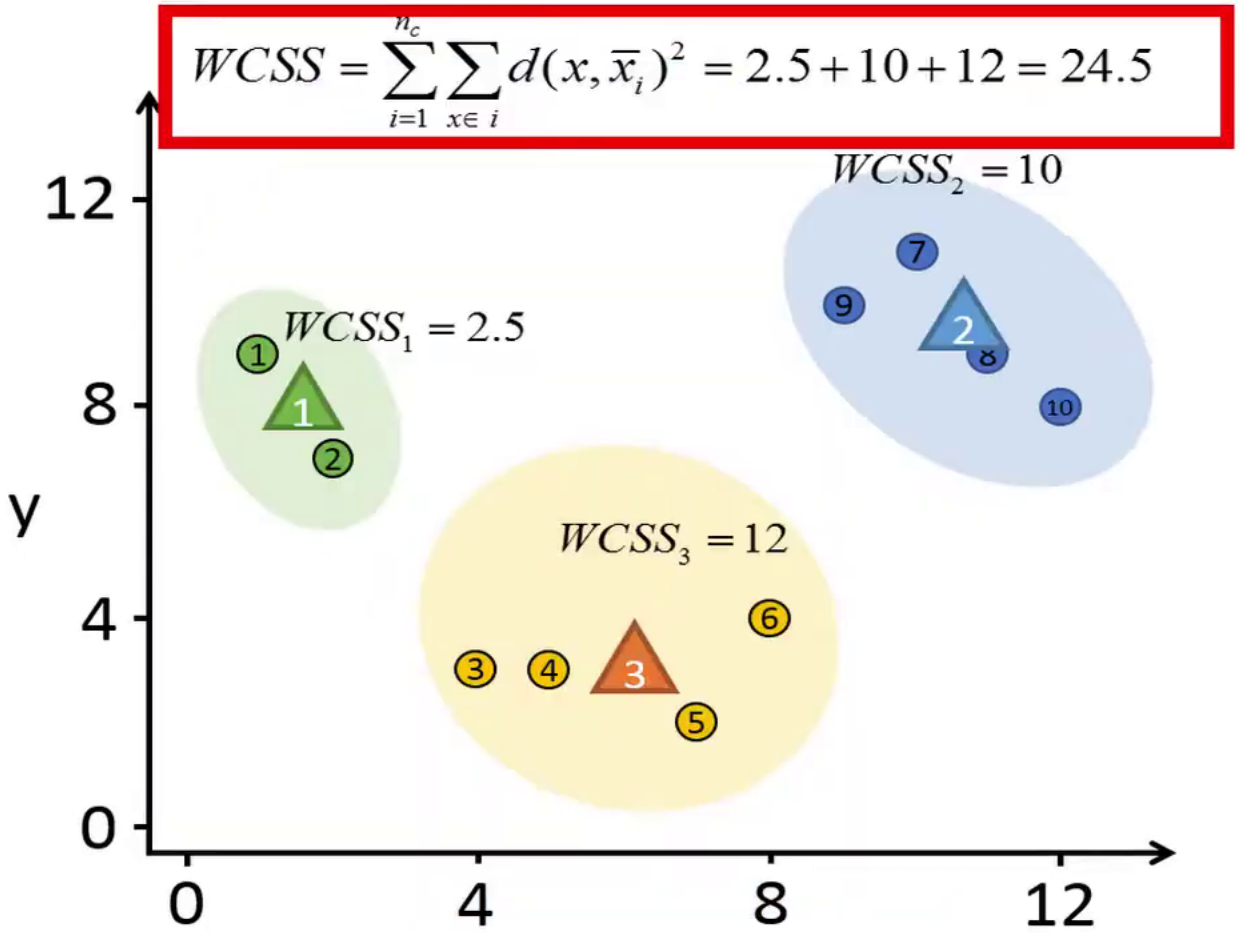
k=2





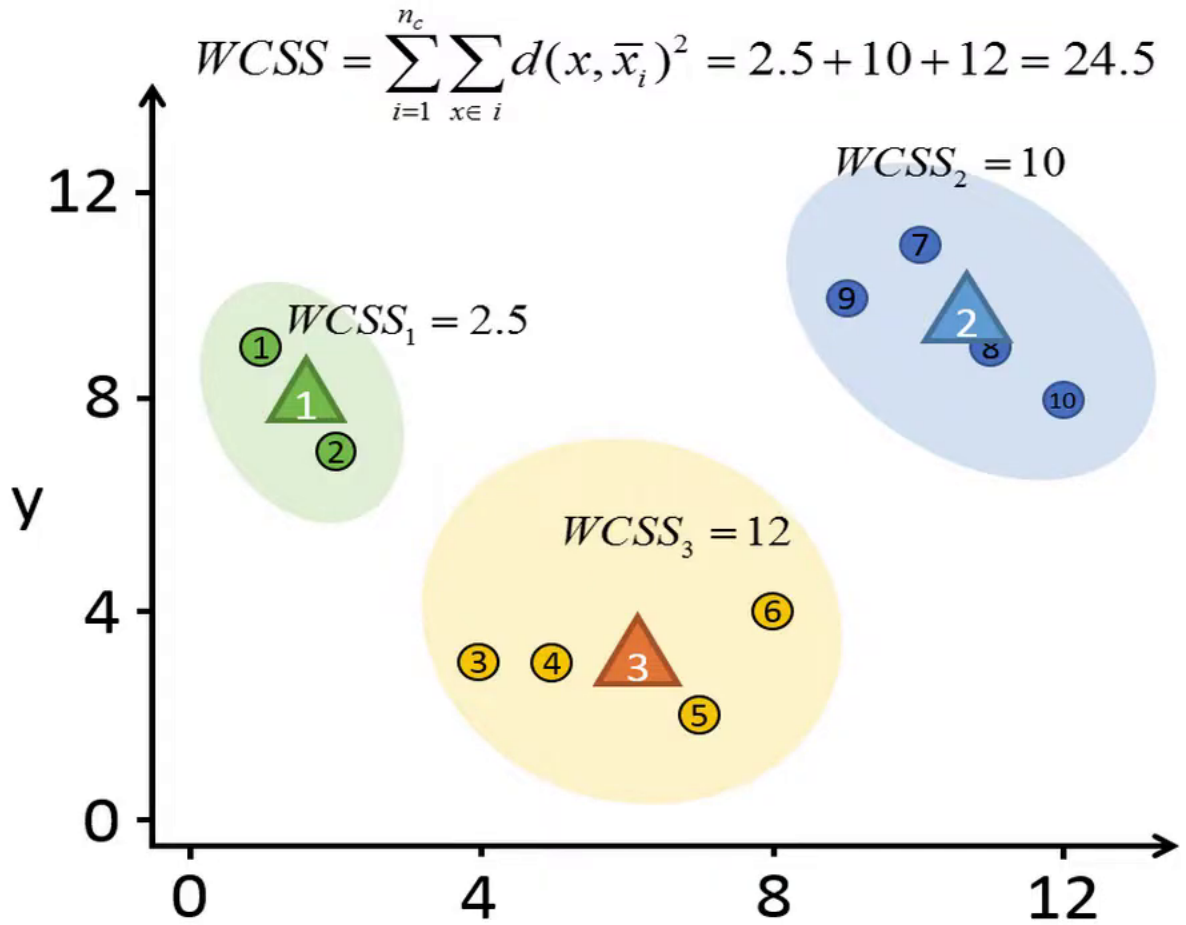
with a within cluster sum of squares that has decreased to 84.83 because the data points are now much closer to the centroids.

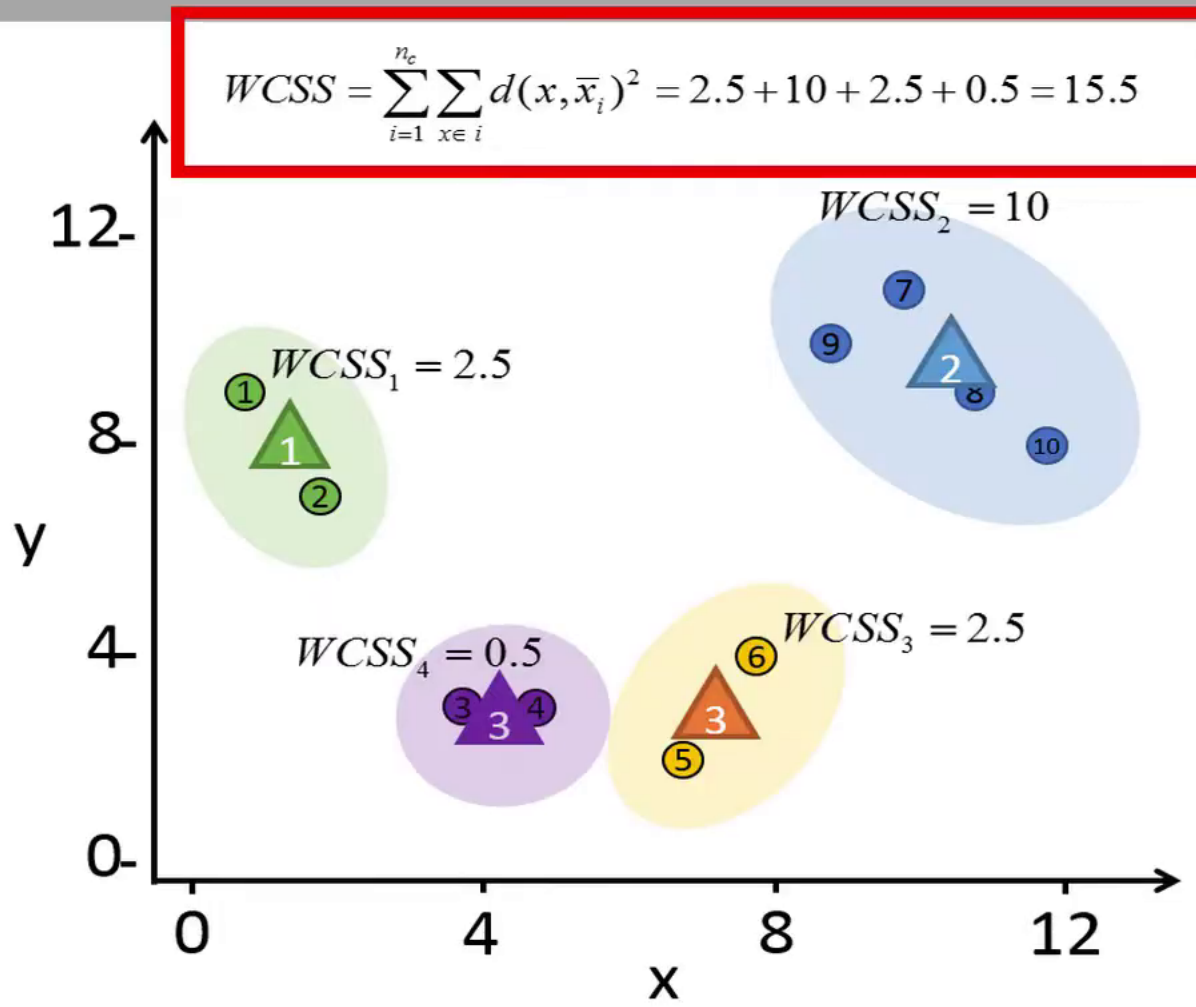
k=3



If we set *k* to three, the within cluster sum of squares has been reduced to 24.5,

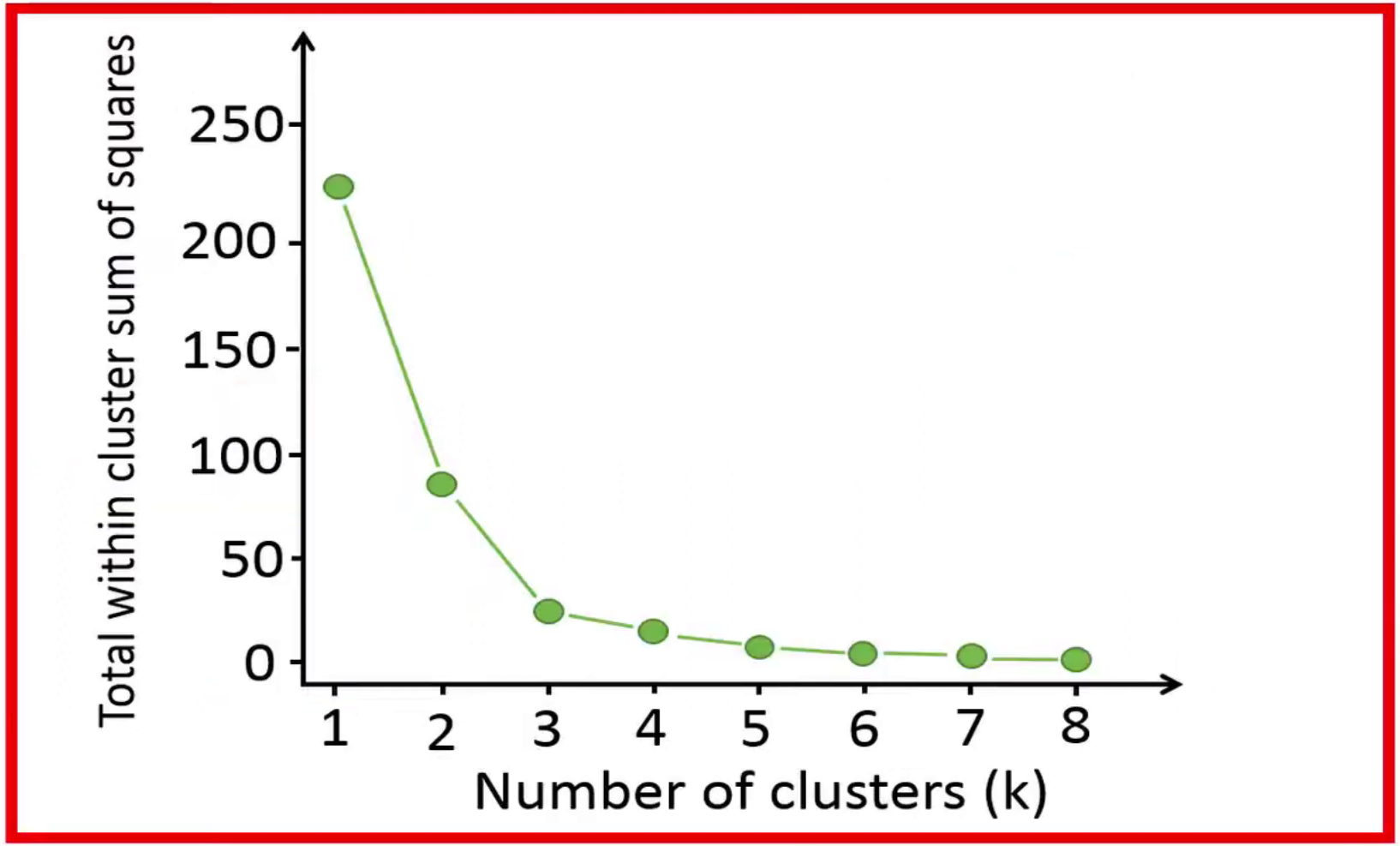
k=4





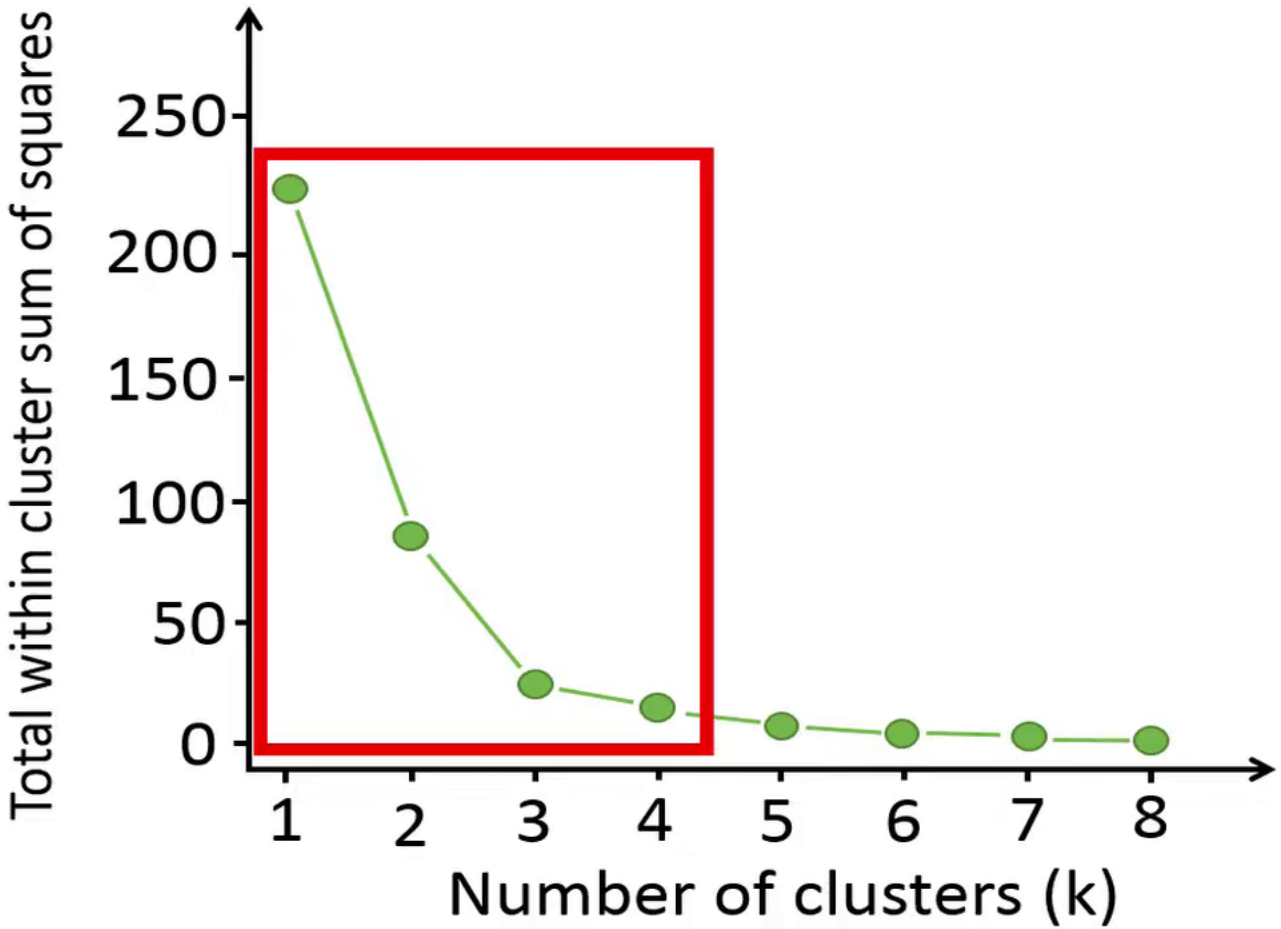
and if we set *k* to four, the within cluster sum of squares has been reduced to 15.5, and so forth.

The elbow method



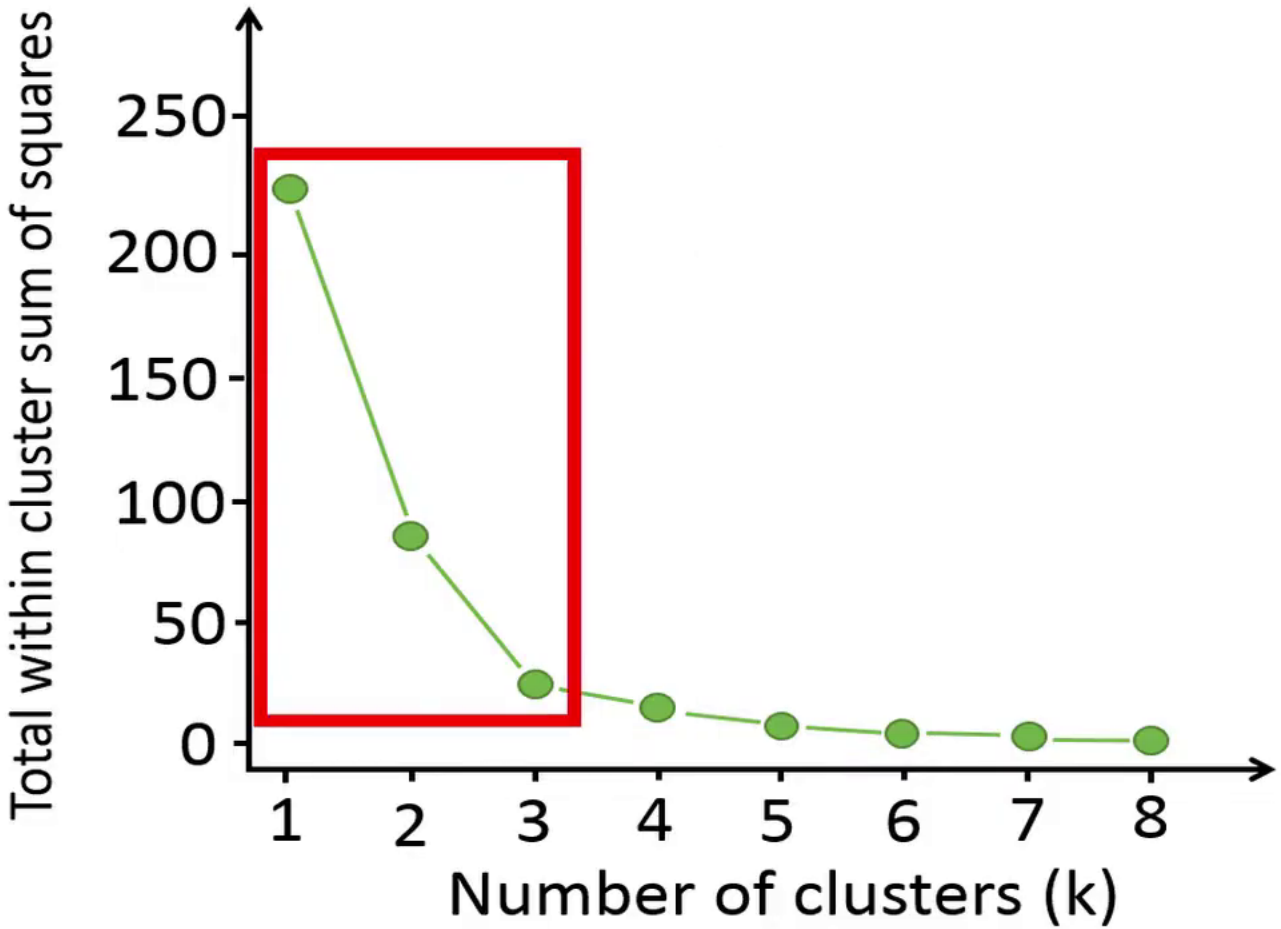
If we plot the total within cluster sum of squares for different values of *k*, we will get the following plot.

The elbow method



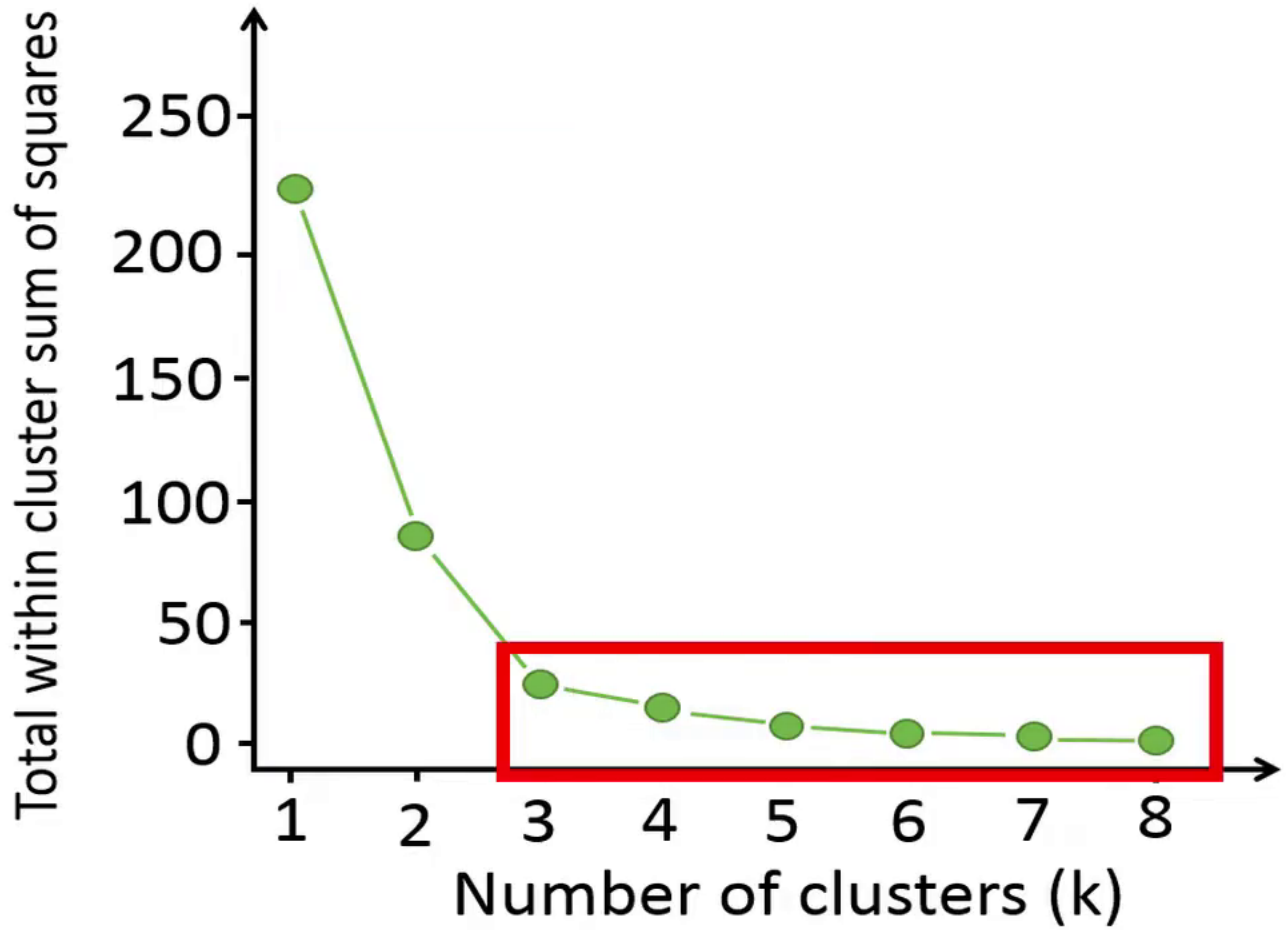
These points represent the total within cluster sum of squares from our previous examples when we tried four different values of *k*.

The elbow method



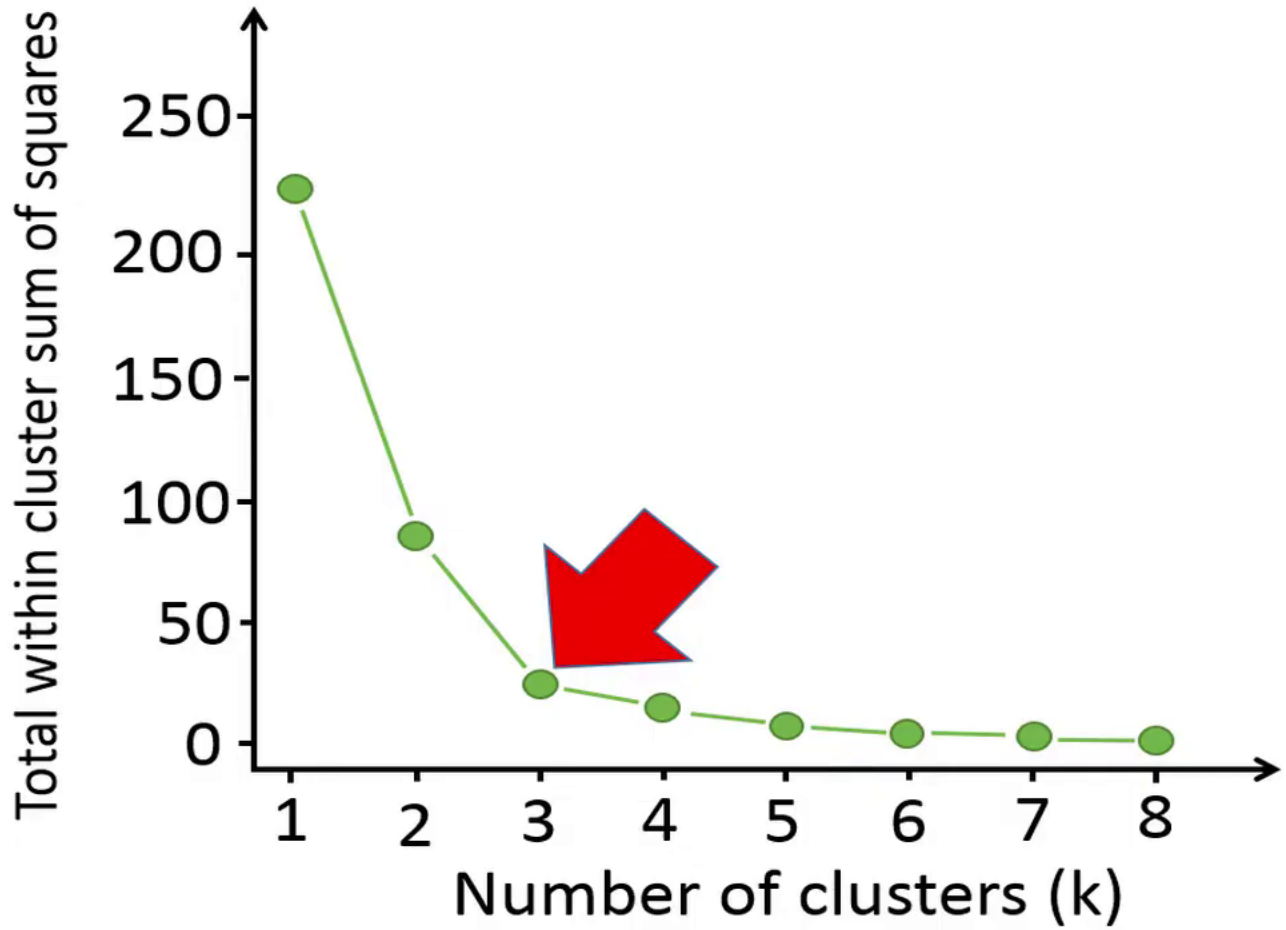
Note that there is a steep decrease in the total within cluster sum of squares when *k* is increased from one to three,

The elbow method



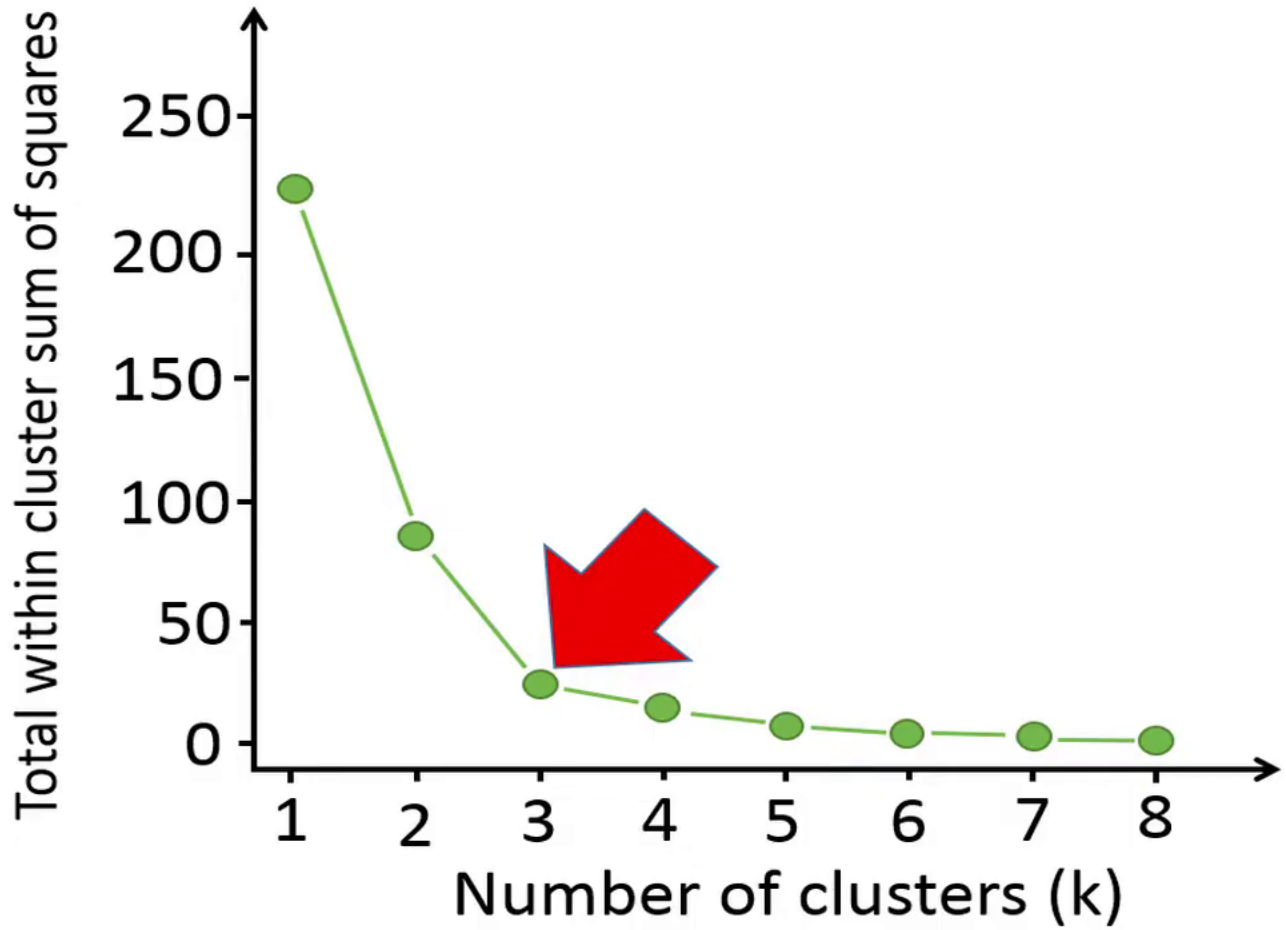
but much less when *k* is increased from three to eight.

The elbow method



The optimal value of *k* is the value at the "elbow" of the curve. The optimal number of clusters for our example data is therefore three. Finding the optimal value of *k* this way is called the elbow method.

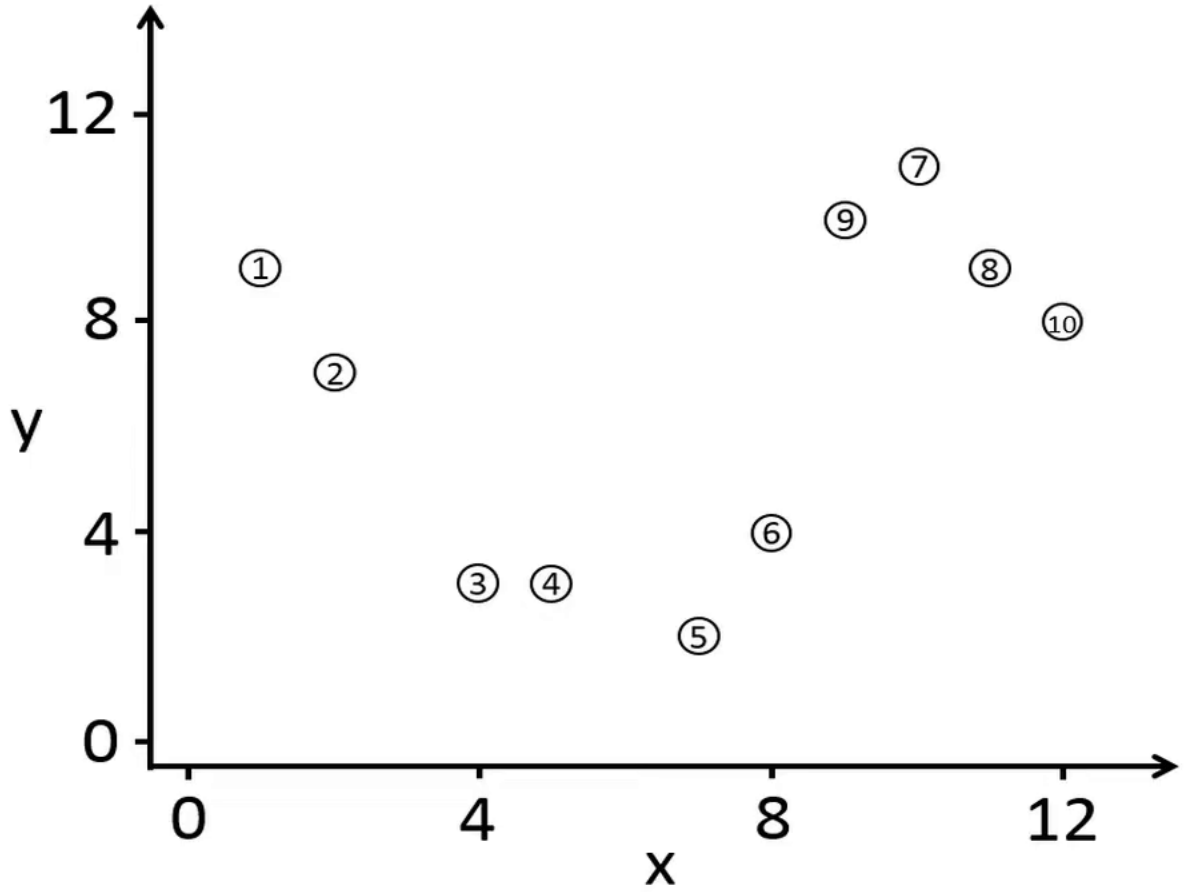
The elbow method



Identifying the elbow of the curve is usually done simply by studying the curve. Note that there are several other methods to find an optimal value of *k* that are not discussed here.

k-means clustering

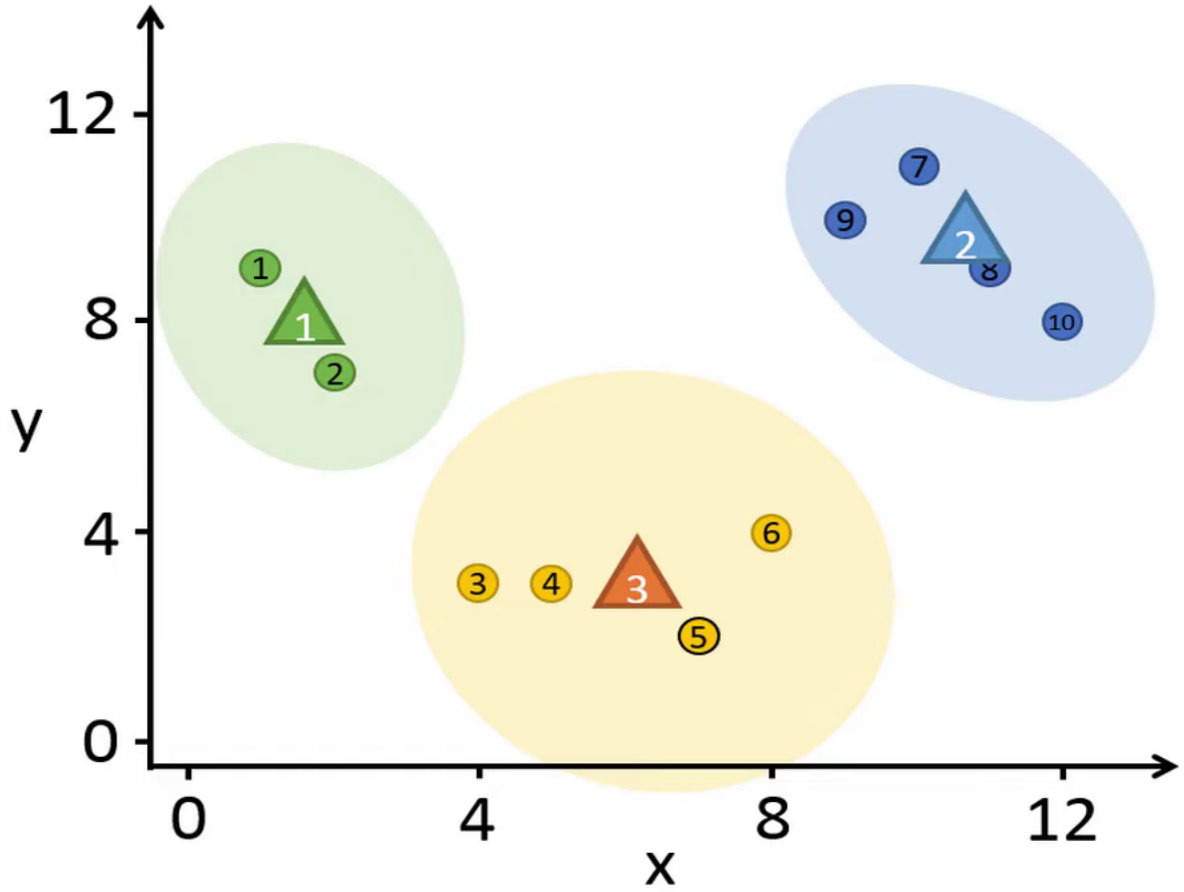
	x	y
1	1	9
2	2	7
3	4	3
4	5	3
5	7	2
6	8	4
7	10	11
8	11	9
9	9	10
10	12	8



Finally, we will try to understand the strength of the k-means clustering. In our previous example, it was quite easy to identify three clusters of this data set by eye.

k-means clustering

	x	y
1	1	9
2	2	7
3	4	3
4	5	3
5	7	2
6	8	4
7	10	11
8	11	9
9	9	10
10	12	8



So, if we can do this by eye, why do we need k-means clustering?

k-means clustering

	x	y	u	v
1	1	9	5	3
2	2	7	4	4
3	4	3	2	5
4	5	3	5	6
5	7	2	2	7
6	8	4	2	12
7	10	11	4	4
8	11	9	6	3
9	9	10	4	2
10	12	8	9	4

Imagine that we now instead would have four variables and would like to find three clusters based on all these four variables. It is now impossible to plot this in a simple way so that we can identify three clusters.

k-means clustering

	x	y	u	v
1	1	9	5	3
2	2	7	4	4
3	4	3	2	5
4	5	3	5	6
5	7	2	2	7
6	8	4	2	12
7	10	11	4	4
8	11	9	6	3
9	9	10	4	2
10	12	8	9	4

$$d = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2 + (u_1 - u_2)^2 + (v_1 - v_2)^2}$$

However, a computer can easily calculate the Euclidean distance in the four-dimensional space and determine which data points in this space that belong to each of the three clusters.